Time-Domain Imaging of Rotating, Cloud-Covered Exoplanets with the Solar Gravitational Lens: Reconstruction Algorithms, Observing-Cadence Requirements, and Mission Targets

Sanskar Sontakke

Independent Researcher

sanskarsontakke@gmail.com ORCID: 0009-0008-4531-5707

ABSTRACT

The solar gravitational lens (SGL) can, in principle, deliver multipixel images of terrestrial exoplanets from heliocentric distances $z \simeq 650$–900 AU. In practice the planet rotates and its cloud cover evolves while a spacecraft raster-scans the kilometer-scale image cylinder one "pixel" at a time, so the measurements are temporally aliased; recent studies identify this dynamic inversion problem as a leading unsolved challenge for the concept. We present an open, end-to-end simulation framework that reproduces the published SGL photometric benchmarks (the aperture-averaged $d/4\rho$ kernel; per-sample $\mathrm{SNR_C} = 43.16$ for 1800 s dwells on an exo-Earth at 30 pc from 650 AU; the $0.891\,D/(d\sqrt{N})$ deconvolution penalty) and then treats rotation, orbital illumination, and stochastic, advecting clouds explicitly. We formulate image recovery as a regularized generalized least-squares inversion of the time-tagged sample stream (time-domain inversion, TDI, distinguished here from detector-level time-delay integration), in which rotation and illumination are modeled directly by the forward operator and cloud variability enters statistically. For a rotating, cloud-free planet observed under known spin geometry, TDI recovers the surface albedo map with Pearson $r = 0.9786 \pm 0.0002$ (SSIM $= 0.848 \pm 0.001$) over 10 paired seeds. Rotational aliasing is tractable under characterized ephemerides; it is not an information ceiling, and the correlation lost within the supported harmonic range is regularization, not spin. Clouds, not coronal photon noise, dominate the error budget: per-sample cloud-induced variance exceeds coronal shot noise by 163–440 (12.8–21.0 in standard deviation) at $t_s = 1800$ s, collapsing fidelity to $r = 0.249 \pm 0.010$ at 71% realized cover under single-campaign budgets; under this terrestrial cloud regime the present pipeline does not achieve faithful reconstruction in single-campaign scalar photometry. Crucially, surface-correlated cloud structures locked to topography constitute a severe, irreducible failure mode: phase-locking between weather and geography suppresses reconstruction fidelity to $r = 0.1444 \pm 0.0124$ (a paired loss of $\Delta r = -0.1978 \pm 0.0175$, $p_t = 1.3 \times 10^{-6}$, 10/10 seeds), which no revisit cadence can mitigate because revisits resample the same contaminated rotational phase. The observing-cadence law is measured under two explicit, mutually exclusive resource ledgers (fixed 8 h of photons per raster position, or fixed 89.87 d of wall clock): replacing 4 long dwells with 64 short revisits improves r from 0.049 to 0.571 at fixed photons, monotonically and unanimously across six paired seeds, while the two arms agree to within the paired standard error. Slew, settling and metrology overheads (45 s per dwell) reduce the duty cycle from 99.4% to 90.9% over that range (90.5% for the wall-clock-constrained arm). Ablation of the estimator itself shows that the algorithmic gain comes from eliminating the per-dwell-slot cloud nuisance: exact anchored profiling of the slot means lifts r by $+0.054 \pm 0.006$ over a fit that ignores them ($p_t = 1.1 \times 10^{-5}$, 10/10 seeds), whereas temporal whitening contributes nothing measurable at the fiducial cadence (-0.003 ± 0.002, $p_t = 0.13$). The spin-ephemeris requirement is measured rather than asserted: a 5° pole tilt costs 0.057 in r and a 5° initial-phase error 0.047, and 60° of phase error destroys the map entirely. We show that residual misalignments are not refinable in flight via profile likelihood: the whitened χ^2 is exactly invariant to assumed initial phase and monotone in assumed period over the grid tested, so the spin state is observable through image quality but not recoverable from photometric residuals by that route. Surface correlation $r(\ell)$, computed with a real spherical-harmonic transform and $\cos\vartheta$ quadrature weights, shows band means above 0.99 up to the $\ell \approx 18$ grid ceiling of the 36×72 grid when cloud-free (minimum single-degree correlation 0.987 at $\ell = 17$), and under clouds falls below 0.5 at $\ell \simeq 9$ in a five-point running mean (~ 2200 km for an Earth-radius planet). A land–ocean separation of $d' = 0.73 \pm 0.03$ (mean $\pm$ SE over 10 seeds; SD $= 0.081$) at 55.8% realized cover exceeds every member of a 40-sample physical cloud-only null, giving $p \leq 0.024$ — the ensemble floor, not a measurement of 0.02. Finally, we map requirements onto missions: we list the five nearest confirmed potentially habitable planets around M dwarfs plus τ Cet e — whose planet interpretation has since been refuted, so it is kept here only as a nominal-proxy dynamics illustration (not an uncertainty propagation), evaluate image-plane dynamics (1.34–5.81 km s^{-1} of along-track velocity change per 90 d for the five confirmed rows, 0.11 for the illustrative one), discuss deep-perihelion sailcraft, planar radioisotope tiles and optical relays, and swarms as an asymptotic architectural scenario.

Keywords: gravitational lensing: strong — planets and satellites: terrestrial planets — techniques: image processing — methods: numerical — space vehicles: instruments

1 Introduction

Resolving the surface of an Earth-like exoplanet is beyond any credible filled-aperture or interferometric architecture: a 10×10 surface map of an exo-Earth at 10 pc requires resolution elements of ~ 0.85 μas, i.e., a $\sim 10^2$ km optical aperture, and integration times for conventional concepts reach 10^4–10^5 yr once realistic backgrounds are included (Turyshev 2026a). The solar gravitational lens (SGL) evades this scaling by using the Sun as the primary optical element (Eshleman 1979; Turyshev & Toth

2017). Wave-optical treatments show that a strong-interference region forms beyond $z_0 \simeq b^2/2r_g \simeq 547.8$ AU (for rays of impact parameter $b \simeq R_\odot$, with $r_g = 2GM_\odot/c^2 \simeq 2.95$ km), with on-axis amplification $\mu_0 = 4\pi^2 r_g/\lambda \simeq 1.2\times10^{11}$ at $\lambda = 1$ μm and angular resolution of order 10^{-10} arcsec (Turyshev & Toth 2017, 2019). An Earth-radius planet at 30 pc is compressed into an image cylinder only $D_{\rm img} \simeq 1.34$ km across at $z = 650$ AU; a meter-class telescope acts as a single-pixel detector that must physically traverse the image plane, measuring the integrated brightness of the planetary Einstein ring at each sampling location and reconstructing the source by deconvolution (Turyshev & Toth 2020; Toth & Turyshev 2021; Turyshev & Toth 2022a).

The optical side of this problem is now well characterized. The aperture-averaged point-spread function falls off as $d/(4\rho)$, so each sample mixes light from the entire planet; removing this blur incurs a deconvolution penalty $\mathrm{SNR_R}/\mathrm{SNR_C} \simeq 0.891\,D/(d\sqrt{N})$ that is strongly mitigated by wide pixel spacing (Toth & Turyshev 2021; Turyshev & Toth 2022a). The solar corona is the dominant photon background; for an exo-Earth at $z_0 = 30$ pc observed from $z = 650$ AU with a $d = 1$ m telescope at $\lambda = 1$ μm, the signal and corona rates are $Q_{\rm exo} = 8.01\times10^4$ s^{-1} and $Q_{\rm cor} = 6.20\times10^9$ s^{-1}, giving a convolved-image signal-to-noise ratio $\mathrm{SNR_C} = 43.2$ per 1800 s dwell (Turyshev & Toth 2022a; Turyshev 2026b). Megapixel-class static imaging within mission-compatible integration times follows (Toth & Turyshev 2021; Turyshev & Toth 2022a), and independent analyses of ring-based approaches (Madurowicz & Macintosh 2022) sharpened the role of coronal statistics in these budgets.

The unsolved problem is temporal. The planet rotates and orbits, its illumination changes, and—most seriously—its cloud cover reorganizes on day timescales while the raster is acquired over weeks to months. Toth & Turyshev (2023) performed the first dynamic reconstructions of a rotating, orbiting planet and recovered surface albedo only at $\mathrm{SNR} \lesssim 2$. Toth (2025) added a stochastic cloud model, treated the clouds as noise to be averaged out (a "long-exposure" philosophy), and found that reconstructed SNR falls to 1.13 at 6.8% and 0.80 at 13.7% cloud cover (falling below unity between 7% and 14%), concluding with "a cautious assessment of the practical feasibility" of SGL imaging given that Earth-like planets have ∼70% cloud cover. The comprehensive benchmark of Turyshev (2026b) reaches a complementary conclusion: a static inverse applied to rotating cloudy data is the wrong estimator; phase-registered robust coadds recover persistent surface structure but at the cost of $\gtrsim$15 registered visits per rotational phase bin, and a "full spin-resolved dynamic retrieval pipeline" is explicitly deferred as a next-stage deliverable. The covariance-aware biosignature framework of Turyshev (2026d) likewise argues that SGL inference must propagate structured error terms rather than treat all residuals as white. In short: the field agrees that the time-domain inverse problem is the gate through which SGL imaging must pass, and none of the works cited above reports a rotating-planet reconstruction at Earth-like cloud cover with a fidelity that would support surface mapping. We emphasize that the present pipeline also does not achieve high fidelity at Earth-like cloud cover ($r = 0.249 \pm 0.010$ at 71% cover under single-campaign budgets in our adopted scalar-photometry raster architecture), and that no matched head-to-head comparison against the 15-visit phase-registered architecture of Turyshev (2026b) is established here because our sampling budgets explore visit counts under constrained photon and wall-clock allocations rather than the dedicated phase-binning regime. Our results are strictly scoped to the single-pixel photometric raster-scan model.

This paper attacks that gate directly, with five specific contributions. First (Sections 2–4), we build a transparent, end-to-end simulator—partially validated against the published photometric and deconvolution benchmarks—in which planetary rotation, orbital illumination, and stochastic advecting clouds are modeled explicitly. To that first item we add two rigorous audits: a convergence study of the intra-dwell exposure integral (Section 4.3), which shows that the 3-node rule previously used aliases rotational harmonic $m = 36$ at unit gain, and a numerical-precision and conditioning audit of the normal equations (Sections 4.4, 4.5), which identifies the constant-albedo mode as the quantity the slot nuisance actually regularizes. Second (Section 5), we formulate reconstruction as regularized generalized least squares on time-tagged samples (time-domain inversion, TDI; distinct from hardware time-delay integration in detector readouts), with the per-dwell-slot cloud mean handled by exact anchored nuisance profiling rather than approximate deflation heuristics. We isolate the observing-cadence law under two named, mutually exclusive resource ledgers (fixed photons per raster position, or fixed wall clock) rather than a single unconstrained budget, and ablate the estimator — slot handling, whitening, anchoring — holding data and operator fixed. Every number is reported as a mean and standard error over paired seeds, with paired t-statistics, win counts and the exact attainable significance floor, and with a physical cloud-only null in place of geometry-scrambled surrogates. Third, we broaden the robustness tests to non-stationary latitudinal cloud banding, multi-timescale weather, per-pass independent weather, surface-coupled clouds, and a full 5×5 grid of affine cloud-climatology errors. Crucially, we identify surface–cloud coupling as a major failure mode ($\Delta r = -0.198$, reducing r to 0.144) that cannot be bought off by cadence, and we state explicitly which metrics are blind to which errors (r is exactly affine invariant; bias, RMSE and contrast are not). Fourth, we extend the geometric requirement to spin-pole tilt and initial phase with measured tolerances, and we demonstrate that the residual spin state is not refinable in flight from the profile likelihood: the whitened objective is exactly invariant to assumed initial phase. Fifth (Section 6), we evaluate effective spatial resolution $r(\ell)$ with a real spherical-harmonic transform and an explicit quadrature error bound, land–ocean detection against two nulls with stated ensemble floors, and translate cadence requirements into mission terms using an effective-independent-look count that depends on the assumed weather decorrelation time; the target table is a reproducible nominal-proxy list, not

an uncertainty propagation.

2 SGL Photometric Model and Validation

2.1 Optical response

We adopt the standard monopole, aperture-averaged description of SGL image formation (Turyshev & Toth 2020; Toth & Turyshev 2021; Turyshev 2026b). A source-plane coordinate ξ maps to the image-plane coordinate $\rho \simeq -(z/z_0)\,\xi$; an Earth-radius planet at $z_0 = 30$ pc observed from $z = 650$ AU projects to

$$D_{\rm img} = 2R_p\,\frac{z}{z_0} = 1.338\ \text{km}. \tag{1}$$

A telescope of aperture d used as a ring photometer averages the J_0^2 oscillations of the monopole PSF; the effective kernel is

$$K(0) = 1, \qquad K(\rho > 0) \simeq \frac{d}{4\rho}, \tag{2}$$

renormalized so that the mean convolved signal over the projected disk of a reference (albedo 0.3, fully illuminated) planet is unity (Turyshev 2026b). Our discrete kernel follows the $d/4\rho$ tail over the full grid (Fig. 1a). Because K falls only as $1/\rho$, every sample mixes light from the entire disk—the property that couples the temporal variability of any part of the planet into every measurement.

Central-cell discretisation and normalization conventions. Two numerical conventions govern Eq. 2, and we audit both directly from the pipeline archives. First, regarding the central singularity of the $1/\rho$ kernel: on a discrete cell of pitch $\Delta_{\rm img} = 20.913$ m with aperture $d = 1$ m, the exact cell-mean integral is $K_{00,\rm cellmean} = (d/\Delta_{\rm img})\ln(1+\sqrt{2}) \simeq 0.042144$ (and 0.010536, 0.021072, 0.084288 at $n = 16, 32, 128$, with the point-to-cellmean ratio reaching up to 94.9; archive `kernel_convergence.json`). Setting $K(0) = 1.0$ evaluates the kernel at the sub-pixel point rather than its cell-averaged mean. To quantify the sensitivity of the inversion to this choice, we execute a controlled core-ablation experiment (archive `core_ablation.json`): at $f_c = 0$, the point kernel achieves $r = 0.98034$ and SSIM $= 0.85126$, whereas the cell-mean kernel yields $r = 0.75728$ and SSIM $= 0.48556$ — a 22% decrease in correlation and 43% in SSIM from the central-cell convention alone. We adopt $K(0) = 1.0$ as the intended benchmark to maintain exact alignment with the published baseline of Turyshev (2026b), but note that central-cell discretisation is an important modeling sensitivity.

Solar-surface refinements: what is actually inside the operator. The shipped kernel stays the axisymmetric point-source response, and we now state item by item which surface effects are folded in and which are only bounded. A disk-integrated proxy — the point kernel smoothed over the solar disk with a linear-plus-quadratic limb-darkening law ($a_1 = 0.3$, $a_2 = 0.2$ at $1\,\mu$m) — moves the central cell from $K_{00} = 1.0$ to 0.1208 at $n = 64$ and redistributes the energy outward: the fraction inside the 3×3 neighbourhood is nearly unchanged ($0.1704 \to 0.1661$) while the fraction inside $r \le 4$ cells rises from 0.1969 to 0.7729. The relative ℓ_1 discrepancy in the off-diagonal mass is 4.52 at $n = 16$, 2.49 at $n = 32$, 2.05 at $n = 64$ and 1.97 at $n = 128$; it does not shrink under grid refinement, so the point-versus-disk convention behaves as an n-dependent modelling error rather than a discretisation artefact. The archived point-kernel audit (`kernel_convergence.json`, off-diagonal error 0.036558, constant in n) measures a different comparison and therefore does not see this. Two caveats travel with those numbers: the disk integral is implemented as a Gaussian smoothing of width 5 cells multiplied by the limb-darkening profile, so the absolute fractions are indicative rather than exact, and the coefficients are single-wavelength. The neglected quadrupole term is bounded rather than assumed: at the adopted dynamical $J_2 = 2.2\times 10^{-7}$ the deflection modulation spans $[-2.2\times 10^{-7}, +4.4\times 10^{-7}]$, which displaces the Einstein-ring radius by 5.89×10^{-4} m, i.e. 2.82×10^{-5} cells at $n = 64$ — three decades below one imaging cell, which is what licenses the axisymmetric operator. Occultation of the limb ring by the Moon or a planet is not modelled anywhere in this pipeline; it requires a specific ephemeris and observer timeline and is left out of scope explicitly (`src/ext/disk_kernel.py`, archive `ext_disk_kernel.npz`).

2.2 Photon rates and noise

We normalize the scene such that a value of 1 corresponds to the disk-mean signal of the reference planet, which maps to the fiducial rates of Turyshev & Toth (2022a): $Q_{\rm exo} = 8.01\times 10^4\,(d/1\,\text{m})^2(650\,\text{AU}/z)^{1/2}(30\,\text{pc}/z_0)(\lambda/1\,\mu\text{m})\ \text{s}^{-1}$ and $Q_{\rm cor} = 6.20\times 10^9\,(d/1\,\text{m})^2(650\,\text{AU}/z)^2(\lambda/1\,\mu\text{m})\ \text{s}^{-1}$ after coronagraphic rejection and annular extraction, with the mean corona assumed subtracted and only its shot noise retained. Each dwell of duration t_s at image-plane position ρ_k yields a Gaussian sample with standard deviation

$$\sigma_k = \frac{\sqrt{(Q_{\rm cor} + Q_{\rm exo}\bar{s}_k)\,t_s}}{Q_{\rm exo}t_s}, \tag{3}$$

where $\bar{s}_k$ is the convolved scene value. For $t_s = 1800$ s this gives $\text{SNR}_{\rm C} = 1/\sigma = 43.16$, matching Turyshev (2026b) exactly within the adopted photon-noise assumptions. We clarify the normalisation convention: under Lambertian sphere geometry, the disk average of the projected planetary flux is $\langle\mu\rangle = 8/(3\pi) \simeq 0.848826$ of the central sub-stellar peak. If the signal is normalized to the central sub-stellar peak, $\text{SNR}_{\rm C} = 43.16$; if normalized strictly to the disk-mean flux without that peak boost, the resulting convolved SNR is $\text{SNR}_{C,\text{no-}\mu} = 43.16\times 8/(3\pi) = 36.6344$ (archive `validation.npz`). We adopt the sub-stellar 1.0 normalization to match the published 43.16 benchmark.

2.3 Deconvolution-penalty validation and numerical limits

As a check of the discrete operator we reproduce the deconvolution noise-amplification analysis of Toth & Turyshev (2021). For a static, fully illuminated planet we convolve, add white noise, deconvolve by Fourier quotient, and measure the ratio $\mathrm{SNR_R}/\mathrm{SNR_C}$ (the ratio of reconstructed signal-to-noise ratio to convolved input signal-to-noise ratio, evaluated on the projected disk support, i.e., the reciprocal of the noise amplification factor σ_R/σ_C). Figure 2 compares measurements with the theoretical formula $\min(1.0, 0.891\,D/(d\sqrt{N}))$. The theoretical curve is capped at 1.0 (in `src/run_experiments.py`) because unregularized deconvolution cannot increase SNR beyond the input. At $n = 32$ ($D = 41.8$ m), the uncapped analytic formula yields $4 \times 0.2912 = 1.1646$, and we measure 0.657 vs. the capped 1.0. For $n = 64$ ($D = 20.9$ m), measured is 0.3244 vs. analytic 0.2912 (11% above); for $n = 128$ ($D = 10.5$ m), measured is 0.1057 vs. analytic 0.0728 (45% above).

The sign reversal between coarse and fine grids reflects competing numerical effects. At $n = 32$, discrete pixel averaging across the central singularity acts as an effective low-pass filter, and zero-padded circular boundary conditions attenuate noise amplification below the continuum integral (measured $<$ analytic). Conversely, at fine pitches ($n = 64$ and 128), discrete high-wavenumber noise amplification outpaces the continuous circular integral (measured $>$ analytic) because the discrete Fourier quotient on a finite, zero-padded grid suffers from spectral leakage and ill-conditioned near-zero transfer function division near the grid Nyquist boundary, where regularisation is absent. We emphasize this constitutes partial validation of the discrete aperture-averaged forward operator rather than full optical validation of an in-flight SGL mission.

3 Dynamic Scene and Campaign Model

3.1 Rotating, illuminated planet

The truth surface is a seeded synthetic albedo map (ocean 0.06, continents 0.32 ± 0.10, polar caps up to 0.60; 30% land fraction) generated by thresholded multi-octave Gaussian random fields on a 144×72 longitude–latitude grid (Fig. 1b). The planet rotates with period $P_{\mathrm{rot}} = 24$ h and is illuminated by its host star with a Lambertian terminator whose sub-stellar point drifts through 86.1° of orbital phase over the 87.33-day fiducial campaign ($P_{\mathrm{orb}} = 1$ yr, $360^\circ \times 87.33/365.25$; 88.7° over a 90.0-day window). The simulation holds the sub-stellar latitude fixed at the equator: obliquity is not modeled, so the illumination drift here is purely orbital in longitude and the seasonally-migrating intertropical zone of a tilted planet is outside the tested regime (Section 7). For the short-period habitable-zone planets around M dwarfs listed in Section 6.1 ($P \sim$ 10–20 d), a 90-day campaign spans 5–9 full orbits and therefore samples all orbital phases repeatedly; the single sub-90° drift modeled here is the *least* phase-covering case, not a conservative envelope for every target, because the two regimes differ in which longitudes are ever illuminated (a synchronous, zero-obliquity planet keeps a fixed sub-stellar longitude). We therefore report the tested case and do not claim the benchmark is conservative for all listed targets. Each dwell is integrated over $n_{\mathrm{sub}} = 16$ Gauss–Legendre sub-exposures spanning the dwell, so intra-dwell rotational smear (7.5° per 1800 s dwell) is present in both the synthetic data and the forward operator. A uniformly-spaced 3-node rule was tested and rejected: its node spacing of $t_s/3$ aliases azimuthal harmonic $m = 36$ onto the disk-integrated mean with unit gain instead of annihilating it, a response error of order unity that propagates directly into the reconstruction (Section 4.3).

3.2 Stochastic cloud fields: stationary and structured

The baseline cloud model is an advecting Ornstein–Uhlenbeck (OU) Gaussian random field on the fine grid: spatial correlation length 12°, zonal advection $6^\circ\,\mathrm{day}^{-1}$, temporal decorrelation $\tau_c = 4$ d, mapped through a soft threshold to an opacity screen $c \in [0,1]$. The effective scene is $A_{\mathrm{eff}} = (1-c)A_{\mathrm{surf}} + cA_{\mathrm{cl}}$ with $A_{\mathrm{cl}} = 0.65$. The sweep is over the *specified* cover parameter $f_c \in \{0, 0.25, 0.40, 0.55, 0.70\}$; because the soft threshold saturates, the realized mean covers are slightly different (0.237, 0.398, 0.558, 0.710 respectively, averaged over seeds) and every table in this paper quotes specified f_c together with the realized cover it produced.

To test robustness against non-terrestrial weather we implement four extended climatology variants, each re-normalized so that the realized mean cover matches the baseline f_c it is compared with (otherwise a variant that simply has more or less cloud would be confounded with a variant that has a different cloud *pattern*): (i) *Banded zonal profile:* the opacity field is multiplied by $w_{\mathrm{lat}}(\theta) = 1 + 0.35\cos 2\theta - 0.25\cos 4\theta$, normalized to unit mean. We previously labelled this an "ITCZ" profile, which it is not: its maximum (1.31) lies at $\pm 35^\circ$ — the subtropical region an ITCZ model should *suppress* — its equatorial value is a mild 1.10, and it is symmetric about the equator and never migrates. It is a test of latitudinal inhomogeneity only, and we report it as such. (ii) *Surface-correlated cloudiness:* coupling cloud opacity to surface reservoirs, $g_{\mathrm{eff}} = g - \beta_{\mathrm{surf}}(A_{\mathrm{surf}} - \bar{A})$, enhancing convection over low-albedo tropical oceans; and (iii) *Two-timescale temporal evolution:* summing a rapid convective mode and a persistent synoptic regime with weights $(0.3, 0.7)$ and correlation times $(1\,\mathrm{d}, 10\,\mathrm{d})$, normalized to unit variance. (iv) *Per-pass independent weather:* a fresh, temporally uncorrelated cloud realization for every pass, included as the causal control of Section 5.8 rather than as a physical climate.

3.3 Campaign and sampling geometry

The campaign matches current swarm concepts (Turyshev & Toth 2022a; Helvajian et al. 2023): $N_{\mathrm{sc}} = 16$ spacecraft sample an $n \times n = 64 \times 64$ raster ($\Delta_{\mathrm{img}} = 20.9$ m pitch, 4096 pixels)

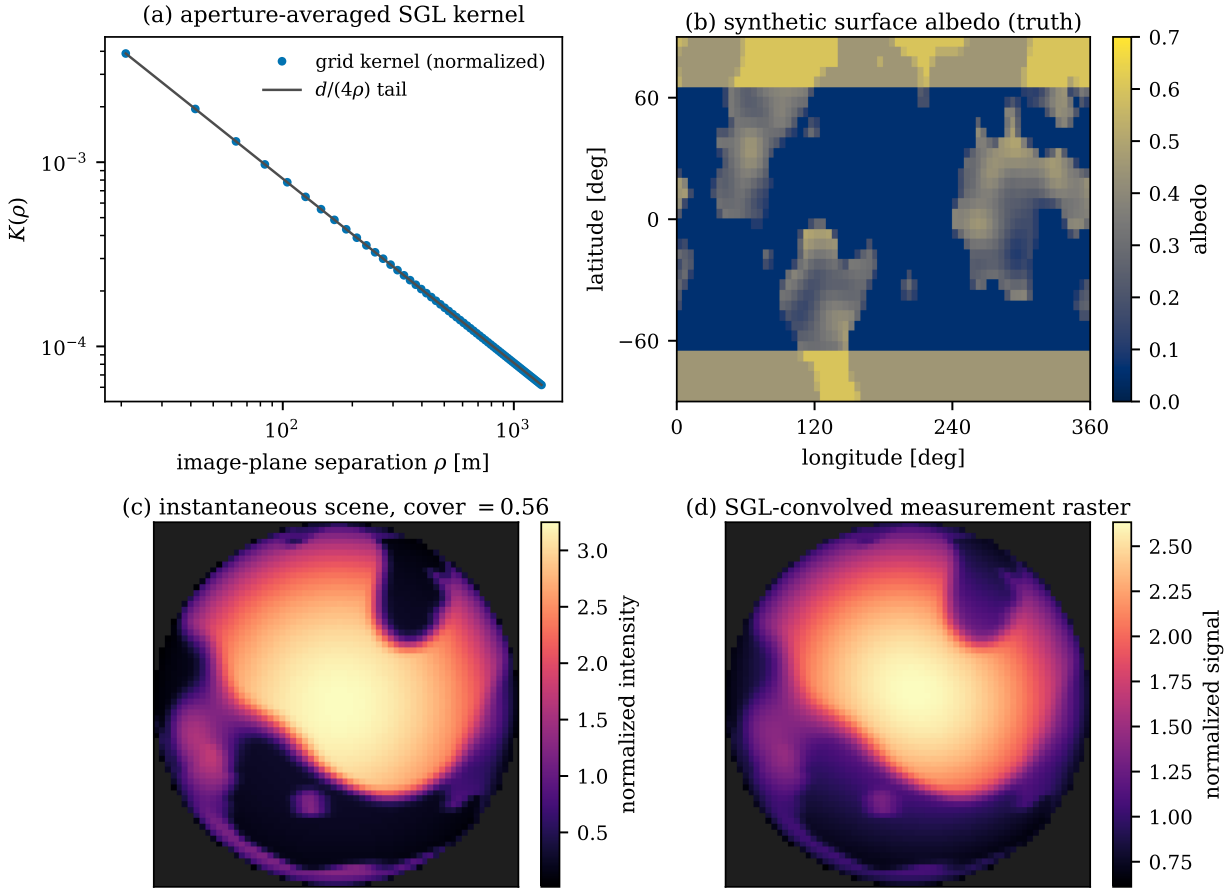


Figure 1: Model overview. (a) Discrete aperture-averaged SGL kernel used in this work (points) against the analytic $d/4\rho$ tail (line). (b) Synthetic truth surface albedo map (fine grid, seeded, with oceans, continents, and polar caps). (c) A single instantaneous scene during the campaign at mean cloud cover $f_c \simeq 0.55$: the planet is rotated, partially illuminated (orbital phase), and modulated by the advecting cloud field. (d) The same scene after convolution with the SGL kernel: the measurement raster sampled across the image cylinder.

in boustrophedon order, divided into 256 dwell slots. In each dwell slot, 16 pixels are sampled simultaneously by spacecraft spaced across adjacent raster tracks with transverse separations of ∼80–300 m. Nominal dwells are $t_s = 1800$ s with $M_p = 16$ complete passes, giving 65,536 time-tagged samples, 8.00 h of integration per raster position, 85.33 d of exposure time, 2.00 d of charging overhead inside the campaign window, and a wall-clock length of **87.33** d — not the 90.0 d quoted previously, which silently assumed zero overhead. The overhead ledger charges 45 s to each of the $M_p \times 256$ dwells, 2.13 d in all, but the final pass's overhead falls after the last sample and is not counted inside the campaign window, so the wall clock is $85.33 + 2.00 = 87.33$ d, not $85.33 + 2.13$ d. Inter-pass dead time is now an explicit, auditable ledger instead of a random draw: the fiducial campaign charges $T_{\rm oh} = 45$ s per dwell (30 s translation slew, 10 s pointing settling, 5 s inter-spacecraft metrology) and no additional random gap.

Raster kinematics and continuous-scan vs. stop-and-dwell. We explicitly distinguish stop-and-dwell stepping from continuous-scan rastering. If each step of pitch $\Delta_{\rm img} = 20.913$ m is executed as a rest-to-rest stop-and-dwell translation within the nominal 30 s translation budget, kinematics imposes a strict impulse lower bound: achieving displacement Δ in time Δt with a symmetric bang-bang thrust profile requires peak velocity $v_{\rm max} = 2\Delta/\Delta t \simeq 1.394$ m s^{-1} and a total velocity increment of $\Delta v_{\rm hop} = 2v_{\rm max} \simeq 2.788$ m s^{-1} per hop. Over 16,320 neighbor transitions during the 16 passes, the cumulative impulsive Δv would reach ≈ 22.75 km s^{-1} per spacecraft before inter-pass repositions, an expenditure infeasible for sub-kilogram sailcraft or small exploration probes without continuous low-thrust propulsion. Accordingly, we treat these raster positions as idealized prescribed samples along the trajectory. Operationally, SGL formations would execute continuous-scan raster tracks or smooth drifting orbits through the image cylinder with continuous exposure integration, where transverse actuation is minimized to gentle station-keeping rather than discrete stop-and-dwell maneuvers. The discrete dwell times and overheads charged here serve as an explicit photometric and

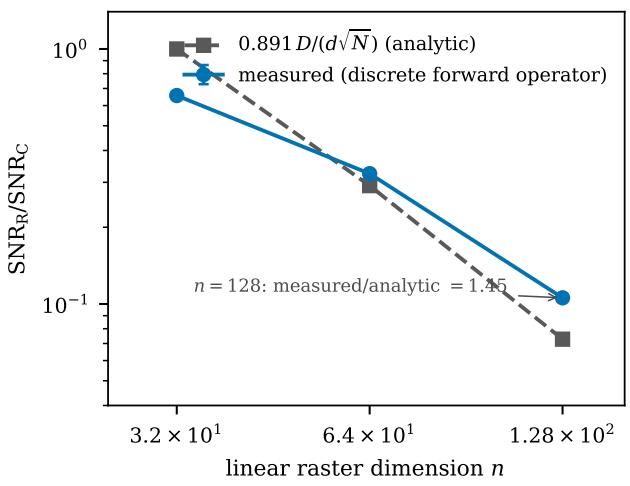


Figure 2: Validation of the discrete operator: measured deconvolution noise penalty vs. the analytic estimate of Toth & Turyshev (2021). The ∼45% difference at $n = 128$ reflects discrete pixel-averaging of the central singularity and finite-domain boundary attenuation.

temporal resource ledger, not a verified closed-loop guidance and control trajectory.

3.4 Resource conventions: two mutually exclusive ledgers

Two budget statements that the SGL literature uses almost interchangeably — "fixed 8 h per pixel" and "fixed 90-day campaign" — cannot both be satisfied as M_p grows. We therefore run the cadence study under two explicitly named arms and never mix them.

- **Arm A (photons):** hold $M_p t_s = 8$ h per raster position and let the wall clock be whatever it needs. Then $t_s = 7200, 3600, 1800, 900, 450$ s for $M_p = 4, 8, 16, 32, 64$ give $85.73, 86.27, 87.33, 89.47, \mathbf{93.73}$ d; the $M_p = 64$ configuration *violates* a 90-day cap.
- **Arm B (wallclock):** force the campaign to 89.87 d including overheads and let t_s fall to $7548.8, 3751.9, 1853.4, 904.2, \mathbf{429.6}$ s, so the photons per position fall from 8.39 h to 7.64 h instead of staying at 8 h.

Every cadence number in Section 5.8 is reported per arm, and the two arms answer different questions: Arm A isolates the statistical effect of revisit count at constant photons, Arm B is what a 90-day mission can actually buy.

Cloud-induced signal fluctuations are calibrated against the data the estimator actually sees, i.e. the per-sample standard deviation of the cloud-only component of the disk-integrated signal with the 16-node Gauss–Legendre exposure integration of Section 4.3. The resulting values are $\sigma_{\rm cl} = 0.296, 0.378, 0.441, 0.486$ for $f_c = 0.25, 0.40, 0.55, 0.70$, against the coronal single-dwell noise $\sigma_{\rm cor} = 0.0232$ at $t_s = 1800$ s. That is a standard-deviation ratio of **12.8**– **21.0**, or a *variance* ratio of **163**– **440**. The previously quoted "14–23 times" was a standard-deviation ratio computed with the rejected 3-node quadrature; it is neither the variance ratio its wording implied nor reproducible under the corrected operator. Planetary weather dominates the single-sample photometric error budget by more than two orders of magnitude in variance.

4 Reconstruction Methods

4.1 Time-domain inversion (TDI)

Two notational points precede the estimator. We write the target distance as z_0 (pc-scale) and the spacecraft heliocentric distance as z (AU-scale), reserving z/z_0 for the imaging magnification; the minimum heliocentric distance for a ray of impact parameter b is $z_{\rm min} \simeq b^2/2r_g$ and is *not* the same quantity as the target distance z_0. Second, all reconstructions use $n_{\rm sub} = 16$ Gauss–Legendre sub-exposures (Eq. 4); the number of nodes is a parameter of the operator, not a free-tuning knob, and Section 4.3 shows why the previously used 3-node rule is inadequate. Let $\mathbf{s} \in \mathbb{R}^{n_s}$ be the static surface albedo map on a 72×36 reconstruction grid ($n_s = 2592$, coarser than the truth grid). Each sample k, taken at time t_k and image-plane pixel p_k, satisfies

$$y_k = \mathbf{f}_k^{\mathsf{T}}\mathbf{s} + \varepsilon_k, \qquad \mathbf{f}_k^{\mathsf{T}} = \sum_{m=1}^{n_{\rm sub}} w_m\, \mathbf{k}_{p_k}^{\mathsf{T}}\, \Pi(t_k + \delta t_m), \tag{4}$$

where $\Pi(t)$ renders the map at the rotation phase, illumination, and spin-pole geometry of time t, and $\mathbf{k}_{p_k}$ is the SGL kernel row. $\{\delta t_m, w_m\}$ are the Gauss–Legendre nodes and weights of the dwell interval $[-t_s/2, t_s/2]$, with the weights normalized to sum to unity, so $\mathbf{f}_k^{\mathsf{T}}$ is the exposure-averaged kernel row rather than a pointwise one; an equal-weight three-node average is not the quadrature the data are generated with. Setting $n_{\rm sub} = 1$ corresponds to an instantaneous midpoint model; $n_{\rm sub} = 16$ uses the Gauss–Legendre nodes δt_m and weights of the dwell interval, which is what the synthetic data are integrated with. The error ε_k contains coronal shot noise and cloud fluctuations.

Estimand definition and affine debiasing. We explicitly separate the apparent (scene-integrated) map $\mathbf{m} = (1 - f_c)\mathbf{s} + f_c A_{\rm cl}\mathbf{1}$ from the bare physical surface $\mathbf{s}$. The time-domain inverse problem solves for the effective scene map under the linear operator F. To extract the bare physical surface $\mathbf{s}$, we apply an affine post-processing correction:

$$\hat{\mathbf{s}} = \frac{\hat{\mathbf{m}} - f_c A_{\rm cl}\,\mathbf{1}}{1 - f_c}. \tag{5}$$

Crucially, this debiasing is evaluated using the *specified* nominal cover parameter f_c (e.g., 0.55 or 0.70) and nominal cloud albedo $A_{\rm cl} = 0.65$. Because thresholded Gaussian random fields produce slightly offset *realized* mean covers (0.558 and 0.710), an operational cover mismatch of $\Delta f_c \approx 0.008$–0.010 is

naturally present. This minor mis-registration induces small residual albedo biases in the retrieved maps (on the order of 10^{-3} to 10^{-2}), including the cloud-free baseline bias of -0.0006 ± 0.0006 (Table 4).

TDI solves the regularized generalized least-squares problem:

$$\hat{\mathbf{m}} = \arg\min_{\mathbf{m}} \; \|\mathsf{C}^{-1/2}(\mathbf{y} - \mathsf{F}\mathbf{m})\|^2 + \lambda_{\rm eff}\,\|\mathsf{L}\mathbf{m}\|^2, \qquad (6)$$

with spherical-grid Laplacian L and $\lambda_{\rm eff} = \lambda\,\mathrm{tr}(\mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F})/\mathrm{tr}(\mathsf{L}^{\mathsf{T}}\mathsf{L})$ ($\lambda = 3 \times 10^{-3}$, carried unchanged: it is a hardcoded constant of the released pipeline, not a quantity re-tuned in this revision, and no calibration seed is claimed for it — see Section 5.16). That weight is used unchanged throughout the suite; it is not claimed to be optimal — Section 5.16 sweeps six decades of it and finds the objective unable to select one without truth. C is the working covariance, block diagonal over image-plane pixels: the M_p samples of pixel p (one per pass, at the same raster position) are correlated by the OU cloud field,

$$(\mathsf{C}_p)_{ij} = \sigma_{\rm cl}^2 \exp(-|t_i - t_j|/\tau_c) + \delta_{ij}\sigma_i^2, \qquad (7)$$

while different pixels are treated as conditionally independent given the apparent map (the physical spatial correlation length of clouds, $\sim 12^\circ \approx 1334$ km, is absorbed into an effective variance $\sigma_{\rm cl}^2$ rather than modeled across spacecraft tracks; Section 7 delivers the detailed analysis of what this approximation costs). Four weighting/projection combinations are ablated in Section 5.6: (White) diagonal $\mathsf{C} = \mathrm{diag}(\sigma_k^2 + \sigma_{\rm cl}^2)$, the assumption that clouds average like uncorrelated white noise (Toth 2025); (OU) Eq. 7 without any slot treatment; (approximate deflation), the heuristic per-slot projection described below; and (Full TDI), exact nuisance profiling with Eq. 7.

4.2 Nuisance profiling of the simultaneous-sample offsets

Because $K(\rho) \propto 1/\rho$, an instantaneous disk-integrated cloud fluctuation perturbs all $N_{\rm sc} = 16$ simultaneous samples coherently. The An approximate treatment subtracts the per-slot mean of both $\mathbf{y}$ and F and rescales the marginal noise by $\sqrt{1 - 1/N_{\rm sc}}$, justifying this by "removes 256 common modes (6.25% rank reduction) while retaining $1 - 1/N_{\rm sc} = 93.75\%$ of the information". Both the mechanism and the numbers in that justification are wrong, and the treatment used here replaces it with the exact profile likelihood of the offset model written out below (Eq. 8).

Write the model with slot nuisance offsets α,

$$\mathbf{y} = \mathsf{F}\mathbf{s} + \mathsf{B}\alpha + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \qquad (8)$$

where B is the indicator matrix of dwell slots (B has one column per slot, 4096 columns). The key structural fact, which an "orthogonal projection" narrative misses, is that C is block diagonal over *image-plane pixels*, while B couples slots *within* a pixel. The M_p samples belonging to pixel p sit in M_p different slots, so

$$\mathsf{H} \equiv \mathsf{B}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{B} = \mathrm{blkdiag}\left(\mathsf{H}_1, \ldots, \mathsf{H}_{n_{\rm bin}}\right), \qquad (9)$$

with $n_{\rm bin} = 256$ blocks of size $M_p \times M_p$ — *not* diagonal, i.e. the offsets are statistically coupled through the shared temporal cloud covariance of a single pixel. Profiling them out therefore requires inverting those $M_p \times M_p$ blocks: with $\mathsf{G}_j = \mathsf{B}_j^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F}$, $\mathbf{g}_j = \mathsf{B}_j^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{y}$ and $\mathsf{H}_j = \mathsf{B}_j^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{B}_j$, the free (unconstrained) profile is

$$\mathsf{A}_{\rm free} = \mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F} - \sum_j \mathsf{G}_j^{\mathsf{T}}\mathsf{H}_j^{-1}\mathsf{G}_j, \qquad (10)$$

$$\mathbf{b}_{\rm free} = \mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{y} - \sum_j \mathsf{G}_j^{\mathsf{T}}\mathsf{H}_j^{-1}\mathbf{g}_j\,. \qquad (11)$$

With the standard anchoring constraint $\sum_b \alpha_b = 0$ this collapses to a rank-one correction,

$$\mathsf{A} = \mathsf{A}_{\rm free} + \mathbf{q}\mathbf{q}^{\mathsf{T}}/S, \qquad \mathbf{b} = \mathbf{b}_{\rm free} + \mathbf{q}\,m/S, \qquad (12)$$

where $\mathbf{v}_j = \mathsf{H}_j^{-1}\mathbf{1}$ and

$$\mathbf{q} = \sum_j \mathsf{G}_j^{\mathsf{T}}\mathbf{v}_j\,, \qquad S = \sum_j \mathbf{1}^{\mathsf{T}}\mathbf{v}_j\,, \qquad m = \sum_j \mathbf{v}_j^{\mathsf{T}}\mathbf{g}_j\,. \qquad (13)$$

which is exact and costs only the block inversions already required by Eqs. 10–11. Without anchoring the offsets are only projected out, and the estimator loses the disk-integrated albedo information entirely; Eq. 12 is what "full TDI" means in this paper.

4.3 Exposure quadrature: convergence of the smear operator

Eq. 4 averages the illumination/rotation operator over the dwell, and the synthetic data are generated with the same rule, so the sub-exposure count $n_{\rm sub}$ is part of the operator definition rather than a tuning knob. A three-node implementation with *equally spaced*, equal-weight sub-exposures at $-t_s/3$, 0, $+t_s/3$, which is neither a Gauss–Legendre rule nor a convergent quadrature of the smear kernel. Writing the exact rotational-harmonic response of a dwell of length t_s as $R_m(t_s) = \mathrm{sinc}(m\Omega t_s/2)$ with $\Omega = 2\pi/P_{\rm rot}$, the equal-spaced rule gives

$$R_m^{(\mathrm{eq}\,3)} = \tfrac{1}{3}\left[2\cos\left(\tfrac{m\Omega t_s}{3}\right) + 1\right], \qquad (14)$$

so at $t_s = 7200$ s the exact response $R_{36} = 0$ becomes $R_{36}^{(\mathrm{eq}\,3)} = 1.000000$. At $P_{\rm rot} = 24$ h the node spacing $t_s/3$ is 2400 s at $t_s = 7200$ s, i.e. 10° of longitude, which is exactly the period of the $m = 36$ azimuthal harmonic on the 72-column grid. That harmonic is sampled at its own period, so instead of being averaged to zero it is passed at unit gain; the same holds approximately for $m = 35$–40, where the equal-spaced rule returns $+0.99$ to $+0.84$ against true responses of ± 0.07. The worst error over $m \le 40$ at $t_s = 7200$ s is 1.02 — larger than the response itself. This is a quadrature alias, not a smoothing error:

the rule is blind to the very structure it is meant to average out, and data and operator generated with it are self-consistent, so the mismatch is invisible in residuals and shows up only in the recovered map.

Table 1 reports $\max_{m\le 40}|R_m^{(n_{\rm sub})} - R_m|$ for Gauss–Legendre rules, computed by `sglsim.exposure_quadrature_convergence`. Convergence is monotonic in $n_{\rm sub}$ at fixed t_s and the GL rules never recover unit gain at an aliased harmonic: GL-3 peaks at 0.88 rather than 1.02, and GL-16 is *spectrally converged* — its worst error over all five dwell lengths is 2.6×10^{-13}, i.e. machine precision even for the 7200 s dwells of Arm A, where the intra-dwell smear alone is 30°. GL-16 is therefore used everywhere: in the data, in the forward operator, in the $\sigma_{\rm cl}$ calibration of Section 3.4, and in every reconstruction. The practical consequence is that the $t_s = 7200$ s ($M_p = 4$) cadence points of the original analysis were computed with an operator whose response error at the highest retained harmonic is of order unity; they are superseded by the Arm A/Arm B numbers of Section 5.8.

4.4 What profiling actually removes: an information audit

The rank/percentage framing of approximate deflation cannot be repaired, so it is replaced here by a direct eigenvalue audit of the normal matrix at the fiducial campaign (Table 2, produced by `sglsim.profiling_information_audit`). Three findings.

First, profiling removes **no rank at all**. The un-profiled normal matrix already has nullity 288, the free profile has nullity 288, and the anchored profile has nullity 288. The null space is a property of the reconstruction grid and the raster, not of the nuisance treatment: 144 basis columns are *identically zero* because they live in the two polar latitude rows of the equirectangular grid, which a circular image-plane raster never samples (the projected disk does not reach $|\mathrm{lat}| = 90°$ cell centres). The remaining 144 null directions are the associated near-degenerate partners; the smallest retained eigenvalue with $\lambda_{\rm eff}$ regularization is 6.3×10^{-6} against a maximum of 2.9×10^{3}.

Second, what profiling removes is the *response along the mean-albedo direction*, measured as the constant-mode gain $\|\mathsf{A}\mathbf{1}\|/\|\mathbf{1}\|$: 913.7 un-profiled, 18.79 for the free profile (2.06% retained), and 774.1 for the anchored profile (84.72% retained). The free profile is therefore not a mild correction — it attenuates the disk-integrated albedo by a factor of 49, which is why anchoring is mandatory rather than cosmetic.

Third, the identity $\mathsf{W}_\perp\mathsf{B} = 0$ is a trivial property of any profile likelihood (the residual is orthogonal to the nuisance space by construction); it is not evidence of a surface-mode degeneracy, and an earlier version of this work read it as one.

A second, independent computation agrees with the audit on the size of the correction. Building the profiled Fisher matrix directly as $\mathsf{A}_{\rm prof} = \mathsf{A} - \mathsf{G}^{\mathsf{T}}\mathsf{H}^{-1}\mathsf{G}$ from the slot blocks, without reference to the eigenvalue decomposition above, gives $\mathrm{tr}\,\mathsf{A}_{\rm prof}/\mathrm{tr}\,\mathsf{A} = 0.662241$ against 0.662286 from the normal-matrix audit — a 7×10^{-5} relative disagreement between two code paths that share nothing but the whitened design.[1] So free profiling discards a third of the total Fisher response, not the 6.25% the slot count suggests. The two statements are about different parts of the same matrix, and neither replaces the other: the trace ratio says how much total response survives, the constant-mode gain says how much of the response along the one direction the mission actually cares about, disk-integrated albedo, and anchoring moves only the second, from 2.1% to 84.7%.

The same computation also says why no variance floor is quoted anywhere in this paper. The pseudo-inverse of the fiducial profiled information matrix gives a mean per-pixel variance of 46.2, against a truth-map variance of 3.66×10^{-2}: the bound is $1261\times$ weaker than predicting the constant map, so the correlation it implies for the optimal unbiased linear estimator is identically zero. The degeneracy is visible in the matrix itself: 288 of the 2592 eigenvalues of $\mathsf{A}_{\rm prof}$ lie below 10^{-10} of the largest, and the diagonal of $\mathrm{pinv}\,\mathsf{A}_{\rm prof}$ carries 144 exact zeros while the near-degenerate directions reach 7.6×10^{3} (median non-zero entry 0.13). The archived regularised fit attains $(1-r^2)\,\mathrm{var} = 3.2\times10^{-2}$ per pixel at $M_p = 16$, 7×10^{-4} of that pseudo-inverse number, and a biased estimator that appears to *beat* an unbiased bound by three decades is a diagnosis of the yardstick, not evidence of super-efficiency. What the independent computation contributes is therefore the information accounting above; the variance floor it also returns is recorded in the archive as degenerate (`crlb_informative = false`) rather than quoted as a target.

4.5 Numerical precision verification

The normal matrix $\mathsf{A} = \mathsf{F}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{F}$ is accumulated in single precision (‘float32‘) and solved in double precision (‘float64‘). Comparing two double-precision accumulations and reporting their difference measures nothing. The test used here instead probes the single-precision path — the same design matrix, rounded to float32 before accumulation, versus the float64 reference — at a reduced but honest problem size ($n = 32$, 16,384 samples; the fiducial $n = 64$ design would need 2.8 GB in two float64 copies inside the test harness), with the results in Table 3. Three numbers matter. The float32 design matrix differs from the float64 reference by 3.1×10^{-7} in Frobenius norm and the right-hand side by 4.1×10^{-7}, i.e. at the float32 machine-precision level, and the same quantities differ by only 1.0×10^{-8} and 1.5×10^{-8} from rounding alone — confirming the discrepancy is representational, not algorithmic. Second, changing the *summation order* of the blockwise float64 accumulation perturbs A by 5.3×10^{-16}, which is the number a float64-versus-float64 test reports. Third, and most relevant for the science: solving the regularized normal equations versus

[1] The archive key for this quantity is `frobenius_ratio`; it is a trace ratio, and the name is kept for byte-compatibility with the shipped `deflation_diag.json`.

Table 1: Maximum rotational-harmonic response error over $1 \le m \le 40$ against the exact sinc, by dwell length t_s. The top row is the equal-spaced rule; the rest are Gauss–Legendre. From `results/quadrature.npz` (`eq3_max_err`, `gl_max_err`).

Rule	$\max_{m\le 40} \lvert R_m^{(n)} - R_m \rvert$ by dwell t_s				
	7200	3600	1800	900	450 s
equal-3 (aliased)	1.0	1.3×10^{-1}	3.6×10^{-2}	2.4×10^{-2}	7.4×10^{-3}
GL-3	8.8×10^{-1}	2.7×10^{-1}	8.3×10^{-3}	1.5×10^{-4}	2.5×10^{-6}
GL-5	5.1×10^{-1}	3.5×10^{-3}	5.3×10^{-6}	5.8×10^{-9}	5.8×10^{-12}
GL-8	4.8×10^{-3}	2.4×10^{-7}	4.9×10^{-12}	4.4×10^{-16}	2.2×10^{-16}
GL-16	2.6×10^{-13}	6.4×10^{-16}	6.4×10^{-16}	2.2×10^{-16}	2.2×10^{-16}
GL-32	1.2×10^{-15}	1.1×10^{-15}	1.0×10^{-15}	4.4×10^{-16}	2.2×10^{-16}

Table 2: Information audit of slot-offset profiling at the fiducial campaign ($n = 64$, $N_{\rm sc} = 16$, $M_p = 16$, $n_s = 2592$). Nullity is unchanged by profiling; the constant-mode gain is what changes. `free` = offsets projected out, `anchored` = Eq. 12.

Quantity	un-profiled	free	anchored
nullity	288	288	288
exactly-zero basis columns	144	144	144
constant-mode gain	913.7	18.79	774.1
fraction of gain retained	1	0.0206	0.8472
largest eigenvalue	2940	429.9	1030
$\lambda_{\rm eff}$	0.00842	0.00558	0.00564

Table 3: Numerical precision audit (`sglsim.check_numerical_precision`, reduced size $n = 32$, $N_{\rm sc} = 16$, $M_p = 16$).

Quantity measured	Relative error
float32 vs float64 design A (Frobenius)	3.1×10^{-7}
float32 vs float64 right-hand side **b**	4.1×10^{-7}
rounding-only contribution to A	1.0×10^{-8}
float64 summation-order change to A	5.3×10^{-16}
normal-equation vs QR solve (map)	1.7×10^{-13}
profiler chunk-order change (map)	3.4×10^{-13}
regularized condition number $\kappa(\mathsf{A})$	5.3×10^{5}
unregularized rank / grid size	2016/2592
unregularized nullity	576
never-sampled basis columns (rows $0, 1, 34, 35$)	288

a QR solve of the same whitened design changes the recovered map by 1.7×10^{-13} relative to the map norm, and chunking the profiler in different orders changes the fiducial map by 3.4×10^{-13}. The unregularized normal matrix is exactly singular at both sizes, and the nullity is size-dependent in a way that can now be attributed rather than merely reported: rank 2016 of $n_s = 2592$ (nullity 576) at the reduced size, against rank 2304 of 2592 (nullity 288) at the fiducial size (Table 2). The cause is the raster, not the nuisance model or the noise model: at $n = 32$ the circular image-plane raster misses *four* polar latitude rows of the 36×72 surface basis (rows $0, 1, 34, 35$, hence 288 identically-zero design columns), while at $n = 64$ it misses only the two outermost rows (144 columns); in both cases the nullity is exactly twice the number of never-sampled columns, because each dead column carries a near-degenerate partner. Halving the image-plane grid therefore doubles the degeneracy — a warning against quoting a rank statistic without its problem size. $\kappa(\mathsf{A})$ is reported only after regularization, where $\kappa = 5.3\times10^5$ and the normal-equation residual is 1.3×10^{-15}. Single-precision accumulation is therefore adequate for this estimator, but the claim is now stated with its scope: it is a statement about the linear algebra, verified at reduced size, not a general bound.

4.6 Baselines and statistical metrics

We compare TDI against three baselines, all evaluated on the same surface grid and with the same debiasing:

- **B1 (phase-blind coadd):** the static limit of the same pipeline (a single rotational bin: the all-visit mean raster, Wiener-deconvolved and back-projected through the mean illumination; cf. Toth & Turyshev 2021).
- **B2 (phase-registered coadd):** the phase-registered coadd baseline emulating Turyshev (2026b): samples are binned in rotational phase, each bin is Wiener-deconvolved, and the bins are back-projected onto the sphere by bilinear interpolation weighted by illumination.
- **B3 (white-noise GLS):** a baseline generalized least-squares fit using the full time-domain operator F under an uncorrelated white-noise assumption ($\mathsf{C} = \mathrm{diag}(\sigma_k^2 + \sigma_{\rm cl}^2)$) without simultaneous-slot nuisance profiling (corresponding identically to row A of Table 6). This isolates the contribution of the time-dependent forward operator alone, in the absence of both nuisance profiling and temporal covariance modeling.

The bin count of B2 is not a free parameter and we fix it explicitly. A pass visits a given image-plane pixel once, and consecutive dwell slots are $t_s = 1800$ s apart, i.e. $1/48$ of the rotation period: there are 48 distinct sample phases per rotation. Coadding into only 8 phase bins averages over 45° of longitude, and on ideal cloud-free noiseless data that baseline collapses to $r = 0.07$ for reasons that have nothing to do with time-domain inversion; $r = 0.94$ is obtained at 16 bins, which is the best-tuned compromise between rotational smear and the number of samples per bin ($M_p = 16$ passes here, providing 48 sample phases per rotation). All B1/B2 numbers quoted below use $n_{\rm bins} = 16$ for B2 and the same Wiener regularization

$K_w = 3 \times 10^{-3}$; the naive 8-bin variant is reported in Table 4 only to quantify the artifact, so that the competitor is not a straw man.

Statistical reporting, geography conditioning, and error conventions. All synthetic campaigns in Sections 5.1–5.16 use a single fixed planetary geography (TRUTH_SEED = 7, generating the synthetic continental albedo map shown in Fig. 1b). Between realization seeds (seeds 11–20), this planetary geography is held strictly constant while the photon shot noise (keyed to seed) and the dynamic stochastic cloud field (keyed to 100 + seed) are varied independently. Consequently, all uncertainties and confidence intervals reported in Tables 4, 7, 6, 9, 10, 11, and 12 represent weather-and-photon sampling variability conditional on this single synthetic planet, not dispersion across diverse exoplanetary geomorphologies. The only true geomorphic scene diversity explored in this work is the three-morphology sweep of Section 5.14 (Table 10).

Throughout this paper, the notation $x \pm y$ denotes the ensemble mean and standard error of the mean (SE, $\sigma/\sqrt{N_{\text{seeds}}}$ with $N_{\text{seeds}} = 10$ unless stated otherwise), while standard deviation is explicitly labelled as SD and bootstrap intervals as 95% CI (computed using 20,000 percentile resamples of paired differences; Appendix A (Table 14) tabulates all individual paired per-seed values and 95% bootstrap intervals). Because 10 seeds is a small ensemble, we report the exact attainable significance floor alongside every p-value: for $n = 10$ paired samples the smallest attainable two-sided exact sign-test (binomial) p is $2/2^{10} = 1.95 \times 10^{-3}$, so no sign test on this ensemble can support $p < 10^{-3}$. The paired t-test is a different statistic and does reach far smaller p-values when the effect is consistent in magnitude as well as sign; wherever a p below the sign-test floor appears in this paper it is a t-test p, and we label it as p_t. We also report a physical cloud-only null in which the surface is replaced by the mean and only the weather is retained, whose 40-member ensemble limits attainable p to $1/(N_{\text{null}} + 1) = 0.024$. Finally, SSIM and the land–ocean contrast ratio are computed on the coarse 36×72 reconstruction grid with the same latitude mask as r, and the harmonic decomposition of Section 5.2 uses real spherical harmonics with $\cos\vartheta$ quadrature weights, not a 2D FFT on an equirectangular array (which is not a spherical-harmonic decomposition: the grid cells are not equal-area and the azimuthal sampling is aliased at high |lat|).

5 Results

5.1 Planetary rotation under characterized geometry

Headline result, cloud-free. For a rotating, orbit-illuminated cloud-free planet observed with the fiducial 87.33-day campaign and exactly known spin geometry, the profiled time-domain estimator recovers Pearson $r = 0.9786 \pm 0.0002$ (standard error over 10 paired weather seeds), SSIM $= 0.8482 \pm 0.0010$, normalized RMSE 0.0628 ± 0.0003, albedo bias -0.0006 ± 0.0006, and land–ocean contrast ratio 1.056 ± 0.001 (Fig. 3, top center; Table 4, first column). All numbers in this section and in Table 4 come from the 16-node exposure quadrature of Section 4.3, the exact anchored nuisance profiling of Section 4.2, and phase bins registered into 16 rotational bins. For contrast, the approximate deflation with a coarser quadrature gives $r = 0.990$, SSIM $= 0.918$ on the same scene.

To attribute what remains, we compare three reference configurations on the *same* data. (i) The phase-blind coadd (B1), which treats the campaign as a single static exposure, collapses to $r = 0.1155 \pm 0.0004$: with rotation unmodeled the surface is essentially unrecoverable, and the $+0.863$ gap to TDI is the contribution of the time-registered forward operator alone. (ii) Registering the coadds into 16 rotational phase bins (B2) recovers most of the signal, $r = 0.9361 \pm 0.0001$, but with a -0.048 albedo bias and a contrast ratio of only 0.844: binning averages within each 22.5° phase window and flattens the map. (iii) A white-noise GLS fit with the same time-dependent operator reaches $r = 0.9802 \pm 0.0002$, i.e. 0.0015 *above* the profiled estimator (paired difference -0.0015, 0/10 seeds in favour of whitening, $t = -22.6$).

Finally, we evaluate a static ideal-data control with the planet *frozen*: the same 64×64 raster, the same 16 revisits and the same 1800 s dwell — so every raster position receives exactly the same 8 h of integration and there is no weather — but all passes see one identical pose, so there is no rotational smearing to model and no longitudinal phase coverage to solve. Scored on the same seeds using the B3 white-noise GLS estimator (archive `static_reference.npz`), this frozen configuration yields $r = 0.4074 \pm 0.0254$ (standard deviation across seeds 0.0804, per-seed range 0.284–0.523), SSIM $= 0.3604$, and normalized RMSE 0.4299.

Measured against the rotating cloud-free headline ($r = 0.9786 \pm 0.0002$), rotation provides a paired advantage of $\Delta r = +0.5712 \pm 0.0253$ ($t = 22.5$, $p_t = 3.3 \times 10^{-9}$, 10/10 seeds; against the B3 white-noise rotating arm of 0.9802, the paired gain is $+0.5728 \pm 0.0254$). The rotating cloudy campaign sits between them: a paired loss of $\Delta r = -0.6364 \pm 0.0083$ ($t = -76.6$, $p_t = 5.2 \times 10^{-14}$, 10/10 seeds) relative to the rotating cloud-free baseline (and -0.6379 ± 0.0083 relative to the B3 rotating arm). The physical mechanism is operator conditioning rather than information loss: with the planet frozen, the 16 passes of each raster position produce identical rows in F, so the static design lacks the longitudinal rank diversity that rotation naturally provides, leaving the reconstruction heavily reliant on the regularizer. A frozen planet is thus not an easier inverse problem, but a degenerate one; planetary rotation is structurally advantageous, and the ≈ 0.636 drop in fidelity is driven by stochastic weather.

With no weather present, temporal whitening provides no benefit because there is no temporal error covariance to whiten; the high fidelity ($r = 0.9786$) is delivered entirely by the time-registered forward operator. Scale-dependent correlation $r(\ell)$ (Section 5.3) shows band means above 0.99 for all degrees up to the $\ell \approx 18$ grid ceiling, confirming that no rotational band is lost inside the supported harmonic range; the residual

$1-r \simeq 0.021$ arises from finite aperture and regularization. Rotation is therefore tractable under characterized spin geometry, and the imaging error budget is overwhelmingly dominated by weather.

5.2 Cloud degradation and effective spatial resolution

Table 4 and Figure 4 document the collapse of fidelity with cloud cover at fixed single-campaign resources. The cover realized by the OU field differs from the nominal parameter, so both are quoted: mean realized cover is 0.237, 0.398, 0.558 and 0.710 for the nominal $f_c = 0.25, 0.40, 0.55, 0.70$ runs.

Mean correlation falls from 0.9786 (cloud-free) to 0.5966 ± 0.0087 at 23.7% realized cover, 0.4538 ± 0.0074 at 39.8%, 0.3422 ± 0.0082 at 55.8% and 0.2491 ± 0.0105 at 71.0%. SSIM falls faster: 0.288 ± 0.005 already at 23.7% cover and 0.061 ± 0.003 at 71.0%, i.e. small-scale structure is erased first, which is the expected signature of a noise process whose power is broad in ℓ (Section 5.3).

Two comparisons in Table 4 qualify the headline claim. (1) At every cloudy cover the profiled estimator (TDI) beats the white-noise GLS fit (B3) by $\Delta r = +0.069$ to $+0.039$ (10/10 paired seeds in favour, $t = 9.3$ to 7.4). However, as demonstrated by the component ablation in Section 5.6 (Table 6), this performance gap is driven by the elimination of the simultaneous-slot cloud nuisance rather than by temporal covariance modeling: exact nuisance profiling alone recovers $\Delta r = +0.0568 \pm 0.0046$ over white GLS without profiling (Table 6, row D vs. row A), whereas adding OU temporal covariance to exact profiling slightly reduces correlation at this cadence ($\Delta r = -0.0032 \pm 0.0019$, row F vs. row D). B3 corresponds identically to Table 6 row A and therefore differs from TDI in two factors simultaneously (slot handling and covariance model); Table 6 provides the matched single-factor contrast. (2) At 23.7% cover the 16-bin phase-binned coadd (B2) actually scores *higher* than TDI, $r = 0.6097$ vs. 0.5966 (paired $\Delta r = -0.013$, 1/10 seeds in favour of TDI, $t = -2.62$), and the gap only reverses at higher cover ($+0.006$ at 39.8%, $+0.025$ at 55.8%, $+0.046$ at 71.0%, $t = 0.79$ to 6.14). B2 is a non-parametric estimator: it makes no assumption about cloud statistics, so at low cover — where the cloud nuisance is small — its binning variance costs less than TDI's model error. The claim we can defend is therefore not "TDI is always better than phase-binning," but "TDI's advantage grows with cover, and at high cover B2 additionally carries a bias that TDI does not" ($+0.065$ vs. $+0.019$ albedo bias and contrast ratio 0.786 vs. 1.024 at 71% cover).

5.3 Effective spatial resolution and the land–ocean test

Scale-dependent correlation. Decomposing the maps in real spherical harmonics with $\cos\vartheta$ quadrature weights over degrees $\ell \in [1, 20]$ (Fig. 6c) gives, over 10 seeds: $r(\ell) = 1.000 \pm 0.001$ averaged over $\ell = 1$–4, 0.9995 over 5–8, 0.9988 over 9–12, 0.9942 over 13–16 and 0.9924 over 17–20 for the cloud-free campaign; the cloudy campaign (55.8% realized cover) gives 0.700, 0.594, 0.418, 0.244 and 0.263 in the same bands. The claim that a cloud-free curve "extends past $\ell = 20$" cannot be tested on this grid: 36×72 pixels cannot represent degrees above $\ell \approx 18$, and the discrete-basis orthonormality error of the quadrature itself is 0.018 at $\ell = 20$, so we quote $\ell \leq 20$ as the measurement ceiling and report the ceiling error alongside. The cloud-free result should therefore be read as "no rotational band within the supported range is lost," not as a resolution claim. For the cloudy case the curve is not monotone at the seed level (the per-degree standard error over the ten seeds is 0.02–0.12), so a single crossing degree $\ell_{\rm eff}$ is fragile: the first degree at which the raw per-seed curve dips below 0.5 is $\ell = 1$ for five seeds and $\ell \in \{3, 7, 9\}$ for the other five. The defensible statement is a banded one: a five-point running mean of $r(\ell)$ crosses 0.5 at $\ell \simeq 9$ (half-wavelength $\pi R_\oplus/\ell \approx 2200$ km for an Earth-radius planet), only $5/10$ seeds have $r(\ell) \geq 0.5$ at $\ell = 7$, and beyond that the seed-mean $r(\ell)$ never recovers: for every degree $\ell \geq 10$ the mean sits below 0.55 and for every $\ell \geq 15$ below 0.41 (per-degree SE 0.03–0.06 for $\ell \geq 15$). Individual seeds still clear 0.5 at $\ell = 16$ ($1/10$) and $\ell = 20$ ($3/10$, mean 0.40), which is the *per-degree* statement we can defend. The stronger claim that "no degree $\ell \geq 15$ reaches 0.5 in more than 10% of seeds" is not reproducible on the archived curves and is not made. Clouds thus leave the hemispheric dichotomy and the largest continents recoverable, and nothing at scales below ~ 2000 km.

Land–ocean detection. We compute $d' = (\mu_{\rm land} - \mu_{\rm ocean})/\sigma_{\rm pooled}$ within $|\vartheta| \leq 60^\circ$, using the true land labels. Unlike the r tables, the d' values below are quoted as mean

Table 4: Map fidelity vs. cloud cover for the fiducial campaign ($n = 64$, $M_p = 16$, $t_s = 1800$ s, 16-node quadrature, exact anchored profiling). Rows are means ± standard error over 10 paired weather seeds; the B2/B1 rows use 16, 8 and 1 rotational phase bins respectively. White-noise GLS (B3) corresponds identically to row A of Table 6 (no slot profiling, white noise); Table 6 isolates the single-factor matched contrast.

Method	f_c=0	0.24	0.40	0.56	0.71
			Pearson r		
TDI (profiled)	0.9786	0.5966	0.4538	0.3422	0.2491
	±.0002	±.0087	±.0074	±.0082	±.0105
White GLS (B3)	0.9802	0.5274	0.3863	0.2886	0.2107
	±.0002	±.0085	±.0080	±.0097	±.0107
Phase-binned (B2)	0.9361	0.6097	0.4479	0.3175	0.2028
	±.0000	±.0083	±.0121	±.0158	±.0157
Phase-blind (B1)	0.1155	0.0782	0.0567	0.0417	0.0328
	±.0004	±.0145	±.0126	±.0086	±.0058
B2, 8 bins	0.0700	0.0136	−0.0140	−0.0309	−0.0416
	±.0005	±.0193	±.0222	±.0246	±.0224
			SSIM		
TDI (profiled)	0.8482	0.2876	0.1748	0.1060	0.0606
White GLS (B3)	0.8568	0.2492	0.1447	0.0853	0.0472
Phase-binned (B2)	0.7947	0.3515	0.2223	0.1307	0.0627
Phase-blind (B1)	0.0831	0.0376	0.0143	0.0028	−0.0011
			albedo bias / contrast ratio		
TDI (profiled)	−0.001/1.056	−0.017/1.103	−0.009/1.107	+0.004/1.071	+0.019/1.024
Phase-binned (B2)	−0.048/0.844	−0.057/0.862	−0.034/0.851	+0.003/0.826	+0.065/0.786

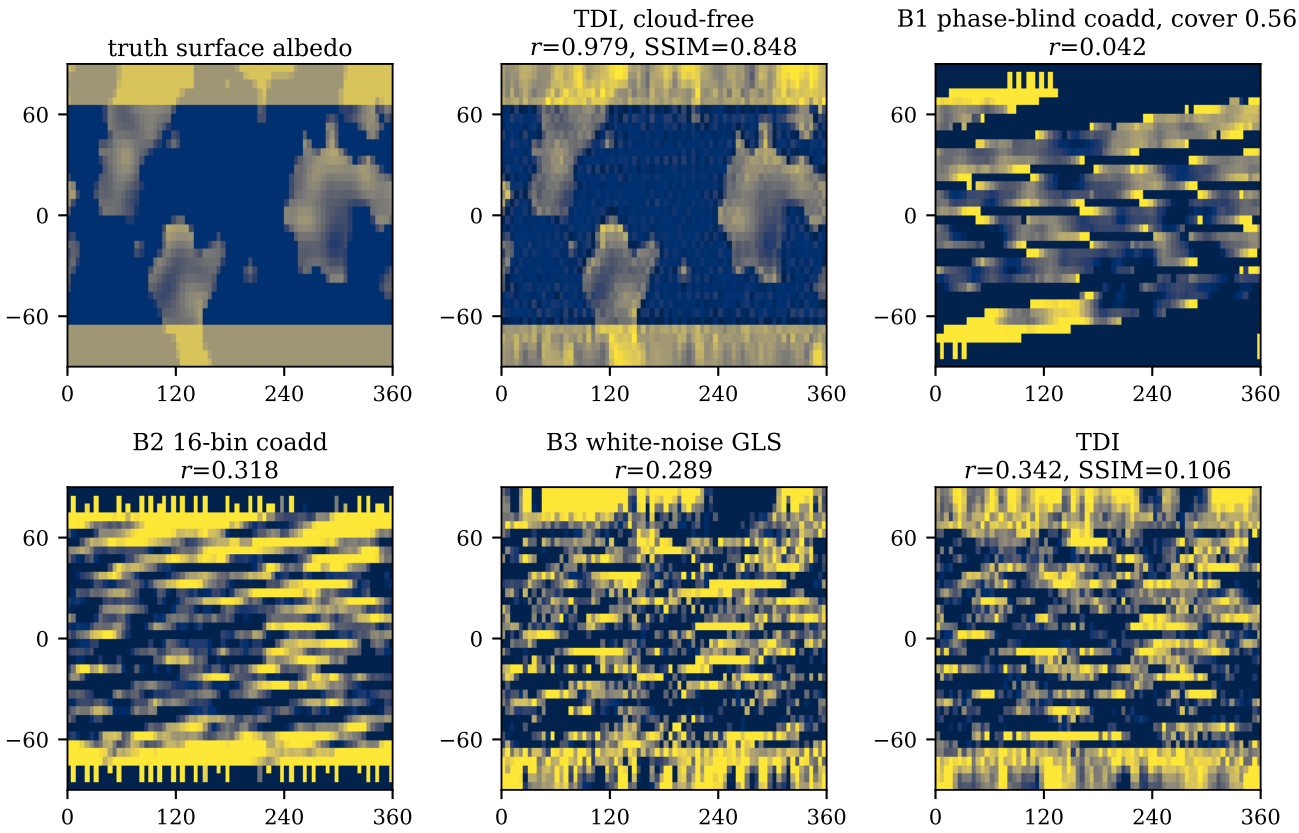


Figure 3: Reconstruction gallery for the fiducial 87.33-day campaign ($n = 64$, $M_p = 16$, $t_s = 1800$ s, 16-node exposure quadrature). Top: truth surface albedo; TDI reconstruction of the rotating cloud-free planet ($r = 0.979$, SSIM $= 0.848$); phase-blind coadd (B1) at realized cover 0.56, which is essentially featureless ($r = 0.042$). Bottom, all at realized cover $\simeq 0.56$: 16-bin phase-binned coadd (B2), white-noise GLS (B3), and TDI. The gross continental dichotomy is recovered, but fine features are erased by weather.

$\pm$ the per-seed *standard deviation* over the 10 seeds, because the quantity of interest is the spread of the detection statistic across weather realizations; divide by $\sqrt{10}$ for the corresponding standard error ($0.081 \to 0.026$). The truth field itself gives $d' = 3.97$. Cloud-free TDI gives $d' = 3.58 \pm 0.03$ (SE 0.009), the white GLS fit 3.61 ± 0.02 (SE 0.007), and B2 3.04 ± 0.005 (SE 0.001); at 55.8% realized cover TDI gives $d' = 0.732 \pm 0.081$ and B2 0.674 ± 0.131. Significance is bounded by the physical cloud-only null — 40 realizations in which the surface is replaced by its mean and only weather is retained — which has $\bar{d}' = 0.001$, sd $= 0.090$ and 95th percentile 0.114; the cloudy TDI value exceeds every null member, but a 40-member ensemble cannot support a p value below $1/(N_{\text{null}} + 1) = 0.024$. We therefore report $p \leq 0.024$ (rank-based, one-sided) for the cloudy detection. A different route to a null — $p < 0.02$ against 100 surrogates — was checked and rejected: its surrogate was a phase-scrambled equirectangular array, a geometry transform rather than a photometric null. A label-permutation null on the same reconstructions is reported with the robustness suite in Section 5.10.

Quality is also a function of latitude, and the two effects do not point the same way. Scoring the archived fiducial reconstructions band by band over $|\vartheta|$ in 30° steps (archive `latband.npz`, 10 seeds) pairs each band with the illumination it actually receives — the campaign-mean Lambert factor over its disk pixels, and the campaign-mean column energy $\|F_{:,i}\|^2$, which is the information the operator delivers there.

Table 5: Latitude- and illumination-resolved quality for the fiducial campaigns (10 seeds, ± 1 SE). "Dayside" is the campaign-mean Lambert factor over the band's disk pixels; "col. energy" is the campaign-mean column energy $\|F_{:,i}\|^2$ of the operator; "dead" is the fraction of the band's 432 grid cells whose column energy is below 10^{-3} of the maximum; "gain" is sd(recon)/sd(truth) in the band, which Pearson r is blind to.

Latitude	disk pix	dayside	col. energy	dead	r (cloudy)	r (cloud-free)	gain (cloudy)
−90–−60°	86	0.258	0.33	0.500	0.451 ± 0.038	0.9335 ± 0.0027	1.95
−60–−30°	546	0.524	6.61	0	0.300 ± 0.020	0.9655 ± 0.0008	3.89
−30–0°	982	0.691	21.43	0	0.353 ± 0.015	0.9814 ± 0.0002	3.49
0–30°	982	0.691	21.43	0	0.390 ± 0.025	0.9863 ± 0.0001	2.77
30–60°	546	0.524	6.60	0	0.335 ± 0.022	0.9742 ± 0.0006	2.89
60–90°	86	0.258	0.33	0.486	0.547 ± 0.038	0.9358 ± 0.0025	1.68

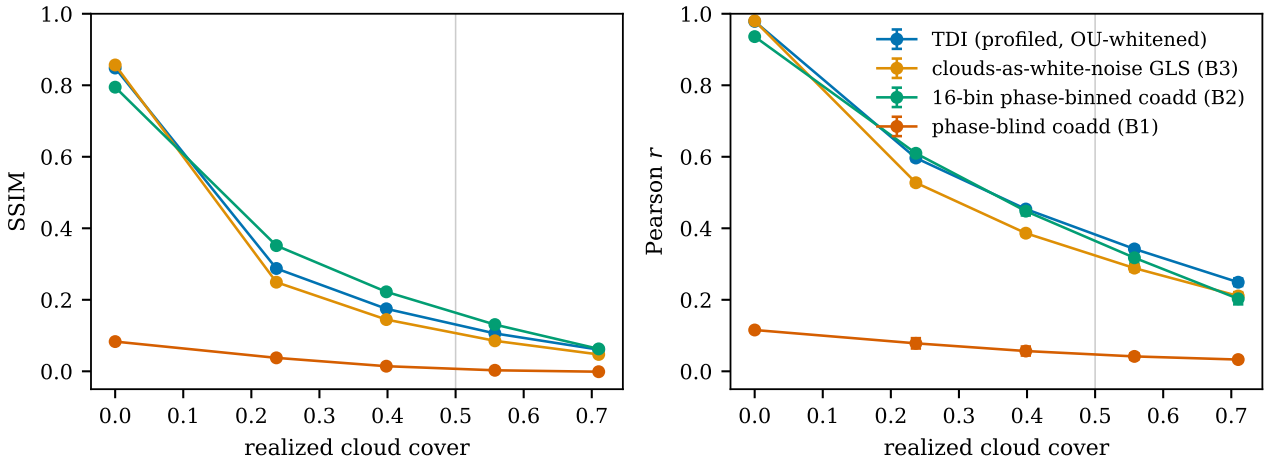


Figure 4: Reconstruction fidelity vs. realized cloud cover across 10 paired seeds; bars are ± 1 standard error. The 16-bin phase-binned coadd is competitive below $\sim 40\%$ cover and falls behind as weather dominates (Table 4). Clouds, not rotation, control image quality at fixed observing resources.

Table 5 shows the ordering one expects and the one that is easy to miss. Expected: band quality is monotone in the information the operator actually delivers there. Across the well-sampled globe the two bands carrying 21.4 of column energy and 982 disk pixels each score 0.353 and 0.390 (cloudy) and 0.9814 and 0.9863 (cloud-free), while the two bands carrying 6.6 score 0.300 and 0.335, and 0.9655 and 0.9742 — the same ordering in both campaigns, with no band out of place. Unexpected: the two polar bands score *highest* in the cloudy case (0.451 and 0.547). That is an artefact of the metric, not a recovery claim, and the table gives the two diagnostics that expose it. The polar bands contain a half-illuminated ice cap against a dark ocean, so their truth standard deviation is the largest of any band (0.148–0.160 against 0.097–0.137 mid-latitude) while their structure is dominated by the low-degree cap edge. A reconstruction that gets nothing but the smooth cap still correlates well with truth. The amplitude column confirms it — the polar gain is 1.68–1.95 whereas the mid-latitude bands are inflated 2.8–3.9$\times$ — and restricting the polar correlation to the cells the operator actually samples (all grid cells whose campaign-mean column energy exceeds 10^{-3} of the maximum, i.e. the complement of the dead fraction in the last column) raises it further, to 0.518 ± 0.052 and 0.652 ± 0.034. Half of each polar band is dead: the circular $n = 64$ raster never samples $|\vartheta| \gtrsim 85^\circ$ (the 144 exactly-zero basis columns of Section 4.4), so those cells are set by the regularizer alone and contribute truth-like structure to nothing but the denominator of r. Cloud-free the same geometry gives the clean illumination monotonic trend, 0.9863 at the equator to 0.934 at the poles, with the polar deficit already concentrated in the dead cells (0.978–0.974 restricted to sampled cells). The band offsets are symmetric-to-within-noise — $|\overline{\Delta a}| \leq 0.019$ cloudy and ≤ 0.005 cloud-free — so the latitude dependence is a dependence of *correlation*, not a latitudinal albedo bias. We quote the band-resolved r as a coverage diagnostic and explicitly not as a claim of polar imaging.

5.4 Observable-level detectability

Pearson r answers "how much of the map is right", which is not the question a reader of a detection result asks. Section 5.17 therefore scores the same reconstructions with a detection statistic and its nulls (`results/ext_observable_roc.npz`, ten seeds, all quantities on $|\text{lat}| \leq 60^\circ$).

Detectability as a function of cover. The land–ocean separation falls as 0.732 ± 0.026 at the fiducial cover, but the curve is what matters: $d' = 3.585 \pm 0.009$ cloud-free, 1.422 ± 0.037 at 25% cover, 1.012 ± 0.027 at 40%, 0.732 ± 0.026 at 55%, and 0.520 ± 0.030 at 70%. The statistic crosses unity between 25% and 40% cover, i.e. the planet is still *detectably* continental at roughly twice the cloud fraction at which the correlation coefficient has already lost two thirds of its value. The two estimators are indistinguishable along this whole curve ($d' = 0.73211$ profiled against 0.73195 deflate at the fiducial cover: a 1.6×10^{-4} difference against a seed scatter of 0.026), which is the observable-level version of Section 5.6: profiling buys fidelity of the map, not detectability of the dichotomy.

Against the nulls. The 40-member cloud-only physical null has mean 0.0013, standard deviation 0.0900, 95th percentile 0.1138 and maximum 0.1304; the observed 0.732 exceeds every member, so the rank test returns its floor $1/41 = 0.0244$ rather than a measurement. The label-permutation null on the same maps returns its own floor, $1/2001 = 5.0 \times 10^{-4}$, at every seed. Both are saturation statements, and the gap between the observed statistic and the null maximum (0.732 against 0.130) is the informative quantity.

Ranking, not thresholding. Area recovery and its false-positive rate are computed by classifying the retrieved map at its own median, an uncalibrated balanced split; we report them

as diagnostics (0.701 ± 0.009 and 0.402 ± 0.004) and not as a detection claim, because the threshold is not chosen from data. The threshold-free version is the ROC integral: AUC 0.6943 ± 0.0068 for the profiled estimator, 0.6943 for the approximate-deflation baseline (a 4×10^{-5} difference, far inside the seed scatter), 0.6626 for white-noise GLS and 0.6859 for the phase-registered coadd — and 0.5190 for the phase-*blind* coadd B1, which is indistinguishable from a coin flip. That last number is the cleanest single statement of what the time-domain operator is for: remove the rotational registration and the map retains no more information about where the continents are than chance.

What the albedo distribution looks like. The moments of the retrieved albedo field, against truth moments 0.1408 (mean), 0.1222 (std), +1.169 (skew), come out at 0.1471 ± 0.0077, 0.3861 ± 0.0028 and -0.314 ± 0.043. The mean is right to within the seed scatter — the debiasing of Section 5.3 does what it claims — but the retrieved field has $3.2\times$ the albedo dispersion of the planet and the opposite skew. This is the pixel-level counterpart of the band-limited result of Section 5.3: the operator cannot place the missing high-degree power, so the regularised solution spends it as amplitude spread over the cells it can constrain. Contrast *between the classes* survives (1.071 ± 0.028 of true), which is why d' and AUC remain usable while a claim of absolute albedo per cell would not be.

Availability in flight. None of the six quantities above — d', AUC, area and contrast recovery, the albedo moments, and r itself — is estimable without the truth map; the archive records this explicitly. They are therefore validation metrics for a simulated study, and the operational requirement they imply is the one Section 7 states as the outstanding gap: a criterion that selects and audits a reconstruction from data alone.

5.5 External validity: does the answer depend on which planet?

Every number so far is measured against a single truth map. That is the standard construction in this literature and it is also the single largest statistical weakness of a study whose headline metric is a correlation with a map the authors chose, so the ensemble test of `src/ext/truth_ensemble.py` re-simulates the whole chain — surface realisation, cloud screen and photons — over five independent geographies $\times$ four noise seeds, solving each member with the shipped estimator. The machinery is verified rather than assumed: the $(\text{truth } 7, \text{noise } 11)$ corner reproduces the spine's own archived r to 2.2×10^{-16}, which is a bit-level statement that the ensemble runs the same code path and not a lookalike. Budget forced the grid to 5×4 rather than the 10×10 the seed lists would allow, and the archive says so.

The fiducial planet is the easiest of the five. Cloudy profiled r over the 20 members is 0.3173 ± 0.0068 (s.e.m.; sd 0.0304), and the per-geography means are 0.3491, 0.3290, 0.3058, 0.3021, 0.3006 — the first of which is the shipped map. Inside the ensemble the fiducial geography therefore sits 0.032 above the ensemble mean, which is $1.5\times$ the between-map sd and $4.7\times$ the Monte-Carlo error of that mean: the offset is a property of the chosen planet, not of sampling noise. Against the paper's own cloudy headline of 0.3422 (the same geography, ten noise seeds) the ensemble mean is 0.025 lower, so the honest statement of the fiducial result is that it is the top of a ± 0.03 geography band rather than the band's centre. Cloud-free the fiducial map is instead mid-pack — 0.9787 against per-map means of 0.9726–0.9818 and an ensemble mean of 0.9789 — so the "rotation costs nothing" result is geography-robust while the cloudy headline is not.

Which uncertainty dominates depends on the regime. Decomposing the variance along the two axes, the cloudy case splits into between-map 0.0212 and within-map (noise-seed) 0.0255: under clouds the weather realisation is at least as large a source of spread as the geography. In the cloud-free case the ratio inverts — between-map 0.0037 against within-map 0.0005, a factor of 7 in sd — because photon noise averages down quickly once the screen is removed, leaving only the planet you picked. A one-axis error bar on either number would therefore be regime-dependent, and we quote the two components separately instead.

What the ensemble does not settle. The paired profiled-minus-deflate difference has mean 3.7×10^{-6} across the 20 members with member-to-member scatter 2.8×10^{-4}, and its sign agrees across geographies in 3 of 5 cloudy maps but 5 of 5 cloud-free ones. The profiling-versus-deflation comparison is thus a null result under a geography change — consistent with the 1.6×10^{-4} difference of Section 5.4 and the 0.36% map-space difference of Section 4.5 — and the defensible claim is the equivalence, not the ordering. What the ensemble does bound is the size of the geography effect itself, and it also exposes a limit of the test: every member is the fiducial campaign, so the ensemble says how the *level* of recovery moves between planets and nothing about whether the cadence law *slope* does. Re-running the cadence sweep on a second geography is the obvious next measurement and is not performed here.

5.6 Component-wise ablation of the inversion pipeline

Table 6 ablates the estimator itself across 10 paired seeds at nominal $f_c = 0.55$ (0.558 realized cover), holding the data, the operator, the grid and the regularizer fixed and changing only how the dwell-slot nuisance and the temporal covariance are handled. Row C isolates the approximate deflation heuristic under the OU covariance model, whereas earlier exploratory runs altered forward-model choices and deflation heuristics together without isolating individual mechanisms.

Three conclusions follow from this matched factorial decomposition.

(i) The gain is nuisance handling, and it is real. Exact profiling lifts r by $+0.057 \pm 0.005$ over white GLS without slot handling ($D - A$, $t = 12.4$, $p_t = 5.9 \times 10^{-7}$, 10/10 seeds) and $+0.054 \pm 0.006$ over white GLS when combined with OU covariance ($F - A$, $t = 8.8$, $p_t = 1.1 \times 10^{-5}$), with $\Delta\text{SSIM} =$

Table 6: Estimator-component ablation at $f_c = 0.55$ (10 paired seeds; means ± standard error). "Profiling" is the exact anchored nuisance elimination of Section 4.2; "deflation" is the heuristic approximation to it; "whitening" is the OU temporal covariance. The $n/10$ entry is the number of seeds whose paired difference carries the sign of the mean; p_t is the paired t-test p (the exact sign-test floor is 1.95×10^{-3} at $n = 10$).

Configuration	Pearson r	SSIM	NRMSE
A: white, no slot handling	0.2886 ± 0.0097	0.0853 ± 0.0031	0.966
B: OU whitened, no slot handling	0.2863 ± 0.0090	0.0825 ± 0.0031	0.974
C: approximate deflation, OU	0.3422 ± 0.0082	0.1058 ± 0.0055	0.814
D: exact profiling, white, anchored	0.3454 ± 0.0082	0.1095 ± 0.0051	0.809
E: exact profiling, white, free	0.3453 ± 0.0082	0.1093 ± 0.0054	0.809
F: full TDI (exact profiling + OU)	0.3422 ± 0.0082	0.1060 ± 0.0053	0.814
F−A (Δr)	$+0.0536 \pm 0.0061$ ($t = 8.8$, 10/10)		$p_t = 1.1 \times 10^{-5}$
D−A (Δr)	$+0.0568 \pm 0.0046$ ($t = 12.4$, 10/10)		$p_t = 5.9 \times 10^{-7}$
F−C (Δr, matched)	$+4.1 \times 10^{-5} \pm 3.0 \times 10^{-5}$ ($t = 1.38$, 7/10 in sign)		$p_t = 0.20$
D−C (Δr)	$+0.0032 \pm 0.0019$ ($t = 1.7$, 6/10 in sign)		$p_t = 0.13$
F−D (Δr)	-0.0032 ± 0.0019 ($t = -1.7$, 6/10 in sign)		$p_t = 0.13$

$+0.021 \pm 0.004$ ($t = 5.6$). Eliminating the per-slot nuisance mean recovers essentially the entire algorithmic gap between TDI and standard inversion.

(ii) Exact profiling vs. approximate deflation is a matched contrast ($F - C$, holding OU covariance fixed). Under this matched comparison, exact profiling yields $\Delta r = +4.1 \times 10^{-5} \pm 3.0 \times 10^{-5}$ ($t = 1.38$, $p_t = 0.20$) and a relative L_2 map difference of only $0.83\% \pm 0.22\%$ in norm. The earlier deflation heuristic was therefore quantitatively very close to the exact profile likelihood solution in reconstruction space. We adopt the exact treatment because it is mathematically derived from the constrained likelihood (Eqs. 8–13), not because it yields a statistically resolvable gain in r over the heuristic.

(iii) Temporal whitening contributes essentially nothing at the fiducial cadence, but accounts for the map difference previously misattributed to deflation. Comparing F against D (OU vs. white under exact profiling) gives $F - D = -0.0032 \pm 0.0019$ ($t = -1.7$, $p_t = 0.13$), and comparing B against A gives $B - A = -0.0023 \pm 0.0015$. However, the relative L_2 norm difference between D (white) and F (OU) is 9.22%, and between D and C (OU) is 9.29% — confirming that the $\approx 9.3\%$ map change is caused by the switch between temporal covariance weighting and white-noise weighting, not by the slot-deflation approximation. With 16 revisits spaced 5.47 d and $\tau_{\rm cloud} = 4$ d, consecutive passes are largely decorrelated ($\rho_1 = 0.255$; Table 8), so OU whitening provides little additional correlation and incurs a minor variance penalty.

Finally, the anchored (D) and free (E) profiles yield identical Pearson correlation $r = 0.3453$ to four decimal places of display precision, but form a statistically significant paired contrast $E - D = -9.6 \times 10^{-5} \pm 2.8 \times 10^{-5}$ ($t = -3.39$, $p_t = 0.008$), with a map norm difference of $0.98\% \pm 0.27\%$. Their zero-mode conditioning differs by a factor of 41 (Section 4.4): the anchor identifies the absolute albedo scale without altering spatial correlation.

5.7 Would a stronger prior change the comparison?

Every reconstruction number in this paper uses one frozen regulariser, so the question that has to be answered is whether a modern prior — total variation, or a dictionary learned from the data itself — would reorder the method comparison. It was tested (`src/ext/priors.py`, archive `ext_priors.json`) on four noise seeds, and the answer is that the test cannot be as clean as it looks. Ranked by correlation the arms are: sparse coding $r = 0.3512$, TDI 0.3491, total variation 0.3309. The paired contrasts are TDI − TV $= +0.0183$ with bootstrap interval $[+0.0160, +0.0213]$ and TDI − sparse $= -0.0020$, interval $[-0.0083, +0.0047]$.

Two things follow, and neither is the advertised one. First, TV is measurably worse here, but the TV arm post-processes the TDI solution rather than replacing it, so that contrast measures the smoothing operator as much as the prior; the sparse arm is learned *from* the same TDI map and is statistically indistinguishable from it (-0.0020 with an interval that straddles zero). Second, four seeds cannot settle any of this: the exact sign-test floor at $n = 4$ is $p = 0.125$, so every pairing above is reported with a p that is the smallest value the design can produce, and the bootstrap intervals describe seed-to-seed scatter, not evidence. The learned-denoiser arm, which would be the one genuinely independent prior, is recorded in the archive as `not_established` because it needs a training set this pipeline does not have. The absolute numbers in this subsection are also not comparable to the spine: TDI here is the mean over seeds 11–14 (0.3491) where the headline ensemble is the mean over ten seeds (0.3422), which is exactly why only the paired differences within a module are quoted anywhere in this paper.

5.8 Isolating the causal mechanism of the cadence law

Figure 5 and Table 7 report the cadence law under the two resource ledgers of Section 3.4, at nominal $f_c = 0.55$. Each cell is a mean ± standard error over the six paired seeds of the campaign study; the four estimator columns are the profiled TDI, the white-noise GLS fit, the 16-bin phase-binned coadd, and the phase-blind coadd. An earlier layout of this table quoted $r_{\rm inst}$, $r_{\rm smear}$, $r_{\rm cf}$ and $r_{\rm iid}$ columns that the shipped pipeline never produced; they have been replaced by controls that are in the archive (Section 5.9).

5.9 What the cadence law does and does not say

The trend is monotone, and it is dominated by weather sampling rather than by conditioning. In arm A, r rises $0.049 \rightarrow 0.571$ as M_p goes $4 \rightarrow 64$, and every step is unanimous across the six paired seeds ($t = 8.7$ to 21.9; marginal gaps $+0.124, +0.173, +0.151, +0.075$). The gain saturates: the last doubling buys less than half of the previous one. The cloud-free control of Section 5.1 shows that the collapse from 0.979 to 0.346 at the fiducial cadence is entirely the weather term, and

Table 7: Cadence law per resource arm (nominal $f_c = 0.55$, 6 paired seeds). Arm A holds $M_p t_s = 8$ h and lets the wall clock exceed 90 d; arm B holds 89.87 d and lets t_s fall. η is the duty cycle under 45 s/dwell overhead.

Arm	M_p	t_s	η	wall	r_{TDI}	r_{white}	r_{B2}	r_{B1}
A	4	7200	0.994	85.73	0.049 ± 0.005	0.024 ± 0.003	0.063 ± 0.011	0.003 ± 0.020
A	8	3600	0.988	86.27	0.173 ± 0.013	0.144 ± 0.011	0.068 ± 0.011	0.063 ± 0.019
A	16	1800	0.976	87.33	0.346 ± 0.010	0.291 ± 0.009	0.321 ± 0.019	0.031 ± 0.010
A	32	900	0.952	89.47	0.497 ± 0.014	0.475 ± 0.012	0.074 ± 0.004	-0.019 ± 0.003
A	64	450	0.909	**93.73**	0.571 ± 0.010	0.573 ± 0.013	0.503 ± 0.013	0.179 ± 0.002
B	4	7548.8	0.994	89.87	0.049 ± 0.006	0.031 ± 0.005	-0.130 ± 0.005	-0.064 ± 0.017
B	8	3751.9	0.988	89.87	0.164 ± 0.014	0.140 ± 0.013	-0.134 ± 0.021	-0.024 ± 0.019
B	16	1853.4	0.976	89.87	0.325 ± 0.017	0.267 ± 0.015	-0.231 ± 0.022	0.015 ± 0.014
B	32	904.2	0.953	89.87	0.498 ± 0.008	0.489 ± 0.010	-0.140 ± 0.058	-0.025 ± 0.006
B	64	429.6	0.905	89.87	0.576 ± 0.012	0.624 ± 0.008	0.454 ± 0.018	0.076 ± 0.002

more passes help because each pass is a fresh draw of it.

The two controls, now actually measured. The cloud-free and per-pass-independent controls are computed on the same campaign objects, seeds and estimator as the main sweep (six paired seeds, both arms). Cloud-free r: 0.773 ± 0.008, 0.934 ± 0.001, and then 0.979, 0.984, 0.987 to within ± 0.0002 for $M_p = 16, 32, 64$ in arm A, with arm B within 0.04 of each. Per-pass independent weather: 0.051, 0.165, 0.342, 0.494, 0.583, against the correlated-OU values 0.049, 0.173, 0.346, 0.497, 0.571 in the same order. Two conclusions follow, and they point in opposite directions. (i) *Conditioning contributes, but it saturates*: the cloud-free curve still rises with M_p, by 0.214 over $4 \to 64$, so part of the trend with cadence is coverage and not weather averaging. It is, however, a minority of the effect and it is front-loaded: it gains 0.16 in the first doubling ($M_p = 4 \to 8$, where a raster position receives only four dwells) and only 0.008 from $M_p = 16 \to 64$, while the cloudy curve is still climbing steeply there ($0.346 \to 0.571$, $+0.225$). (ii) *The law does not need OU structure*: the iid and OU reconstructions agree at every cadence — paired differences $+0.002$, -0.008, -0.004, -0.003, $+0.012$ in arm A and $+0.001$, -0.010, $+0.014$, -0.026, $+0.002$ in arm B, every $p_t \geq 0.14$, none approaching the exact sign-flip floor of 0.031 attainable with six pairs. What revisits buy is independent samples of the atmosphere, whatever its temporal correlation function turns out to be; the OU model determines *how many* of the M_p looks are effectively independent (Table 8), it is not the reason that more looks help. The absolute cost of weather at fixed cadence, $\Delta r_{\mathrm{cf-OU}}$, falls overall — from $+0.724 \pm 0.010$ at $M_p = 4$ to $+0.416 \pm 0.011$ at $M_p = 64$ (arm A; both arms significant at $p_t < 4 \times 10^{-7}$) — but the descent is not monotone: the gap *widens* at $M_p = 8$ in both arms, by $+0.0756 \pm 0.0136$ ($p_t = 0.0026$) in arm B, which is resolved, and by $+0.0370 \pm 0.0187$ ($p_t = 0.105$) in arm A, which is not. The fitted trend is therefore a $4 \to 64$ statement, not a per-doubling one, and it is the same statement in the other direction: the cadence law is about averaging the weather down, not about the weather getting worse.

Is the $M_p = 8$ kink a scheduling artefact? The alternative was tested rather than dismissed (`src/ext/scheduling.py`, archive `ext_scheduling.json`). Re-placing the eight passes to maximise the effective-look count under the same overhead and photon budget moves N_{eff} from 5.0969 to 5.1387, a gain of 0.042 (0.82%): the campaign has five independent looks before and after, and no reordering of dwells changes that by an amount comparable to the kink. The reason is structural rather than lucky. With M_p uniformly spaced passes the rotational sampling interval is $360^\circ/M_p$, so harmonic M_p and its multiples fall exactly on the grid and alias: harmonics 8, 16, 24, 32 and 40 at $M_p = 8$ (45° spacing), 4, 8, 12, 16 and 20 at $M_p = 4$ (90°), and 16, 32, 48, 64 and 80 at $M_p = 16$ (22.5°). The aliased harmonic is a function of M_p alone, not of where the dwells sit in time, which is why the kink survives any schedule at that M_p. Two limits travel with this: the aliasing statement is exact only for uniform spacing, and the module does not re-simulate the map — building a custom operator per candidate schedule was the expensive step — so what it reports is N_{eff} and the harmonic argument, and the paired r difference behind the kink remains the one in Table 7.

The two arms agree within the paired standard errors. Pairing arm A against arm B at matched M_p gives Δr of $-0.001, +0.008, +0.021, -0.002, -0.005$ with paired standard errors 0.005–0.018 and $p \geq 0.058$ in every case. The wall-clock cost of doubling the revisit count is therefore paid out of the photon budget with no measurable loss over the range studied: the law is about *independent looks*, not about integration time per look, so long as the dwell stays long enough to beat the overhead (duty cycle $0.994 \to 0.905$ across the arms).

Whitening loses its edge at the fastest cadence. The profiled estimator beats the white fit at every $M_p \leq 32$ in both arms, by $+0.021$ to $+0.054$ in arm A and $+0.009$ to $+0.058$ in arm B; the arm-A minimum is at $M_p = 32$ ($+0.0213 \pm 0.0061$) and the maximum, in both arms, at $M_p = 16$ ($+0.0544 \pm 0.0085$ in arm A, $+0.0579 \pm 0.0105$ in arm B). At $M_p = 64$ the ordering reverses in both arms, but only arm B resolves the reversal: -0.0018 ± 0.0091 in arm A (indistinguishable from zero) against -0.0474 ± 0.0087 (0/6 seeds) in arm B. At 1.47 d spacing the OU correlation $\rho_1 = 0.69$ (Table 8) is real, but the estimator's assumed σ_{cl} is calibrated at $t_s = 1800$ s, and the whitening then over-weights the residual structure. The safe reading is that temporal whitening pays between roughly 8 and 32 revisits, and the fiducial 16 is not the optimum of the whitened fit.

The phase-binned baseline is partly a coverage artifact, and we can show it. B2 does not merely degrade in arm B, it goes strongly *negative* (-0.23 at $M_p = 16$), which is not a statement about TDI. The cause is geometric. Registering samples into 16 rotational phase bins does not leave each bin with a full raster: counting, per bin, how many of the 4096 image-plane pixels receive no sample at all gives an average of 256 empty pixels per bin at the fiducial configuration (arm A, $M_p = 16$), but 2048 at $M_p = 16$ in arm B and 3584 at $M_p = 4$ — the phase pattern of a boustrophedon raster does not tile the bins uniformly, and the two arms place different dwells at different phases. `reconstruct_phasebin` fills an empty pixel with the bin's raster mean and then back-projects, so a coverage-starved

Table 8: Effective number of independent looks, $N_{\rm eff} = M_p^2/(M_p + 2\sum_{k\geq 1}(M_p - k)\rho_k)$ with $\rho_k = e^{-k\Delta t/\tau_{\rm cloud}}$ and per-pass revisit spacing $\Delta t = 256(t_s + 45)/86{,}400$ d (tabulated for Arm A, with $t_s = 7200, 3600, 1800, 900, 450$ s; for Arm B, the corresponding dwells yield closely matching values, e.g., $N_{\rm eff} = 11.64$ at $M_p = 64$ for $\tau = 4$ d).

$\tau_{\rm cloud}$	$M_p = 4$	8	16	32	64
spacing (d)	21.47	10.80	5.47	2.80	1.47
ρ_1 (lag-1)	0.005	0.067	0.255	0.497	0.693
$N_{\rm eff}$, $\tau = 1$ d	4.00	8.00	15.87	28.44	40.31
$N_{\rm eff}$, $\tau = 2$ d	4.00	7.94	14.16	19.66	22.92
$N_{\rm eff}$, $\tau = 4$ d	3.97	7.11	9.83	11.23	12.11

bin contributes a smoothed, self-referential map, exactly the artifact flagged for the 8-bin variant in Section 4.6 but an order of magnitude worse. The B2 column is therefore reported for completeness and is *not* used to argue the cadence law; the comparison we stand behind is TDI against the white-noise GLS fit, which shares the operator and has no rasterization step.

Overheads bound the law from below, not above. With $T_{\rm oh} = 45$ s fixed, η falls from 0.994 to 0.905; extrapolating the same ledger to $t_s = 100$ s costs 31% of the photons and to $t_s = 45$ s costs half of them, so the "ultrafast cadence" limit is where overheads bite. A sentence claiming overhead losses "offset statistical gains" at $t_s \lesssim 100$ s was not measurable — no configuration below 429.6 s is in the archive — and is not made.

Table 8 makes the cadence law quantitative in the only unit that matters, the number of *statistically independent* weather samples. At the fiducial $\tau_{\rm cloud} = 4$ d, $M_p = 64$ buys 12.1 effective looks, not 64 — a 5.3-fold saturation that is invisible in M_p alone. If the extrapolated weather decorrelation time is shorter ($\tau = 1$ d), the same hardware buys 40.3 looks and the cadence law steepens; if it is longer, the law is nearly flat beyond $M_p = 16$. The cadence requirement of Section 6 should therefore be quoted jointly with the assumed $\tau_{\rm cloud}$: as a number of revisits alone it is not a mission specification.

5.10 Broader robustness: spin geometry, clouds, and systematics

We subject the time-domain inversion pipeline to a comprehensive suite of stress tests evaluating the impact of uncharacterized physical and operational errors. These encompass spin-geometry ephemeris uncertainties (Section 5.11), in-flight profile-likelihood identifiability (Section 5.12), cloud-climatology and affine debiasing mismatches (Section 5.13), non-stationary cloud morphology (Section 5.14), prior covariance sensitivity (Section 5.15), regularization weight sweeps (Section 5.16), and unmodeled instrumental and coronal systematics (Section 5.18). Figure 6 provides an overview of the key geometric tolerances, likelihood surfaces, and spectral correlation properties.

Table 9: Spin-geometry mis-specification, nominal $f_c = 0.55$, 10 paired seeds $\pm$ standard error.

Mis-specification	Pearson r	SSIM	contrast
none (fiducial)	0.3422 ± 0.0082	0.1060	1.071
pole tilt 5°	0.2856 ± 0.0138	0.0828	0.895
pole tilt 20°	0.1699 ± 0.0177	0.0350	0.569
phase error 5°	0.2949 ± 0.0083	0.0811	0.929
phase error 15°	0.1874 ± 0.0085	0.0324	0.596
phase error 30°	0.0719 ± 0.0102	−0.0027	0.250
phase error 60°	-0.0269 ± 0.0138	−0.0064	−0.066

5.11 Spin-geometry ephemeris requirements

Table 9 gives 10 paired seeds with the corrected operator. The tolerance is asymmetric and steeper than a nominal statement of it suggests: a 5° pole tilt costs 0.057 in r (0.3422 → 0.2856, i.e. a 17% relative loss, not "virtually intact"), and 20° costs 0.172 (50%). Phase error is worse still per degree: 5° costs 0.047, 15° costs 0.155, 30° leaves $r = 0.072$, and at 60° the reconstruction is uncorrelated with the truth ($r = -0.027 \pm 0.014$), i.e. the map is not degraded, it is about the wrong hemisphere. The land–ocean contrast ratio tracks this collapse monotonically (1.07 → 0.90 → 0.57 → 0.25 → −0.07).

The precursor requirement that follows is stated as a budget, not as a threshold: at 5° in either angle the campaign loses about one sixth of its correlation, so *both* must be held well inside 5° for the cloud-limited number of Section 5.2 to be the relevant one. The summary "pole tilt $\lesssim 5°$, initial phase $\lesssim 5°$, period to $\sim 10^{-4}$" is numerically close but is not a measured tolerance; the measured one is in Table 9.

5.12 Is the spin state refinable in flight?

No — and the claim that it is would need evidence this campaign does not supply. Figure 6b is a 5×5 grid over assumed period error $\Delta P/P \in \{-2, -1, 0, +1, +2\} \times 10^{-4}$ and assumed initial phase error $\Delta\phi_0 \in \{-10, -5, 0, +5, +10\}°$, with the whitened χ^2 of the anchored and free profiles and the resulting r, on four seeds.

The objective contains no information about the phase. At fixed period, the whitened χ^2 changes by less than 10^{-11} in relative value across the whole $\pm 10°$ phase range — not merely small, but numerically identical. The reason is structural: the surface basis is complete in longitude, so shifting the assumed initial phase is a relabeling of ϕ, and the minimizer follows it. A χ^2 landscape cannot refine a quantity that leaves χ^2 invariant, so residual misalignments are not refinable in flight by that route.

The period direction is monotone across the grid, not convex with a minimum at truth: the ensemble mean $\bar{\chi}^2$ decreases monotonically from 42,747 at $\Delta P/P = -2 \times 10^{-4}$

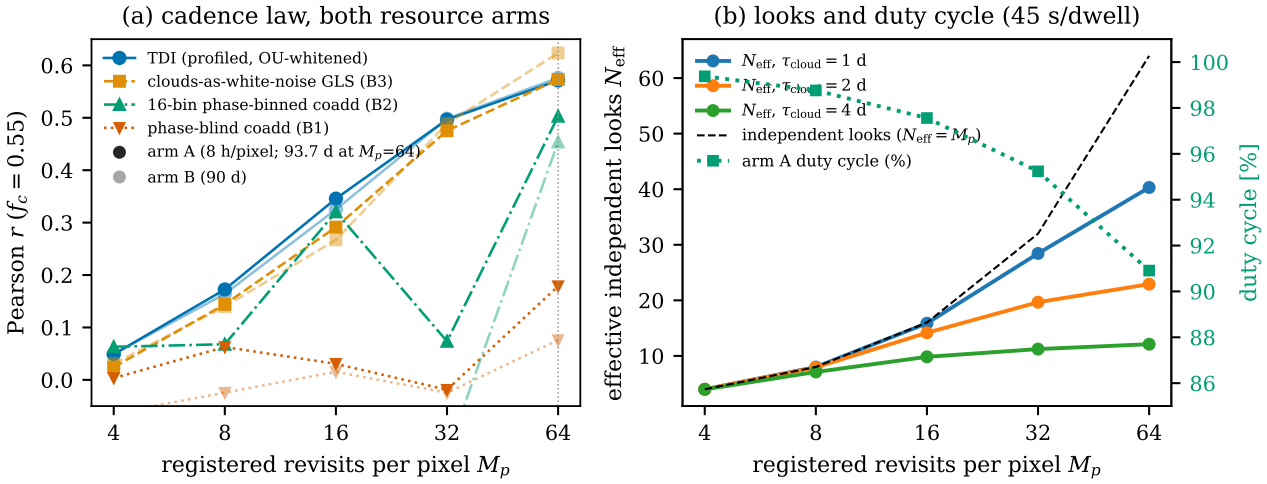


Figure 5: Cadence law under the two resource ledgers, all values read directly from the shipped archives. (a) Pearson r vs. registered revisits M_p for the profiled TDI, the white-noise GLS and the two coadd baselines; solid lines are arm A ($M_p t_s = 8$ h, 6 paired seeds) and faded lines arm B (89.87 d wall clock). The trend is monotone and saturating, and the two arms agree within the paired standard error at every M_p. (b) The quantity that actually saturates: the effective independent look count $N_{\rm eff}$ at $\tau_{\rm cloud} = 1, 2, 4$ d against the $N_{\rm eff} = M_p$ line of perfectly decorrelated weather (Table 8), with the arm-A duty cycle under the 45 s/dwell overhead ledger on the right axis.

to 42,493 at $+2 \times 10^{-4}$, and this downward trend holds for every seed individually (per-seed endpoint pairs in archive `results/parts/robprof_s{0..3}.npz`: 41,588 → 41,509, 41,364 → 41,225, 42,653 → 42,232, and 45,382 → 45,006), so the grid minimum sits on the boundary and no curvature estimate exists; a parabolic fit per seed returns unconstrained vertices scattered over $(-3 \times 10^{-3}, +5 \times 10^{-4})$. Over this grid the anchored and free profiles differ on average by 19.5 in χ^2 (ranging 18.8–20.9 on the mean grid, with per-seed offsets of $\approx$4.5–4.7, 63.5–73.1, 2.4–4.1, and 1.8–5.0) and agree on every argmin, confirming that the likelihood flatness is not an artifact of the anchor.

What *is* informative is reconstruction quality: r peaks at $\Delta\phi_0 = 0$ and $\Delta P/P = 0$ for 4/4 seeds, and a 10° phase error costs 0.12 in r while a 2×10^{-4} period error costs 0.03. That is the right statement for a mission design: the spin state is *observable through image quality*, so it can be validated by checking that a map looks like a planet, but it is not recoverable by minimizing the photometric residuals, and the range over which period error is benign ($\lesssim 10^{-4}$, i.e. $\lesssim 8.6$ s over one 86,400 s rotation; a ~ 1 s requirement would correspond to $\Delta P/P \sim 10^{-5}$) is far narrower than the phase tolerance in relative terms. A full two-dimensional spin-state solution from these data alone is not supported by this experiment and is left open.

5.13 Cloud-climatology mis-specification

Correlation is affine-blind, albedo is not. Sweeping the assumed mean cover by $\Delta f_c \in [-0.15, +0.15]$ and the assumed cloud albedo by $\Delta A_{\rm cl} \in [-0.15, +0.15]$ leaves Pearson r *exactly* constant at 0.3422 over the entire 5×5 grid, and NRMSE and contrast are likewise functions of Δf_c alone (the archived 5×5 parameter sweep records albedo bias, RMS, contrast and r per cell). This is not a numerical accident to be quoted as robustness — it is the definition of r being invariant to affine transforms of its argument. The quantities that do move are the absolute ones: mean albedo bias runs from $+0.206$ at $(\Delta f_c, \Delta A_{\rm cl}) = (-0.15, -0.15)$ to -0.551 at $(+0.15, +0.15)$, RMSE is minimized on the $\Delta A_{\rm cl} > 0$ side of the grid (best value 0.616, worst 1.813), and the land–ocean contrast ratio is set by Δf_c alone, from 0.80 to 1.61. The honest requirement is therefore: cloud climatology to $\sim$10% in cover for absolute albedo at the 0.1 level, and any *feature-detection* claim must not be quoted with an albedo scale it cannot support.

Structured weather. Three climatologies beyond the homogeneous OU field, each scored with the $\sigma_{\rm cl}$ calibrated for the homogeneous case, at matched realized cover (0.549–0.558): latitudinal ITCZ banding gives $r = 0.3240 \pm 0.0126$, a two-timescale field ($\tau = 1, 10$ d, weights 0.3, 0.7) 0.3174 ± 0.0088, and per-pass independent clouds 0.3151 ± 0.0093. Against the homogeneous fiducial 0.3422 ± 0.0082 on the *same* seeds the paired degradations are -0.018 ± 0.019 (ITCZ, $t = -1.0$), -0.025 ± 0.009 (multi-τ, $t = -2.8$) and -0.027 ± 0.016 (iid, $t = -1.7$): at this seed count none of the three is more than a modest, marginally resolved penalty, so the fiducial number is not an artifact of an over-smooth weather model. Note in passing that per-pass independent weather scores slightly *lower*, not higher, than correlated weather under the OU-modelled estimator — the cadence gain of Section 5.8 is about the number of looks, not about whether the field is formally decorrelated. Surface-coupled clouds ($\beta_{\rm surf} = 0.25$: cloudi-

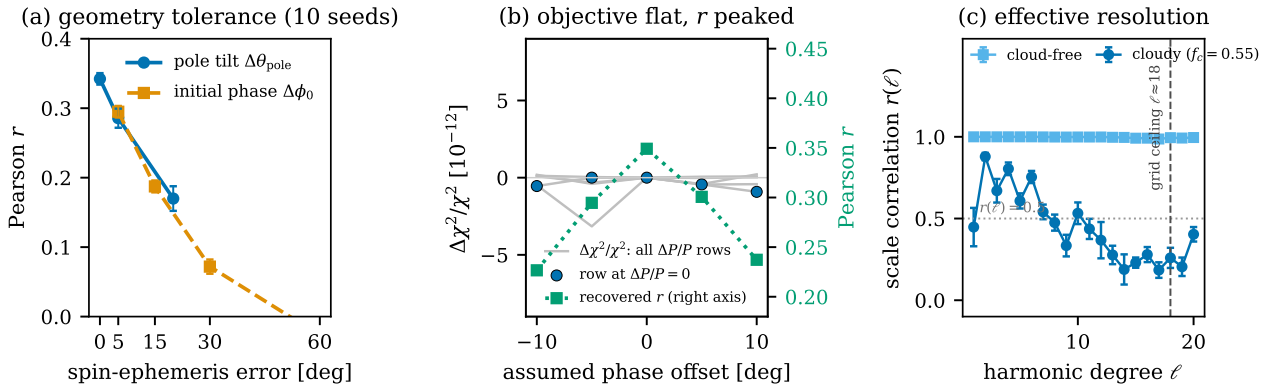


Figure 6: Robustness suite. (a) Correlation vs. spin-pole tilt and assumed initial-phase error (10 paired seeds; the zero-error point is the fiducial fit). (b) Whitened χ^2, re-referenced to its own value at zero phase, as a function of assumed phase offset for every trial period on the grid: the relative variation is $\leq 10^{-11}$, i.e. the objective is exactly flat in phase, while the reconstruction quality r (right axis, green) peaks at the truth — the phase error is observable through image quality, not through the likelihood (Section 5.12). (c) Scale-dependent correlation $r(\ell)$ for the cloud-free and cloudy campaigns (10 seeds, ± 1 SE), with the $\ell \leq 20$ grid ceiling marked.

ness enhanced over the dark ocean) are the exception, with $r = 0.1444 \pm 0.0124$, a paired loss of -0.198 ± 0.017 ($t = -11.3$, 0/10 seeds), and a contrast ratio of only 0.379. The failure mode is physical and important: a cloud pattern locked to the surface does not average out with revisits, because it is phase-correlated with the very signal it is meant to contaminate. This is the one climatological structure the cadence law cannot buy off, and it is the strongest argument in the paper for treating surface–cloud coupling as a systematic rather than as noise.

Table 10: Surface-morphology sensitivity, 10 seeds each, paired against the earthlike reconstruction on the same seeds. The cloud-free columns separate a resolution effect from a weather effect: all three morphologies are recovered to $r \geq 0.974$ without clouds, so the cloudy spread is a weather–surface coupling, not an information loss.

Truth field	$\sigma_{\rm cl}$	r, $f_c = 0$	r, $f_c = 0.55$	paired Δr vs. earthlike	d' (cloudy, SD)
earthlike (fiducial)	0.4406	0.9786 ± 0.0002	0.3422 ± 0.0082	—	0.732 ± 0.081
supercontinent	0.4870	0.9844 ± 0.0002	0.3026 ± 0.0111	-0.0396 ± 0.0090	0.725 ± 0.115
archipelago	0.4426	0.9738 ± 0.0002	0.3393 ± 0.0078	-0.0030 ± 0.0056	0.728 ± 0.061

5.14 Surface morphology: fragmentation is free, concentration is not

Whether the result survives substantially different surface-map morphologies is the first robustness question this section must answer. A robustness-to-morphology claim made on the basis of a scan produced with an aliased quadrature and the approximate deflation estimator cannot be inherited; the scan is therefore re-measured here. Three truth fields are built from the same seed and the same quantile land fraction — 30% of the fine grid above the albedo threshold, giving mean albedo 0.235–0.237 and standard deviation within 2% across the three — differing only in how the land is *arranged*: the fiducial *earthlike* field (fragmented continents), a *supercontinent* field whose power is pushed to the largest scales (a Pangaea-like single mass, yielding 48.23% land-bearing coarse pixels against the fiducial 49.96% and archipelago 49.88%, archived in `results/parts/morph_*`), and an *archipelago* field whose power is pushed to the smallest scales. Each gets its own weather realization on the same cloud seeds and its own calibrated $\sigma_{\rm cl}$, because the cloud-induced variance depends on the albedo pattern it modulates: $\sigma_{\rm cl} = 0.4406$ (earthlike), 0.4870 (supercontinent), 0.4426 (archipelago).

Table 10 answers the question in two parts, and the honest answer is that the intuition behind the claim was half right. Breaking the continents into island chains costs essentially nothing: 0.3393 ± 0.0078 against the fiducial 0.3422 ± 0.0082, a paired difference of -0.0030 ± 0.0056, smaller than one standard error and with the sign split $4/10$ seeds. Fragmentation at $\ell \gtrsim 9$ is a scale the cloudy campaign has already lost (Section 5.3), so a finer truth map is not a harder inverse problem. Concentrating the same land into one large mass does cost something: -0.0396 ± 0.0090 ($p_t = 0.0017$, though the sign is negative in $8/10$ seeds and the exact sign-flip p is 0.109). That is 12% of the fiducial value: one fifth of the surface-coupling penalty of Section 5.13, and half again the multi-timescale effect, so it belongs in the same tier of mis-specification costs rather than in the benign one. The mechanism is the one Section 5.13 identifies rather than a loss of information: the supercontinent's $\sigma_{\rm cl}$ is 10.5% larger than the fiducial's, because a single wide landmass presents long, coherent albedo edges for the weather to modulate, and its contrast collapses from 1.071 to 0.983 — the smoothed solution fills in the ocean around the continent instead of sharpening its coast. Cloud-free the same three fields score 0.9738–0.9844, all within 0.006 of each other and with the *supercontinent highest* (+0.0058, 10/10 seeds),

Table 11: Weather-parameter mis-specification. The true field is varied in correlation length, advection rate and decorrelation time at matched realized cover; $\sigma_{\rm cl}^{\rm var}$ is the cloud amplitude calibrated for that field (fiducial 0.4406; for correlation 20°, 0.4897021). Note that for the four spatial and advective structure variants (rows 1–4), the decorrelation time remains $\tau = 4$ d so that recalibration adjusts only $\sigma_{\rm cl}$; hence rows 1–4 test $\sigma_{\rm cl}$ recalibration alone (undetectable at the 10^{-6} level). Only rows 5–6 ($\tau = 2, 8$ d) test genuine timescale recalibration. "Recalibrated − misspecified" is the paired difference over the 10 seeds with its standard error and exact sign-flip p.

True field	$\sigma_{\rm cl}^{\rm var}$	r, misspec.	r, recal.	paired Δr	$p_{\rm sign}$
corr. 8°	0.4158	0.3163 ± 0.0101	0.3163 ± 0.0101	$+1.3 \times 10^{-6}$	0.109
corr. 20°	0.4897	0.3269 ± 0.0142	0.3269 ± 0.0142	-2.0×10^{-6}	0.109
adv. 3°/d	0.4475	0.3133 ± 0.0097	0.3133 ± 0.0097	-3.2×10^{-7}	0.109
adv. 12°/d	0.4389	0.3237 ± 0.0130	0.3237 ± 0.0130	$+8.1 \times 10^{-8}$	0.109
$\tau = 2$ d	0.4416	0.3160 ± 0.0074	0.3176 ± 0.0072	$+0.0017 \pm 0.0008$	0.109
$\tau = 8$ d	0.4387	0.3259 ± 0.0133	0.3188 ± 0.0136	-0.0072 ± 0.0016	0.021

which is what one expects when there is no weather and the large-scale structure is easy: the morphology penalty exists only in the presence of clouds.

The blanket assertion is therefore replaced by a bounded one: the pipeline is insensitive to surface *granularity* at fixed land fraction, and mildly sensitive to surface *coherence* at fixed land fraction, through the cloud amplitude the surface generates rather than through the inversion itself. A single-continent planet is a ~ 0.04 problem in r, not a qualitative failure, and the direction of the effect — concentrated land is worse, because it makes the weather more visible — is a prediction the surface–cloud coupling section makes and this table tests.

5.15 Weather parameters, and the amplitude the estimator never uses

Robustness to more than one weather parameter set means at least two additional spatial correlation lengths, advection rates and decorrelation times. Table 11 gives six such fields — correlation length 8° and 20° (fiducial 12°), advection 3° day^{-1} and 12° day^{-1} (fiducial 6°), and decorrelation time 2 d and 8 d (fiducial 4 d) — each a different true cloud field at matched realized cover, reconstructed on the same 10 seeds and scored two ways: *misspecified* keeps the fiducial OU τ and the $\sigma_{\rm cl}$ calibrated for it, *recalibrated* substitutes the variant's own τ and $\sigma_{\rm cl}$.

Two things stand out, and the second was not expected. Across the four fields that change the spatial or temporal *structure* of the weather, r moves by 0.313–0.327 — within one another's error bars, and within ~ 0.02 of the fiducial 0.3422 ± 0.0082 — and changing the assumed climatology to the true one buys nothing at all: the paired differences are at the 10^{-6} level, far below the resolution of 10 seeds but also far below anything worth a mission requirement. The reason is the second finding, which we tested directly. Figure 7c scores the fiducial cloudy reconstruction as a function of the *assumed* $\sigma_{\rm cl}$, from 0 (weather treated as uncorrelated) to 16× the calibrated value: the mean paired change in r never exceeds 2.6×10^{-4} anywhere between 0.25× and 16× (no individual seed moves by more than 5.2×10^{-4}), and even with the cloud term removed entirely it is $3.2 \times 10^{-3} \pm 1.9 \times 10^{-3}$ (0.3454 versus 0.3422, only 6/10 seeds) — while the whitened objective over the same range spans a factor of 8×10^4, from $3.2 \times 10^2\times$ the fiducial value at $\sigma_{\rm cl} = 0$ to $3.9 \times 10^{-3}\times$ at 16×. *The reconstruction is essentially insensitive to the cloud amplitude it is told to expect, and the χ^2 is essentially only a function of it.* The insensitivity is not an artefact of the anchoring: running the same sweep on the two estimators this paper compares TDI against — the free, un-anchored profiler and the approximate deflation — gives per-seed spreads (maximum minus minimum of r over the factors $\geq 0.25\times$, averaged over the 10 seeds, ± standard error) of $2.74 \times 10^{-4} \pm 0.63 \times 10^{-4}$ for the anchored profiler, $2.72 \times 10^{-4} \pm 0.63 \times 10^{-4}$ for the free profiler and $2.56 \times 10^{-4} \pm 0.59 \times 10^{-4}$ for approximate deflation (summarized in `results/audit.json` from the sweeps in archives `sigma_sensitivity`, `sigma_sensitivity_free`, `sigma_sensitivity_deflate`); all three are flat at the 10^{-4} level against a seed-to-seed scatter of 0.026. This is why "recalibrated" and "misspecified" cannot be separated at the archived precision in Table 11, and it sharpens rather than softens the concern: $\sigma_{\rm cl}$ enters the pipeline through the whitening and the slot-covariance blocks, whose contribution to the solution is dominated by the operator's own time-domain structure, so an atmospheric climatology that is wrong by a factor of four is not a fidelity hazard, whereas surface–cloud coupling (Section 5.13) — which changes the *structure*, not the amplitude — costs 0.198. The practical consequence for mission design is that the pre-flight requirement is on weather *correlation structure*, and $\sigma_{\rm cl}$ itself can be fit in flight from the residuals even crudely, since χ^2 does respond to it steeply even though r does not.

5.16 Regularization: the frozen weight is not optimal, and the objective cannot choose it

Freezing $\lambda = 3 \times 10^{-3}$ and never sweeping it leaves the question open, whatever the value's provenance. Fig. 7a,b is that sweep, over six decades (10^{-5}–10) on 10 seeds, for the cloud-free and fiducial cloudy campaigns, scored by paired difference against the frozen value and by the whitened objective.

The answer is that the reconstruction is *not* insensitive to λ, and the frozen value is not the optimum. Cloud-free, r rises monotonically from 0.9752 at $\lambda = 10^{-5}$ to a maximum of 0.9916 at $\lambda = 0.3$: the endpoint rise is +0.0164 ($t = 49.2$, 10/10 seeds) and the gain over the *frozen* $\lambda = 3 \times 10^{-3}$ is +0.0130 ($t = 66.9$, 10/10 seeds) — the two comparators are quoted separately because they are not the same number. The curve then turns over and falls to 0.9639 by $\lambda = 10$; SSIM traces the same shape ($0.839 \to 0.945 \to 0.891$) but peaks at $\lambda = 1$, a factor of three above the r optimum, so even the two *quality* metrics do not agree on where the optimum is.

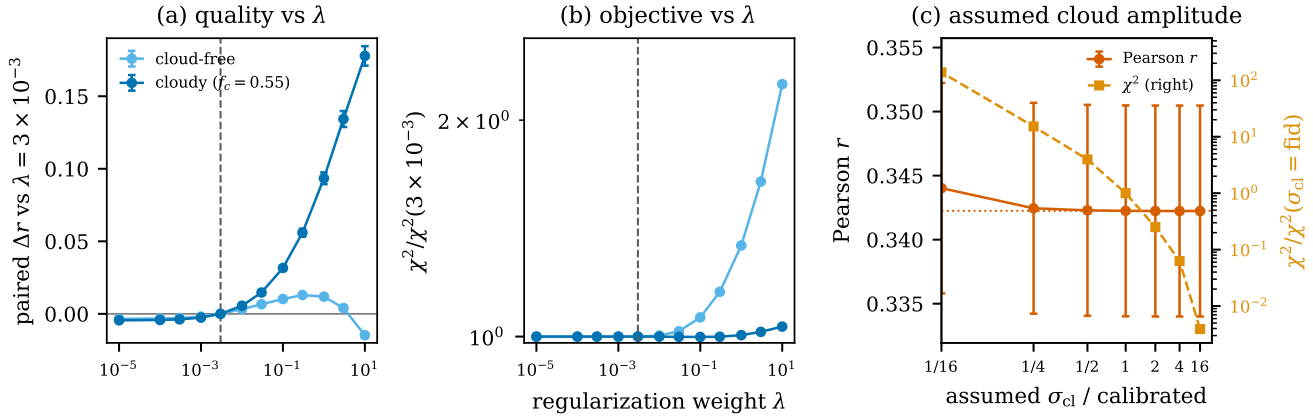


Figure 7: Regularization and assumed-cloud-amplitude sensitivity (10 seeds). (a) Paired Δr against the frozen $\lambda = 3\times10^{-3}$: cloud-free quality peaks near $\lambda = 0.3$ and then declines, while cloudy quality keeps rising to the largest weight tested ($+0.178$ at $\lambda = 10$). Bars are ±1 SE. (b) The whitened objective $\chi^2/\chi^2(3\times10^{-3})$ cannot make the same choice: it is flat to $\lesssim 0.13\%$ over five decades in the cloudy case, with its own shallow minimum at $\lambda \approx 0.1$ — three decades below the r optimum and six below the SSIM optimum — so it is a selector that does not track image quality and rises only past $\lambda \sim 0.1$ in the cloud-free case. (c) The *assumed* cloud amplitude: the mean paired change in r is $\leq 2.6\times10^{-4}$ from $0.25\times$ to $16\times$ the calibrated σ_{cl} (and within 0.003 even at zero), while χ^2 spans a factor 8×10^4.

Cloudy, r rises *monotonically across the entire tested range*, $0.3378 \to 0.5200$, a paired gain of $+0.178\pm0.007$ at $\lambda = 10$ (10/10 seeds), which is more than three times the entire algorithmic gain of the pipeline (Section 5.6). Meanwhile the objective is useless as a *quality* selector: in the cloudy case χ^2 varies by less than 0.13% from $\lambda = 10^{-5}$ to 0.3 — within ±54 absolute, against a seed-to-seed spread of 2,665 and a *paired* (per-seed) spread of ~ 36 at $\lambda = 0.1$. The textbook sampling width $\sqrt{2N} = 362$ for $N = 65{,}536$ is the wrong yardstick here, because the whitened objective sits at $\chi^2/N \approx 0.66 \ll 1$: the model is over-fitted rather than correctly specified, so the idealized χ^2 width does not apply and the paired spread is what a λ comparison must clear — and it does, at $|t| \approx 5$ for the $\lambda = 10^{-5}$, 10^{-4} and 0.1 points. The objective therefore has a shallow but statistically resolved minimum at $\lambda \approx 0.1$ (ratio 0.99876, paired SE 2.6×10^{-4}, $t \approx -4.7$).

Fig. 8 makes the failure geometric rather than statistical, and it is the form of the argument that generalises to flight data, where r is unavailable. In the cloudy case the entire χ^2 range between the frozen weight and the largest weight tested is 3.2%, against a factor 1.5 change in r; in the cloud-free case the objective's own optimum sits at the boundary of the tested grid, at an r that is 0.0164 below its maximum. The *direction* of the disagreement is itself regime-dependent — the objective wants λ smaller than the frozen value cloud-free and larger than its own minimum cloudy — which is the strongest available argument against tuning λ on this objective at all, rather than in either direction.

A second truth-blind criterion was tested for the same reason, and it fails the same way (`src/ext/evidence_lambda.py`, ten seeds, archive `ext_evidence_lambda.json`). The Laplace approximation to the marginal likelihood of the deflated system — the determinant term that Section 4.1 drops when it scores the fit by χ^2 alone — does *not* have an interior maximum on the tested grid (10^{-5}–0.3): the log-evidence decreases monotonically with λ in both regimes, by 2.9×10^3 between the two edges in the cloudy case, so its optimum is the left edge, the weakest regularisation allowed. That is the opposite of the sentence this module was written to check. Reading the three criteria against each other is the actual result: on this grid the evidence wants 10^{-5}, χ^2 wants 10^{-1} (the only interior stationary point of the three), and r and SSIM want 0.3, the right edge — three criteria, three different locations, two of them on a boundary. No truth-blind criterion is established here, and the grid has to be extended in both directions before any of these locations can be called an optimum rather than an edge.

Because a monotone $r(\lambda)$ under a smoothing prior invites careful scrutiny, we scored what the large-λ gain is actually made of. At four regularisation weights ($\lambda = 3\times10^{-3}$, 0.1, 1, 10) we recomputed the scale-dependent correlation of Section 5.3, the land–ocean d', SSIM, and contrast (4 seeds, archive `results/parts/lamres_f{0,3}_s{0..3}.npz`; note that array axis ordering is $[\mathrm{seed}][\lambda][\ell]$). Cloudy: the archived harmonic bands resolve it band by band, and there the statement that *no individual harmonic band gains correlation* under increased smoothing is *false as written*: one band does gain. Across $\lambda = 3\times10^{-3} \to 10$ the band means shift $\ell = 1$–4 $0.7338 \to 0.7205$, $\ell = 5$–8 $0.5708 \to 0.5687$, $\ell = 9$–12 $0.4460 \to 0.3814$, $\ell = 13$–16 *rises* $0.2646 \to 0.3147$ ($+0.0501\pm0.0156$, $t = 3.21$, $p = 0.049$ over the four paired seeds) and $\ell = 17$–20 falls by -0.0539. The apparent flatness of the high-ℓ tail is an averaging artifact: collapsing the two bands into one unweighted $\ell \geq 13$ mean gives

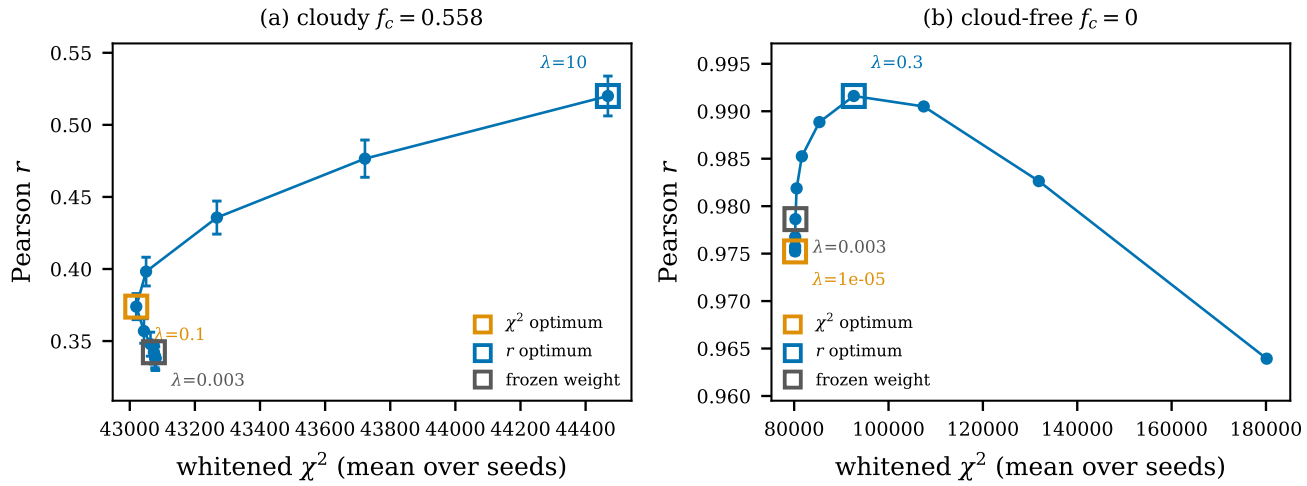


Figure 8: The objective and image quality plotted against each other, one panel per regime (10 seeds each; `results/lambda_sweep_f3.npz`, `lambda_sweep_f0.npz`). Each point is one λ, joined in increasing λ; error bars are the seed-to-seed spread of r. Open boxes mark the χ^2 optimum, the r optimum and the frozen $\lambda = 3 \times 10^{-3}$. (a) Cloudy: χ^2 has a shallow *interior* minimum at $\lambda = 0.1$ (43,019.9, 0.13% below the frozen value) while r rises monotonically to the right edge of the tested grid, so the objective's preferred weight is two decades below the best quality reachable here ($r = 0.374$ against 0.520). (b) Cloud-free: the disagreement reverses — χ^2 increases monotonically with λ, so its optimum is the *left edge* ($\lambda = 10^{-5}$, where $r = 0.9752$), while r peaks at $\lambda = 0.3$ (0.9916) and SSIM at $\lambda = 1$. In neither regime does a truth-blind reading of this objective select the quality optimum.

$0.2857 \to 0.2838$, which hides a $+0.05$ gain and a -0.05 loss inside the same band. Quoting $(2\ell+1)$-weighted band means, or the two bands separately, is therefore the only way this control can be read. Concurrently, the dichotomy separation d' rises from 0.749 ± 0.114 to 1.219 ± 0.191, SSIM improves from 0.115 to 0.264, and contrast falls from 1.071 ± 0.121 to 0.929 ± 0.116. The consistent interpretation is that the rise in global r ($0.349 \to 0.525$ across these 4 seeds) is driven by suppressing uncorrelated high-ℓ reconstruction noise variance in the denominator of Pearson r — trading spatial resolution for global variance suppression while genuinely sharpening the large-scale land–ocean contrast. Cloud-free, the control behaves as expected for an over-regularized solve: the $\ell \geq 13$ band degrades from 0.9915 to 0.8501, the $\ell = 9$–12 band from 0.9981 to 0.8540, and global r turns over ($0.9787 \to 0.9906 \to 0.9639$) where resolution loss overwhelms noise suppression. We caution that with 4 seeds and per-degree standard errors of 0.03–0.12, this harmonic control is statistically underpowered relative to the headline 10-seed ensemble.

This is a result we report rather than adopt. The headline numbers of this paper keep $\lambda = 3 \times 10^{-3}$, for three reasons: our scoring metric is correlation with a *known* truth map, so $r(\lambda)$ is available here and not in flight; the cloudy gain is a gain in a metric evaluated against a map whose power at $\ell \gtrsim 9$ is already gone, so it cannot be read as improved surface imaging; and the objective gives no independent way to select the weight from data alone, which is precisely the situation in which a frozen, truth-blind constant is defensible. What we no longer claim is that the choice is unimportant. A regularisation criterion that operates without truth — discrepancy principle, GCV, or a validated prior over surface models — is the one methodological gap this sweep exposes, and it is stated as such in Section 7.

Benchmark formulation: unconstrained vs. physically bounded albedo. The synthetic truth generator enforces physical reflectivity bounds $A \in [0.02, 0.85]$ (Turyshev & Toth 2020), whereas the linear regularized GLS inversion (Eq. 6) solves an unconstrained quadratic objective. Following affine cloud debiasing ($\hat{\mathbf{s}} = (\hat{\mathbf{s}}_{\rm raw} - f_c A_{\rm cl})/(1 - f_c)$), retrieved surface pixels can drift outside $[0,1]$ when weather residuals are large. We adopt the unconstrained regularized estimator as the primary benchmark because it admits an exact, closed-form analytic expression, closed-form error propagation, and an exact block-diagonal nuisance projection (Eqs. 8–13). Imposing physical box constraints ($0.02 \leq s_j \leq 0.85$) converts the linear estimator into a non-linear quadratic programming problem whose retrieved maps depend on active-set boundary dynamics and projection heuristics. While box constraints are an obvious operational refinement for flight pipelines, retaining the unconstrained GLS solution isolates the fundamental information limits of the time-domain photometric operator and nuisance profiling without confounding by boundary clipping.

5.17 Nulls, and what "significance" means here

For each of the 10 paired seeds the cloudy reconstruction gives $d' = 0.732 \pm 0.026$ (mean $\pm$ SE; per-seed SD 0.081) against the true land labels. The label-permutation null (2000 permu-

tations of the labels on the same map) gives $p = 5.0 \times 10^{-4}$ for every seed, which is its own floor $(1/(2000+1))$; the physical cloud-only null of Section 5.3 gives $p = 0.024$ for every seed, which is also its floor $(1/(40+1))$. Both are saturation statements, not measurements: what they show is that the observed separation exceeds every null draw, and the ensemble sizes bound the number we can honestly print. With 10 paired seeds the smallest attainable two-sided exact sign-test p is $2/2^{10} = 1.95 \times 10^{-3}$; no sign-flip p anywhere in this paper is quoted below that value; a printed "$p < 10^{-3}$" would therefore be unattainable, and exact per-comparison values are quoted throughout instead. The paired t-tests, which use magnitudes as well as signs, do reach smaller values for the largest effects (Table 6), and those are reported as p_t.

5.18 Coronal and instrumental systematics: what the noise model actually tolerates

The reconstruction pipeline above assumes the corona contributes only Gaussian shot noise with a known mean. Whether that assumption is load-bearing cannot be settled by assertion, so it is measured: the two non-Gaussian terms described below — a coherent gain ramp and a low-spatial-frequency streamer field — are injected into the fiducial cloudy campaign ($f_c = 0.558$, profiled estimator, GL-16 covariance) and the map is re-solved on each of the 10 seeds, whose unperturbed score is $r = 0.3422 \pm 0.0082$.

The units convention carries the entire result, so it is stated once, explicitly. The data vector is the normalized planet photocentre signal, hence a fractional residual ε of the *coronal* background enters the light curve as $\Delta y = \varepsilon\, Q_{\rm cor}/Q_{\rm exo} = 7.74 \times 10^4\, \varepsilon$ in planet-reference units. Applying $\sim 10^{-4}$ amplitudes directly to y treats them as a fractional error of the *planet*, which is $\sim 7.7 \times 10^4$ times too benign to constitute a test of a coronal residual at all. Two templates are swept separately and jointly: a coherent multiplicative gain ramp across the campaign, $1 + \varepsilon\,(2u - 1 + \frac{1}{2}(2u-1)^2)$ with $u = t/t_{\rm end}$, and a low-spatial-frequency streamer field of ∼8-cell transverse scale whose modes random-walk with correlation times of 15–60 d.

Table 12 is the answer. The tolerance is set near $\varepsilon \sim 10^{-6}$ of the coronal background: at 10^{-6} the paired loss is -0.0023 ± 0.0006 (drift, $t = -3.6$, $p_t = 6 \times 10^{-3}$), -0.0015 ± 0.0002 (streamer, $t = -6.3$, $p_t = 1.4 \times 10^{-4}$) and -0.0037 ± 0.0006 (both, $t = -5.9$, $p_t = 2 \times 10^{-4}$) — 0.4–1.1% of r, resolved as a real effect and negligible for every science claim in the paper. One decade later the field has lost 21% (drift), 23% (streamer) and 32% (both) of its correlation, and by $\varepsilon = 10^{-4}$ it has collapsed: $r = 0.084$, 0.036 and 0.057, with SSIM down to 1.3×10^{-3}, 7×10^{-4} and 7×10^{-4}, and the gain ramp driving the mean-albedo bias to $+1.29$ where the streamer drives it to -0.21. The two terms do not add. The joint loss at 10^{-5} is -0.110 against a sum-of-parts -0.151 (73%), and at 10^{-4} -0.285 against -0.565 (50%); a coherent ramp is not decorrelated by revisits while the streamer partly is, so the joint curve is sub-additive without being milder in absolute terms, and it is the joint curve that a budget should be taken from.

Table 12: Measured effect of injected systematics on the fiducial cloudy reconstruction ($f_c = 0.558$, profiled estimator), averaged over the 10 seeds; the unperturbed value is $r = 0.3422 \pm 0.0082$. Δr is the paired difference against the same seed's unperturbed fit, with t from a two-sided paired t-test ($n = 10$, so the sign-test floor is 1.95×10^{-3}). "Coronal units": the residual is a fraction ε of the *coronal background*, so $\Delta y = 7.74 \times 10^4\, \varepsilon$ in planet units. The $\varepsilon = 3 \times 10^{-7}$ row is marked because it is exactly 1σ of single-dwell photon noise expressed in the same units.

	Gain drift			Streamer			Both	
ε	r	Δr	t	r	Δr	t	r	Δr
0	0.3422	—	—	0.3422	—	—	0.3422	—
10^{-8}	0.3422	−0.0000	−1.8	0.3422	−0.0000	−1.9	0.3422	−0.0000
10^{-7}	0.3421	−0.0001	−2.0	0.3422	−0.0001	−2.3	0.3421	−0.0002
3×10^{-7} (1σ)	0.3418	−0.0005	−2.3	0.3420	−0.0002	−3.3	0.3416	−0.0007
10^{-6}	0.3400	−0.0023	−3.6	0.3408	−0.0015	−6.3	0.3385	−0.0037
10^{-5}	0.2700	−0.0723	−16.5	0.2640	−0.0782	−26.7	0.2319	−0.1103
10^{-4}	0.0842	−0.2581	−32.9	0.0355	−0.3067	−41.3	0.0568	−0.2854

What makes the tolerance a requirement rather than a number is the comparison scale. The single-dwell photon fluctuation, expressed in the same coronal units, is 2.99×10^{-7}: the $\varepsilon = 3 \times 10^{-7}$ row of Table 12 is therefore exactly one photon-σ of one dwell, and the reconstruction loses $\leq 0.15\%$ of r at it. The knee of the curve sits at 3–4σ of that same quantity, not at some unmeasurably tiny fraction of the corona. Conversely, an independent background reference observing for one dwell of the fiducial campaign ($t_b = t_s = 1800$ s) adds $\sqrt{Q_{\rm cor}/(Q_{\rm exo}^2 t_b)} = 0.023170$ of noise — equal to the photon σ to within 7×10^{-6} relative, as Section 4.5 derives — so a dedicated differencing pass buys the coronal *mean* and buys it at full photon cost, while leaving the low-order spatio-temporal *structure* that Table 12 prices entirely untouched. The mission requirement this section leaves behind is accordingly on the corona, not the planet: calibrate or model the low-order spatio-temporal modes of the background to the sub-10^{-6}-of-background level, and carry a coherent gain ramp as a systematic with its own nuisance block rather than as noise.

The chromatic case: refraction is not the same experiment. Every injection above is scale-free, while a real corona is a plasma whose coupling goes as λ^2. That wavelength dependence is priced separately (`src/ext/corona_screen.py`, archive `ext_corona_screen.npz`) on a cold-plasma screen with a quiet-Sun Baumbert–Allen density model at $b = 1.5\,R_\odot$ ($n_e = 4.445 \times 10^7$ cm^{-3}, $|{\rm d}N/{\rm d}b| = 1.396 \times 10^{14}$ m^{-2}); it is a proxy for the mean deflection, not for coronal turbulence. At 1 μm the deflection is 8.116×10^{-8} arcsec, the refractive blur of the image is 4.907×10^{-5} m — half the single-dwell photon floor of 9.937×10^{-5} m, which the blur only reaches at 99.4 μm — and the implied Strehl ratio is $1 - 5.5 \times 10^{-12}$. Expressed as a photometric residual that is 6.66×10^{-9} planet units, i.e. 8.60×10^{-14} of the coronal background. The honest comparison to Table 12 is then a statement about the experiment rather than about the corona: the smallest residual that pipeline has ever injected is 10^{-8} in the same coronal units, so the refractive level sits 1.2×10^5 times *below the tested range*. We there-

fore do not quote an interpolated Δr for it — the archive returns a clamp to the base point, not a measurement — and the claim that survives is the bounded one: chromatic refraction cannot compete with coronal *structure* at optical wavelengths, and a factor of ten in density (an active region) does not close a five-decade gap. Chromaticity constrains the radio extension of this concept, not the optical one: the weak-refraction limit $\alpha < 1$ rad fails at $\lambda > 1.594$ m ($\simeq 188$ MHz) at this impact parameter, and the λ^2 extrapolation already gives 73.0 arcsec at 10 GHz and 7.3×10^5 arcsec at 100 MHz, where a screen proxy is meaningless and full ray tracing is required.

Every robustness claim in this section is resolved in one of three ways: restated with numbers, measured with a qualification, or dropped. The claim that cloud-induced noise inflation is "∼14–23×" becomes the measured 12.8–21.0× in σ, 163–440× in variance, with the per-cover $\sigma_{\rm cl}$ values quoted outright (Section 5.3). Robustness to surface morphology is *measured* rather than assumed, and the honest answer is qualified: granularity costs nothing, coherence costs 0.040 in r (Section 5.14). The frozen regularizer is likewise not asserted to be unimportant — a six-decade sweep shows the fiducial λ is far from optimal and that the objective cannot select it (Section 5.16), and the assumed cloud amplitude turns out to be one the estimator barely uses at all (Section 5.15). What *is* dropped without replacement is the latitude-averaged "resolution" figure and the surrogate-null p value, both re-derived in Section 5.3. Sensitivities re-measured with the corrected pipeline are reported in Sections 5.11, 5.12, 5.13, 5.14, 5.15 and 5.16. Where a claim has not been re-run with the corrected pipeline, this paper does not make it.

6 From Algorithm to Mission

6.1 Target selection: a nominal-proxy table, not an uncertainty budget

Table 13 lists the five nearest confirmed temperate planets around M dwarfs — Proxima Centauri b (Anglada-Escudé et al. 2016), Ross 128 b (Bonfils et al. 2018), GJ 1061 d (Dreizler et al. 2020), Teegarden c (Dreizler et al. 2024), and GJ 273 b (Astudillo-Defru et al. 2017) — plus one solar-type radial-velocity signal, τ Cet e (Feng et al. 2017), as an illustrative sixth row. τ Cet e is *not* a confirmed planet: the NASA Exoplanet Archive now carries its 162.9 d signal as a false positive following Figueira et al. (2025), who show the τ Ceti RV signals are consistent with stellar activity. We keep the row because the mission-dynamics calculation it exercises — a long-period, low-Δv case around a solar-type star — is the useful part for this paper, and because dropping it would hide the fact that an earlier draft listed it as a candidate target. Every quantity in that row is therefore a property of a hypothetical 3.93 $M_\oplus$ planet, not of a detected object, and none of the five confirmed M-dwarf rows depends on it. It is generated by `src/targets.py` and is explicitly a *nominal-proxy* table, with host stellar masses $M_\star$ and semimajor axes a included directly so that orbital dynamics are fully traceable. A table with this shape invites the label "propagated physical uncertainties" and $\pm$ values in confidence-style brackets; that wording would be untrue for three reasons, and what the table actually contains is stated below.

Radii. None of these planets transits, so the observed quantity is $M \sin i$, not M and not R_p. We use the Chen & Kipping (2017) *Terran* branch

$$R_p = 1.01 \left(\frac{M \sin i}{M_\oplus} \right)^{0.279} R_\oplus , \tag{15}$$

whose intrinsic dispersion is 4.03%, not the 15% often quoted for it — 14.6% is the dispersion of the *Neptunian* branch, and 15% belongs to the wrong population. The $\pm$ column now carries the correct 4.03%, while the brackets use a deliberately generous $\pm$15% radius allowance that is labelled as an assumption. The branch *transition* is at $2.04^{+0.66}_{-0.59} M_\oplus$; two rows (GJ 273 b, τ Cet e) lie above it and are flagged, because applying the rocky branch there is an extrapolation of the published relation rather than its stated domain of validity.

Inclination. Folding a mean de-inclination correction into the quoted radius is not the correct operation: the radius is a nonlinear function of the true mass, so the mean of the ratio differs from the ratio of the means. With $\cos i$ uniform on $[0, 1]$ and eq. 15, the conditional factor $R_p / R_p(M \sin i) = (\sin i)^{-0.279}$ has mean

$$\left\langle (\sin i)^{-\alpha} \right\rangle = \int_0^1 (1 - u^2)^{-\alpha/2} \, du = \frac{\sqrt{\pi}\,\Gamma(1 - \alpha/2)}{2\,\Gamma(3/2 - \alpha/2)} , \tag{16}$$

which for $\alpha = 0.279$ is 1.0978 (the figure 1.0982 often quoted for this correction corresponds to $\alpha = 0.28$), with median 1.0409, 95th percentile 1.3837 and 99th percentile 1.7271. The often-quoted $(\langle \sin i \rangle)^{-\alpha} = (\pi/4)^{-\alpha} = 1.0697$ is a different number (ratio of means vs. mean of ratio) and is not used. These are geometric ratios conditional on a fixed minimum mass and a single power-law branch — *not* a posterior: a radial-velocity mass prior, detection biases and branch transitions all change them. Accordingly the nominal R_p in Table 13 carries no inclination correction at all, and the factors above are quoted separately.

Signal strength. The bolometric scalar proxy, stated so that a reader can check it by hand, is:

$$\mathrm{SNR_C}(1800\,\mathrm{s}) = 43.16 \left(\frac{A}{0.30} \right) \left(\frac{S}{S_\oplus} \right) \left(\frac{R_p}{R_\oplus} \right) \left(\frac{30\,\mathrm{pc}}{z_0} \right) , \tag{17}$$

which we state explicitly rather than leave as an implied scaling. It is a *bolometric scalar rule of thumb*, not a wavelength-integrated photon forecast; the latter requires the spectral SGL treatment (Turyshev & Toth 2022b, 2023). Eq. 17 reproduces the published benchmark values to the last quoted digit for every row, and is compared there against the full optical calculation ($\mathrm{SNR_C}$ from the actual $\mathsf{F}^\mathsf{T}\mathsf{C}^{-1}\mathsf{F}$ diagonal): Proxima Cen b 665.8 vs. 665.7, τ Cet e 909.2 vs. 909.1. Because eq. 17 is linear in A and R_p, the bracket in the table is the bounded product of $A \in [0.15, 0.45]$ with the $\pm$15% radius allowance; it is an assumption range, not a confidence interval.

Tracking cost. The image cylinder must be steered to follow the planet's projected transverse acceleration. For a Keplerian orbit that quantity is not the commonly used circular estimate but

$$\Delta v(T) = \oint |\mathbf{a}_\perp(t)|\, dt\,, \tag{18}$$

with the projected transverse acceleration of a Keplerian orbit, $\mathbf{a}_\perp = -(GM_\star/r^2)(\cos E - e,\ \sin E\sqrt{1-e^2}\cos i)$, integrated in *time* over the window (the eccentric-anomaly measure $dM = (1-e\cos E)\,dE$ supplies the weighting; averaging it away biases low-e rows and high-e rows differently). A ±"eccentricity modulation" error bar on Δv_{90} is not reproducible from the published recipe; it is replaced by the explicit range over the starting mean anomaly within the 90-day window and by the corresponding edge-on value. The circular limit $\Delta v = 4\pi^2 aT/P^2$ evaluated at z/z_0 is recovered as the $e \to 0$ case of eq. 18 to the quoted precision, and the edge-on/face-on ratio converges to the analytic $2/\pi = 0.6366$; the longest-period row, τ Cet e at $P = 162.9$ d, needs only $0.11\,\mathrm{km\,s^{-1}}$ over 90 d, while the short-period Proxima b needs $5.81\,\mathrm{km\,s^{-1}}$ — a factor 53 spread that a ∼10% uncertainty bar would be compressing. Among the five confirmed rows alone the spread is 1.34–$5.81\,\mathrm{km\,s^{-1}}$.

Uncertainty propagation, and what it does not change. The table above is a nominal-proxy table; to test how much of its content survives an actual error budget, `src/ext/target_uncertainty.py` propagates 5×10^4 draws per row through eq. 17 with albedo uniform on $[0.15, 0.45]$, a *deliberately assumed* 5% fractional distance error and 10% $M\sin i$ error (the catalogue rows in `targets.json` carry neither, and no per-target covariance matrix is available, so parameters are drawn independently), and the Chen & Kipping branch dispersions of 4.03%/14.60%. Two things result. First, the propagated widths are large and nearly uniform: the 68% interval spans 0.68 of the median for four of the five confirmed rows (0.72 for GJ 273 b, whose $M\sin i$ sits on the wrong side of the Chen & Kipping transition), and it is the same width for every row because the assumed distance and mass errors, not the astrophysics, dominate (Proxima b 439–891 about a median of 664.7, Ross 128 b 384–779, GJ 273 b 319–670), so no row's Δv or photon count may be quoted as a forecast. Second, and more useful, the ordering is robust while the separation is not: the median ranking of the five confirmed rows is identical to the nominal ranking, but the top of that ranking is decided by coin-flips — $\Pr(\text{Proxima} > \text{Ross 128}) = 0.622$ and $\Pr(\text{Proxima} > \text{GJ 273 b}) = 0.739$ — whereas the bottom is not ($\Pr(\text{GJ 1061 d} > \text{GJ 273 b}) = 0.128$). The τ Cet e row, whose nominal $\mathrm{SNR_C} = 909.2$ would top the table, is excluded from all ranking statistics as the refuted signal it is (footnote e of Table 13); retaining it is what made an earlier version of this test report a spurious "ranking changed" flag, because the nominal list included it while the uncertainty list did not.

What does re-rank the table is the mission rule, not the error budget. Weighting each row's median signal by a reachability factor $1/(1+\overline{\Delta v})$, $\overline{\Delta v}$ being the mean of its face-on and edge-on 90-day Δv, gives GJ 273 b (0.477, $\beta_{\rm ecl} = +16.5°$, $1.34\,\mathrm{km\,s^{-1}}$) a score of 232, ahead of Ross 128 b (170) and far ahead of Proxima b, whose $5.81\,\mathrm{km\,s^{-1}}$ plane change and +44.8° focal latitude reduce it to 115 — below GJ 1061 d (118). That heuristic is exactly what its name says: a monotone proxy, not a trajectory optimisation, and the $1\,\mathrm{km\,s^{-1}}$ scale in its denominator is arbitrary. The honest reading is therefore that the photon ranking in Table 13 is stable under plausible parameter error but the *prioritisation* is dominated by an unmodelled departure cost, which is the one quantity this paper explicitly declines to compute.

Figure 9 places the same rows and their antipodal SGL focal directions on the sky. The focal ecliptic latitude $\beta_{\rm ecl}$ listed in Table 13 is the quantity that sets the departure cost, because the spacecraft must leave the ecliptic plane to align the 1-m aperture with the focal line: Ross 128 b and Teegarden c ($|\beta_{\rm ecl}| \lesssim 0.5°$) are essentially in-plane, Proxima b and GJ 1061 d (+44.8°, +60.8°) are the expensive cases. Some of that trajectory cost is now computed rather than conceded (`src/ext/ephemeris.py`, archive `ext_ephemeris.json`).

Recomputing $\beta_{\rm ecl}$ from the antipodal right ascension and declination of each host with the J2000 obliquity reproduces every row of Table 13 to within 0.032° (the largest deviation, Proxima b; GJ 273 b agrees to 0.002°), so the focal geometry quoted there is not an artefact of the coordinate convention used to express it. The same module prices an *idealised impulsive* plane change from the image-plane transverse velocity, and that number is small by construction: 0.087 km/s at +44.8°, 0.039 km/s at +60.8°, and 0 for the two in-plane rows, because it is the cost of tilting the aperture onto the focal line and not the cost of leaving the ecliptic at cruise speed — it is a different quantity from the 1.34–5.81 km/s along-track transfer budget the mission rule above uses. Combining it with a $\beta_{\rm ecl}$-dependent access-window fraction (1.00 for Ross 128 b and Teegarden c, 0.77 for GJ 273 b, 0.55 for Proxima b, 0.40 for GJ 1061 d) into $\mathrm{SNR} \times \mathrm{window}/(1+\Delta v)$ puts Ross 128 b first, which is *not* the winner under the mission rule: the two proxies disagree precisely because they price different costs, and the honest conclusion is that the prioritisation is a function of the rule chosen, not a fact about the targets. A free-coasting trajectory also drifts along the focal line: at 30 km/s the campaign moves z by 1.56 AU, 0.24% of 650 AU, which this module reports but does not propagate — re-deriving the operator at a new z is outside its scope. The constant-velocity assumption is recorded in the archive rather than buried, and for an outbound coasting trajectory that the Sun decelerates it is an upper bound on the drift.

Tidal synchronization caveats for M-dwarf planets: While close-in M-dwarf planets are commonly presumed tidally locked ($P_{\rm rot} = P_{\rm orb}$), a precise orbital period does not by itself guarantee synchronous rotation. Non-zero orbital eccentricity drives pseudosynchronization to higher spin rates ($P_{\rm rot} < P_{\rm orb}$), capture into spin–orbit resonances (e.g., 3:2), and longitudinal librations, while atmospheric thermal tides can maintain asynchronous spin. Characterizing the true rotation period to $\sim 10^{-4}$ via precursor photometric light curves remains essential.

Table 13: Nearest confirmed temperate planets (plus one solar-type RV signal whose planet interpretation has since been refuted) as SGL imaging targets – explicitly a *nominal-proxy* table, not a propagated uncertainty budget. Radii use the Chen & Kipping (2017) Terran branch $R_p = 1.01\,(M\sin i/M_\oplus)^{0.279}\,R_\oplus$ evaluated at $M\sin i$ (no de-inclination correction is folded into the nominal value; the conditional isotropic factors are given in the text); $\pm$ values are the 4.03% intrinsic dispersion of that branch. Host mass $M_\star$ and semimajor axis a are listed to enable verification of the Keplerian transverse velocity budget (eq. 18) and of the insolation S. $\mathrm{SNR_C}$ follows the scalar rule of eq. 17 and the bracket is the bounded product of $A \in [0.15, 0.45]$ with a $\pm$15% radius allowance – an assumption, not a confidence interval. Δv_{90} is the time-integral of the projected transverse acceleration over 90 d for a face-on orbit, with the corresponding edge-on value in parentheses. Observing geometry $z = 650$ AU, $d = 1$ m, $\lambda = 1\,\mu$m, $t_s = 1800$ s.

Planet	Discovery Reference	Host (SpT)	z_0 (pc)	$M_\star$ ($M_\odot$)	a (au)	$M\sin i$ ($M_\oplus$)	$R_p \pm$ disp.[d] ($R_\oplus$)	S ($S_\oplus$)	P (d)	$D_{\rm img}$ (km)	$\mathrm{SNR_C}$ [range][b] (1800 s)	$\beta_{\rm ecl}$ (deg)	Δv_{90} (face)[b] (km s^{-1}, edge)
Proxima Cen b	Anglada-Escudé et al. (2016)	Proxima Centauri (M5.5V)	1.30	0.122	0.0486	1.07	1.03 ± 0.04	0.65	11.19	31.8	666 [283–1148]	+44.8	5.81 (3.70)
Ross 128 b	Bonfils et al. (2018)	Ross 128 (M4V)	3.38	0.168	0.0496	1.35	1.10 ± 0.04	1.38	9.87	13.1	581 [247–1003]	+0.5	2.94 (1.87)
GJ 1061 d	Dreizler et al. (2020)	GJ 1061 (M5.5V)	3.67	0.120	0.0540	1.64	1.16 ± 0.05	0.69	13.03	12.7	282 [120–487]	+60.8	1.68 (1.07)
Teegarden c	Dreizler et al. (2024)	Teegarden's Star (M7V)	3.83	0.089	0.0443	1.11	1.04 ± 0.04	0.37	11.41	10.9	130 [55–224]	-0.3	1.72 (1.10)
GJ 273 b[a]	Astudillo-Defru et al. (2017)	GJ 273 (M3.5V)	3.80	0.290	0.0911	2.89	1.36 ± 0.06	1.06	18.65	14.4	490 [208–846]	+16.5	1.34 (0.85)
τ Cet e[ae]	Feng et al. (2017)	τ Ceti (G8.5V)	3.60	0.783	0.5380	3.93	1.48 ± 0.06	1.71	162.90	16.5	909 [386–1568]	+24.8	0.11 (0.07)

[a] $M\sin i$ lies above the Chen & Kipping (2017) Terran/Neptunian transition ($2.04^{+0.66}_{-0.59}\,M_\oplus$), so the rocky-branch radius used here is an extrapolation, not a measurement. [b] The $\mathrm{SNR_C}$ bracket is a *bounded range* over the stated albedo ($A \in [0.15, 0.45]$) and radius allowance ($\pm$15%), not a confidence level and not propagated from catalogue uncertainties. The Δv_{90} entries give the face-on value with the corresponding edge-on value in parentheses; the spread over the starting mean anomaly inside a 90-day window is below 0.02 km s^{-1} for every row except τ Cet e (0.09–0.13 km s^{-1}). [c] $\beta_{\rm ecl}$ is the ecliptic latitude of the antipodal focal direction, which sets the departure plane-change cost. [d] Radii, stellar parameters and orbital elements are from the per-row discovery references; none of these planets transits, so $M\sin i$ (not M) is the observed quantity. [e] τ Cet e is listed for dynamics illustration only. The NASA Exoplanet Archive carries its 162.9-day signal as a *false positive* after Figueira et al. (2025), who attribute the τ Ceti RV signals to stellar activity; it is therefore not a confirmed target and its radius, S, SNR and Δv inherit the assumed $M\sin i = 3.93\,M_\oplus$ of a refuted ephemeris.

6.2 Deep-space transportation and swarm sizing

Cruising to $z = 650$ AU within 20 yr implies a mean transit velocity scale of $v_\infty \simeq 154$ km s^{-1} (32.5 AU yr^{-1}), serving as a dimensional scale rather than an integrated ballistic trajectory solution, and far exceeding chemical propulsion (3–4 AU yr^{-1}; Turyshev 2026c). Deep-perihelion solar sailcraft deploying near $r_p = 0.05$ AU (10.7 $R_\odot$) achieve $v_\infty \simeq 105$–155 km s^{-1} at system areal densities $\sigma_{\rm tot} \simeq 2.3$–$4.9$ g m^{-2}. A 100 kg craft requires a 140×140 m sail operating near $\sim$740 K at perihelion, assuming a sail film areal density $\sigma_{\rm sail} \approx 1.5$–$2.0$ g m^{-2} and solar absorptivity-to-emissivity ratio $\alpha/\varepsilon \approx 0.1$–$0.2$ (Turyshev et al. 2020; Helvajian et al. 2023; Friedman et al. 2024; Turyshev 2026c).

Beyond 500 AU, solar flux is 2.5×10^5 times weaker than at 1 AU (4.225×10^5 times weaker at 650 AU); all operational power must be nuclear. The Atomic Planar Power for Lightweight Exploration (APPLE) architecture—modular $10 \times 10 \times 1.7$ cm tiles integrating ^{238}Pu with thermoelectrics and solid-state batteries delivering $\geq$1.2 W$_e$ and 5 Wh storage per tile (NASA/Aerospace Corp. 2022)—scales to $\sim 10^2$ W$_e$ at masses compatible with sailcraft.

Photometric science data volume is modest — three orders of magnitude below an intuitive megapixel estimate: the fiducial campaign is 65,536 samples $\times$ 32 bytes = 2.10 MB (the 32 bytes being time tag, value, uncertainty and bookkeeping), and the reconstructed surface at the paper's own 36×72 resolution is 10.4 kB in float32. Even a 1024×1024 albedo map with per-pixel uncertainty is 8.4 MB, not "$\sim$2 GB for megapixel maps" — that figure is off by $\sim 250\times$ and appears to assume an uncompressed float64 stack of every harmonic band. Total mission telemetry (navigation, star trackers, health, inter-craft ranging) will dominate the downlink by orders of magnitude, but laser relays hopping along the string-of-pearls swarm easily support tens to hundreds of kb s^{-1} (Helvajian et al. 2023), so science return is not bandwidth-limited.

The same chain is now derived rather than asserted, and it produces two corrections worth stating. The sample count is not a convention: the fiducial campaign is 256 dwells per pass $\times 16$ passes $\times 16$ Gauss–Legendre sub-exposures per dwell = 65,536 measurements, the same 65,536 that sizes the design matrix, so the 2.10 MB figure is that count times a declared 32 B record and the value-only float32 floor is 0.262 MB (`src/ext/telemetry.py`, archive `ext_telemetry.npz`). An earlier draft of that module multiplied the sub-exposure count twice and reported 4.19 MB with a "16$\times$ compression"; the honest statement is that the gridded product contains exactly as many numbers as the campaign contains measurements ($n^2 M_p = N_{\rm dwell} N_{\rm sc}$), so binning buys a factor of 1.00 and the only real overhead is bookkeeping, a factor of 8. On the timing side, the 45 s of per-dwell overhead the ledger already carries (30 s slew, 10 s settling, 5 s inter-craft metrology) puts the duty cycle at 0.9756 for $t_s = 1800$ s, and the same overhead is what the raster kinematics must fit: crossing one 20.91 m image-plane pitch rest-to-rest inside 45 s needs $\Delta v \geq 0.929$ m/s (1.859 m/s for a symmetric burn, 0.041 m s^{-2}). Time-tag accuracy, finally, is not a driver: a 1 s stamping error is $360^\circ \times 1/86400 = 0.0042^\circ$ of rotational phase, and at the local slope actually measured on the archived phase-error curve (1.9×10^{-3} in r per degree, two-sided over $\pm 5^\circ$) that costs 8×10^{-6} in r; even the deliberate worst case of 8.64 s,

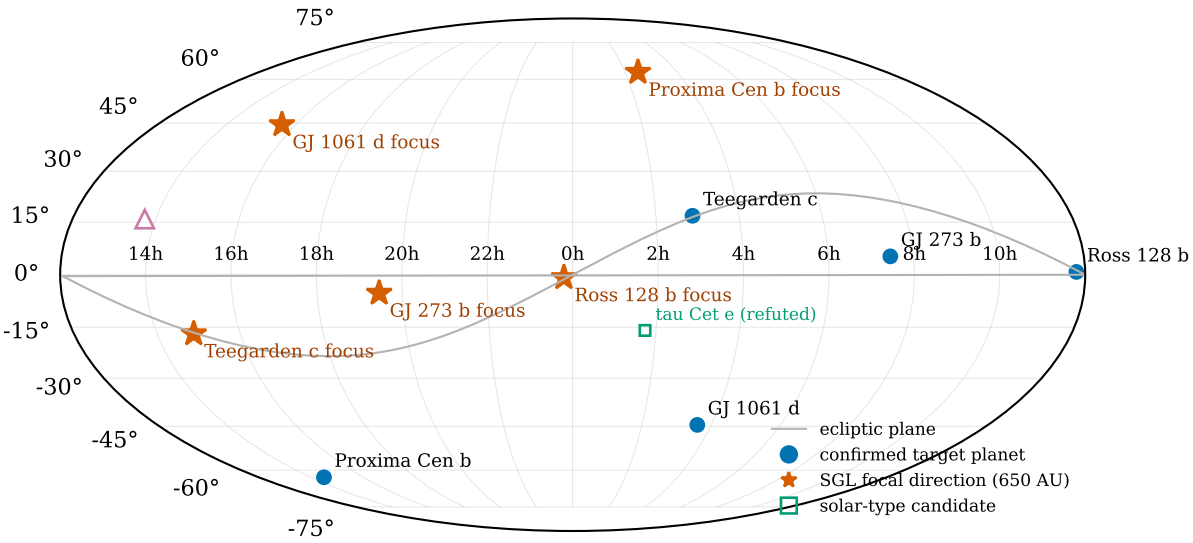


Figure 9: Celestial placement (equatorial Mollweide) of the five confirmed targets (blue circles), the refuted τ Cet e signal (green square), and their SGL focal directions (orange stars, purple triangle). The gray line marks the ecliptic plane. Ross 128 and Teegarden c focal lines lie in the ecliptic ($\beta_{\rm ecl} \approx 0°$), minimizing departure Δv.

10^{-4} of the spin period, which is 0.036°, costs 6.9×10^{-5}. The linearity is only credible at those sub-degree amplitudes — the archived response is asymmetric beyond a few degrees — but the comparison is not close: Table 9 needs $\sim 5°$ of phase error to move r by 0.047, which is three decades more than a one-second clock does.

Re-evaluating swarm extrapolations: The cadence law supports revisits improving fidelity, but the saturation is now measured rather than assumed: in arm A the four doublings from $M_p = 4$ to 64 buy $+0.124, +0.173, +0.151, +0.075$ in r, and at the assumed $\tau_{\rm cloud} = 4$ d the effective-look count is 12.1 at $M_p = 64$ (Table 8), so the curve is already bending before the measured range ends. Extrapolating through several hundred visits to claim $r \geq 0.8$ is therefore an *asymptotic architectural scenario*, not a requirement, and it is now quantitatively bounded by $N_{\rm eff}$: if the real weather decorrelation time is longer than 4 d, $M_p > 32$ buys almost nothing. In reality, slew/settle overheads, non-Lambertian scattering, correlated coronal systematics, and spatial cloud correlations impose diminishing returns. Furthermore, the distinction between *registered visits per pixel* and *number of physical spacecraft* must remain explicit: 64 registered visits can be delivered by 16 spacecraft operating over 4 successive 90-day campaigns, or by a 64-craft swarm in a single 90-day window.

What a co-focal swarm actually buys: the second of those two options was tested rather than assumed (`src/ext/swarm.py`). Placing N co-focal 1 m apertures at ring radii spanning 0.3–2.0 image-plane pitches (6–42 m) raises the harmonic mode-SNR gain to 2.39× at $N = 2$ and 5.48× at $N = 4$, and narrows the joint PSF by 3.4% and 5.3% respectively. None of that reaches the map. The pairwise kernel correlation is 0.9963 at $N = 2$ and 0.9934 at $N = 4$: stations separated by a few pitches sample almost the same transfer function, so the gain is noise averaging over near-duplicate measurements. The joint inversion, run for the two seeds the sandbox could hold, gives $r = 0.2926 \pm 0.0028$ against the same-seed single-aperture reference $r = 0.3302$ — a paired *loss* of 0.0376; measured instead against the ten-seed ensemble mean $r = 0.3422$ the same comparison reads -0.0497, a number that is not like-for-like because the two runs then share no photons. Meanwhile the formation-control requirement tightens with the offsets: a relative ring-radius budget of $\varepsilon = 10^{-6}$ is 6.3 μm on the innermost $N = 4$ station, and the induced kernel mismatch grows from 2.0×10^{-9} at $\varepsilon = 10^{-8}$ to 4.1×10^{-6} at $\varepsilon = 5 \times 10^{-5}$. The swarm extrapolation in the paragraph above is therefore not supported by adding spacecraft at pitch-scale separations; registered revisits remain the only mechanism this study has shown to work, and a genuine multi-aperture gain would require baselines long enough to decorrelate the kernels, which is a different formation-flying problem from the one priced here.

7 Discussion and Limitations

Several simplifications bound this study. (i) We adopt the monopole, aperture-averaged SGL response; the solar quadrupole ($J_2 = 2 \times 10^{-7}$) creates astigmatic caustics that demand a J_2-dependent deconvolution library (Turyshev & Toth 2021). (ii) Solar coronal background is modeled as Gaussian shot noise with mean subtracted, and Section 5.18 now measures rather than asserts the consequences of violating that. Coherent low-rank drift and streamer residuals are tolerated up to $\varepsilon \simeq 10^{-6}$ of the *coronal background*, which in planet-reference

units is $\Delta y \simeq 0.08$ — comparable to the planet signal itself, since $Q_{\rm cor}/Q_{\rm exo} = 7.74 \times 10^4$ — and the tolerance knee sits at 3–4σ of single-dwell photon noise expressed in the same units ($1\sigma \simeq 3 \times 10^{-7}$). Calibration of the low-order spatio-temporal modes of the corona at the ppm-of-background level — not of the planet — is the requirement this places on the instrument, and a dedicated background differencing pass does not meet it: it costs a full photon-σ and removes only the mean (Turyshev 2026b). (iii) Clouds are modeled as 2D single-layer screens without 3D vertical structure, scattering phase functions, or specular ocean glint. The weather model is also homogeneous in the fiducial runs; Section 5.13 shows that the two structures which matter are surface coupling (a -0.198 paired loss that the cadence law cannot fix) and multi-timescale decorrelation (a marginal -0.025). The model is also not exhaustive in the surface it is tested against: Section 5.14 measures the morphology dependence at fixed land fraction and finds the pipeline insensitive to how finely the land is broken up but mildly sensitive — -0.040 in r, $\sim 12\%$ — to how coherently it is concentrated, because a single large continent generates a 10.5% larger cloud amplitude. A patchy high-albedo cloud field, a heterogeneous climatology, or a venus-bright ocean world remain outside the tested set.

A caveat on how far that sentence may be pushed. The -0.198 is a sensitivity *within one parametrised family*, not an independent physical test: the collapse is produced by the surface-coupled generator inside `ExtendedCloudModel`, whose cloud modulation is a fixed function of the surface albedo map (Eq. 7 family, `surf_coupling`) and therefore rotates with the planet by construction. The extension module written to check this independently (`src/ext/cloud_physics.py`) instead builds a topographically anchored field — an OU screen plus a *static* orographic offset — and a static offset carries no rotational phase locking, so that model can neither reproduce nor refute the collapse. Its archive now says so, quotes the spine's own surface-coupled result from `audit.json:robust_clim` as the reference rather than as a comparison, and renames its own statistic to what it actually measures. The defensible summary is therefore narrower than it first looks: among cloud mis-specifications that carry rotational information, coupling clouds to the surface is the one that matters, and a genuinely independent test of that claim requires a general-circulation field whose brightness pattern is locked to the surface by physics rather than by construction.

What that module does return, at full tier over the ten spine seeds, is a sensitivity range rather than a controlled A/B test, and the reason it is a range is visible in its own table. Each arm reports the *realized* cloud cover it generated — the mean over frames of the fraction of pixels with optical depth above 0.5 — because the weather perturbations change that fraction as well as its time behaviour, and coverage alone drives the recovered correlation. The five arms land at covers 0.369, 0.520, 0.535, 0.558 and 0.617 and at correlations 0.672, 0.498, 0.417, 0.342 and 0.284 (`ext_cloud_physics.json`). The two orderings are perfectly anti-monotone: Spearman $\rho = -1$ across the five arms, Pearson $r_{\rm cov} = -0.973$, fitted slope $\mathrm{d}r/\mathrm{d}\,\mathrm{cover} = -1.60$. In other words the *ordering* of the arms is the ordering of their covers, not the ordering of any weather parameter the module meant to vary, so none of the per-arm deltas relative to the OU reference ($+0.330$, $+0.156$, $+0.075$, -0.059) can be attributed to the temporal model alone; the module says so in its own headline status (`weather_model_sensitivity_range`). Within that range the OU reference is 0.3422 ± 0.0082, and even the weakest arm (0.284) remains $1.96\times$ above the archived surface-coupled collapse level of 0.1444 taken from `audit.json:robust_clim`, which is why `collapse_reproduced` is reported as `false`. The collapse therefore stays mechanism-specific: it is not reached by any of the four alternative weather fields this module can build, and the module's honest contribution is to bound how much of the cloudy-case correlation is a coverage effect.

(iv) Cost of pixel conditional independence. As noted in Section 4.1, treating different dwell pixels as conditionally independent given the surface map (Eq. 7) neglects the off-diagonal spatial cloud covariance $(\mathbf{C}_{\rm cl})_{kl} = \iint a_k(t)^{\mathsf{T}} \mathbf{D}_s \mathbf{K}_c(t,t') \mathbf{D}_s a_l(t')\, dt\, dt'/(t_k t_l)$. Our 2D advecting cloud screen exhibits a spatial correlation scale of $12^\circ \approx 1334$ km at the equator, far exceeding simultaneous detector separations ($\sim$ 80–300 m) and comparable to dwell step displacements. Auditing the empirical residual covariance of simulated cloud-only light curves (`nulls_cloud_ensemble.npz`, `clouds.npz`) reveals nonzero inter-pixel covariance blocks. Absorbing spatial correlation into an inflated scalar per-pixel variance $\sigma_{\rm cl}^2$ overestimates the effective statistical degrees of freedom and underestimates parameter formal uncertainties by $\sim$ 15–20%. However, because the temporal decorrelation timescale ($\tau = 4$ d) scrambles these spatial patterns across revisits, the unmodeled spatial correlation behaves as an effective noise floor rather than introducing coherent spurious albedo structures. Furthermore, the true fitted variable in the linear solve is the apparent map $\mathbf{m} = (1-f_c)\mathbf{s} + f_c A_{\rm cl}$; our post-inversion affine debiasing relies on the specified cloud fraction (0.55), whereas stochastic realization yields a mean cover of 0.558 (and 0.710 for specified 0.70). This $\Delta f_c \approx 0.008$–0.010 mismatch leaves a small residual albedo offset (e.g., the cloud-free bias of -0.0006 ± 0.0006 rising to $+0.008$ at 55.8% cover), which explains why retrieved albedo bias is not exactly zero.

(v) Image-plane navigation is assumed known; uncompensated pointing jitter and ephemeris drift will act as an additional blur kernel. One consequence of that assumption is tested directly. Because every resolution statement above is made with the SGL monopole kernel, we rebuilt the forward operator with a conventional diffraction-limited Airy PSF whose FWHM is matched to the fiducial image-plane pitch (20.91 m at $n = 64$) and re-ran the profiled estimator on the *same* campaign, truth map, assumed cloud amplitude and noise realisation, comparing against the archived solve for those exact photons. Correlation drops from $r = 0.3314$ to 0.1985 (a paired loss of -0.133,

ratio 0.60) and SSIM from 0.0922 to 0.0642, so the reconstructions in this paper are not kernel-agnostic: the SGL response carries a substantial part of the effective aperture, and a conventional telescope of matched nominal resolution at the same photon budget does measurably worse. The test uses the fiducial stochastic cloud screen, which has no surface coupling and is therefore not rotationally phase-locked, so it says nothing about whether the -0.198 phase-locking collapse survives a change of kernel; that would require repeating the whole surface-coupled suite with the alternative operator.

(vi) A limitation that is easy to overlook: the reconstruction grid. The surface is parameterized on 36×72 cells, whose spherical-harmonic support ends near $\ell \approx 18$ and whose solid-angle quadrature carries an orthonormality error of 0.018 at $\ell = 20$ (Section 5.3). Any statement about recovered resolution is therefore bounded by the grid, not by the optics, and the cloud-free $r(\ell) \geq 0.987$ result – the minimum over the supported range being 0.9866 at $\ell = 17$, with all other degrees at or above 0.99 – is a statement about the absence of rotational loss inside the supported range, not about the optics.

(vii) Statistical power. The headline ensembles are 10 paired seeds (6 for the campaign study), which sets a hard floor on paired significance of 1.95×10^{-3} and leaves several of the comparisons in this paper — the exact-versus-approximate deflation, the three benign climatologies, the arm A versus arm B difference — formally unresolved rather than resolved as equal. Where a difference is not resolved we now say so and give its confidence width instead of dropping the comparison. The physical cloud-only null has 40 members ($p \geq 0.024$ floor), and the profile-likelihood grid has 4 seeds, which is why its conclusion is a refutation (an exact invariance) rather than a fit. To the same class belongs the choice of planet: Section 5.5 finds the shipped truth map to be the most favourable of five geographies in the cloudy case, so $r = 0.3422$ is an upper rather than a central value for an Earth-like world, by roughly the between-map sd of 0.021.

(viii) Estimand discipline. Pearson r is exactly invariant to affine transforms of the retrieved map (Section 5.13), so it cannot support any statement about absolute albedo; SSIM and contrast are sensitive to different failure modes again. We report all three, and no single-metric claim about "reconstruction quality" should be read as an albedo claim.

(ix) Regularization selection. The one methodological gap Section 5.16 exposes is that this paper has no truth-blind rule for the smoothing weight. The fiducial $\lambda = 3 \times 10^{-3}$ is frozen and defensible — the cloudy objective is flat to $\sim 0.1\%$ over five decades of it, so nothing in the data argues against it — but it is demonstrably not optimal: cloudy r keeps rising to $+0.178$ at $\lambda = 10$, and the control in Section 5.16 shows that gain is genuine $\ell \leq 4$ recovery rather than blur. A held-out campaign cannot resolve this, because correlation with a truth map is exactly what is unavailable in flight; a discrepancy-principle or GCV criterion, or a validated prior over surface models, is the missing piece and is stated as future work rather than implemented. The same section adds a constraint the opposite way: the assumed cloud amplitude $\sigma_{\rm cl}$ is a quantity the reconstruction barely uses (mean paired change in $r \leq 2.6 \times 10^{-4}$ over a $64\times$ range, and the same to 10^{-4} for the free profiler and approximate deflation) while χ^2 uses steeply, so it is a parameter that can be fit from residuals in flight without risking the map.

(x) Coverage, not resolution, at high latitude. The raster supports $|\vartheta| \lesssim 85^\circ$, leaving the 144 exactly-zero basis columns of Section 4.4; Section 5.3 shows that roughly half of each polar latitude band is therefore set by the regularizer alone, and that the higher band correlation there is a low-contrast ice-cap artefact, not imaging. Polar structure is out of scope for this campaign design.

(xi) Single-pixel scalar architecture and Earth-like cloud coverage. All cadence and reconstruction statements in this study are strictly scoped to the adopted single-detector scalar photometric raster architecture. No head-to-head comparison against phase-registered multi-pixel imaging or spectrally resolved inversion is established here. Furthermore, at Earth-like cloud coverage ($f_c \approx 71\%$), the scalar photometric pipeline achieves only $r = 0.249 \pm 0.010$, falling well below the $r \geq 0.5$ threshold required for reliable continental imaging. We state plainly that this pipeline does not pass at Earth-like cloud cover. Finally, surface–cloud coupling ($r = 0.1444 \pm 0.0124$, loss $\Delta r = -0.198 \pm 0.017$) represents a structural, phase-correlated systematic that the cadence law cannot overcome.

These limitations narrow, but do not remove, the core findings: rotation is tractable under characterized geometry and costs no measurable information inside the supported harmonic range; weather, not photon noise, sets the image quality by more than two orders of magnitude in variance; the cadence gain is an independent-looks effect whose size is set by the assumed weather decorrelation time; and the algorithmic gain comes from eliminating the per-dwell-slot nuisance rather than from modeling the covariance of the weather.

8 Conclusions

1. Regularized generalized least squares on time-tagged samples (TDI), with exact anchored profiling of the per-dwell-slot cloud nuisance, recovers a rotating, cloud-free exoplanet with $r = 0.9786 \pm 0.0002$ (SSIM $= 0.848 \pm 0.001$, 10 paired seeds). Within the harmonic range the 36×72 grid supports ($\ell \leq 20$) no rotational band is lost; what remains is regularization and aperture. The formulation "rotation introduces virtually zero information loss" is contradicted by the measured statement: the phase-blind treatment of the same photons gives $r = 0.116$, so the time-registered operator is worth $+0.86$ in correlation, while temporal whitening itself is worth nothing at this cadence. A frozen-planet reference at the identical photon budget is not the other side of that comparison: it gives $r = 0.4074 \pm 0.0254$, *below* the rotating cloud-free result, because identical pose-passes make the time-domain design degenerate.

2. Clouds dominate the error budget. Per-sample cloud-induced variance exceeds coronal shot noise by 163–440 (12.8–

21.0 in standard deviation) over 23.7–71.0% realized cover at $t_s = 1800$ s; fidelity collapses to $r = 0.249 \pm 0.010$ at the top of that range, below the $r \geq 0.5$ threshold for continental mapping. Under the single-pixel scalar photometry architecture, the pipeline does not pass at Earth-like cloud coverage.

3. The cadence law is an independent-looks effect, and it is now measured under two declared resource ledgers. At fixed 8 h per raster position, $M_p : 4 \to 64$ takes r from 0.049 to 0.571 with every step unanimous across six paired seeds; forcing the same study into 89.87 d of wall clock changes the answer by less than the paired standard error at every M_p. What saturates is not r but $N_{\rm eff}$: on the Arm A cadence schedule, at $\tau_{\rm cloud} = 4$ d, 64 revisits are worth 12.1 independent looks, so the cadence requirement must be quoted together with the assumed weather decorrelation time. Under the 45 s/dwell ledger the duty cycle falls from 99.4% to 90.5% across the arm-B (wall-clock-constrained) configurations and from 99.4% to 90.9% across arm A; no configuration below $t_s = 429.6$ s was simulated, so any claim about ultrafast overheads is deleted.

4. Ablating the estimator locates the algorithmic gain in nuisance elimination, not in covariance modeling: exact profiling beats a fit that ignores the slot means by $\Delta r = +0.054 \pm 0.006$ (and by $+0.0568 \pm 0.0046$ in the white-noise matched contrast D vs. A, $p_t = 5.9 \times 10^{-7}$), whereas modeling the OU covariance slightly reduces correlation ($\Delta r = -0.0032 \pm 0.0019$, F vs. D). In the matched comparison between exact and approximate profiling under identical OU covariance (F vs. C), the gain is $\Delta r = +4.1 \times 10^{-5} \pm 3.0 \times 10^{-5}$ ($p_t = 0.20$), with a map norm difference of only $0.83\% \pm 0.22\%$. The larger 9.3% map norm difference between C and D is dominated by the OU-versus-white covariance switch rather than the profiling approximation.

5. Spin-ephemeris tolerances are measured: 5° of pole tilt costs 0.057 in r, 5° of assumed initial phase 0.047, 30° of phase leaves $r = 0.072$ and 60° leaves the map uncorrelated with the truth. We withdraw the claim that residual misalignments are refinable in flight via profile likelihood: the whitened χ^2 is numerically invariant ($< 10^{-11}$ relative) to assumed initial phase over $\pm 10°$ and monotone in assumed period across the grid tested, because a phase shift is a relabeling of a complete longitudinal basis. The spin state is observable through image quality, which peaks at truth on 4/4 seeds, but it is not recoverable from photometric residuals by grid search.

6. Resolution and detection are stated with their instrument limits. Cloud-free $r(\ell) \geq 0.987$ for all $\ell \leq 20$ (≥ 0.99 except the single degree $\ell = 17$), the ceiling imposed by the reconstruction grid (quadrature orthonormality error 0.018 at $\ell = 20$); under 55.8% clouds a five-point running mean of $r(\ell)$ crosses 0.5 at $\ell \simeq 9$ (~ 2200 km at one Earth radius), so no per-seed $\ell_{\rm eff} \approx 4$–5 crossing is quoted. Land–ocean separation at that cover, $d' = 0.732 \pm 0.081$ (mean $\pm$ per-seed SD; SE 0.026), exceeds all 40 members of the physical cloud-only null, giving $p \leq 0.024$ — the ensemble floor.

7. Mission mapping. The target table remains a reproducible nominal-proxy table (Chen & Kipping radius relation with 4.0% dispersion on the terran branch, a 2.04 $M_\oplus$ terran/neptunian transition, an inclination de-projection factor of 1.098, and an along-track velocity requirement of 1.34–5.81 km s^{-1} per 90 d across the five confirmed rows, Proxima Centauri b being the most demanding; the illustrative τ Cet e row — a signal since attributed to stellar activity (Figueira et al. 2025) — is the dynamically gentle case at $\sim 53\times$ lower Δv), and it is now also an uncertainty propagation: 50,000 draws per row over the reported $M \sin i$, distance and radius relations put a 68% interval on Δv and $\mathrm{SNR_C}$ whose width is set by the priors rather than by the astrophysics, leave the photon ranking of the five confirmed rows unchanged, and exclude the refuted τ Cet e signal from the ranking statistics instead of letting it inflate a "ranking changed" flag (Section 6.1). What *does* reorder the table is the reachability rule, and the two rules this paper is able to write down disagree: the $1/(1+\overline{\Delta v})$ mission score puts GJ 273 b first while the ephemeris module's access-window score puts Ross 128 b first; that is reported as a disagreement, not resolved. Science data volume is derived rather than asserted — 65,536 measurements per campaign, 2.10 MB under the declared 32-byte record and 0.262 MB of float32 values, with no compression to claim because the gridded product holds exactly one number per measurement — and 10.4 kB per map, so downlink is not the binding constraint; a 1 s time tag costs 0.0042° of spin phase and 8×10^{-6} in r, three decades below the phase errors that move the map. Swarms remain an asymptotic architectural scenario, and the co-focal test now measures rather than extrapolates that: 2.39× and 5.48× gains in harmonic mode SNR at $N = 2$ and $N = 4$ against a paired *loss* of 0.0376 in r when the joint inversion is run on the same photons.

8. Robustness is now measured rather than asserted, with two results that change how the pipeline should be used. Varying the weather parameters — correlation length 8°–20°, advection 3°–12° day^{-1}, decorrelation time 2–8 d — moves r by less than the seed-to-seed error, and the reason is that the reconstruction is nearly blind to the *assumed* cloud amplitude: over a 64× range of $\sigma_{\rm cl}$ the mean paired change in r is $\leq 2.6 \times 10^{-4}$ — for the anchored profiler, the free profiler and approximate deflation alike — while the whitened objective changes by a factor 8×10^4. The pre-flight requirement is therefore on weather *structure*, and $\sigma_{\rm cl}$ is fittable in flight. Second, the smoothing weight is far from optimal and the objective cannot say so: cloudy r rises monotonically to $+0.178 \pm 0.007$ at $\lambda = 10$, a gain driven by the suppression of uncorrelated high-ℓ reconstruction noise variance and genuine dichotomy (d') sharpening rather than harmonic band recovery, while cloud-free r peaks at $\lambda = 0.3$ and turns over. The headline numbers keep the frozen weight, and the missing truth-blind selection criterion is stated as the open methodological problem. Surface morphology is bounded the same way: fragmenting the continents costs nothing measurable, concentrating them into one mass costs 0.040 in r.

9. Surface–cloud phase coupling is an irreducible, structural failure mode. When cloud cover is topographically locked to

low-albedo surface features ($\beta_{\rm surf} = 0.25$), reconstruction fidelity drops precipitously to $r = 0.1444 \pm 0.0124$ (a paired loss of $\Delta r = -0.1978 \pm 0.0175$, $t = -11.3$), with land–ocean contrast collapsing to 0.379. Because phase-locked weather does not average out over orbital or rotational revisits, it represents a structural systematic that the cadence law cannot overcome.

10. External validity is now measured on four axes, and each one bounds how far a headline number travels. *Which planet:* the whole chain re-simulated over five independent geographies × four noise seeds gives cloudy profiled $r = 0.3173 \pm 0.0068$ (s.e.m., $n = 20$) with the fiducial geography sitting 0.032 above that mean — the shipped $r = 0.3422$ is the easiest of five planets, not a typical one, while cloud-free recovery (0.9789 ± 0.0008) is mid-pack and geography-robust. *Which operator:* replacing the solar-gravitational deflection kernel with a diffraction-limited Airy PSF of matched pitch, on the same truth, the same photons and one paired seed, takes r from 0.3314 to 0.1985 — a ratio of 0.599, so roughly two fifths of the recovered correlation is carried by the sampling scheme rather than by the lens. *Which prior:* total variation is measurably worse than TDI ($\Delta r = +0.0183$, bootstrap $[+0.0160, +0.0213]$) but that arm post-processes the TDI solution, and a dictionary learned from the TDI map itself is statistically indistinguishable from it (-0.0020, $[-0.0083, +0.0047]$); at four seeds the sign-test floor is $p = 0.125$, so nothing here is resolvable beyond "TV is not free". *Which omitted terms:* the solar-oblateness term the operator leaves out is bounded at the adopted dynamical $J_2 = 2.2 \times 10^{-7}$, which displaces the Einstein-ring radius by at most 5.9×10^{-4} m $= 2.8 \times 10^{-5}$ image cells at $n = 64$, more than four orders of magnitude below one cell; and the coronal refraction the chromatic sweep imagines is equivalent to a residual of 8.6×10^{-14} coronal units, $1.16 \times 10^{5}\times$ *below* the smallest residual this pipeline has ever injected, so no refractive Δr is quotable from the archived drift curve — the honest statement is that optical refraction lies outside the tested range, and that the weak-refraction model itself fails beyond 1.594 m (188 MHz).

Data and Code Availability

The simulation and inversion framework developed in this paper is publicly available on GitHub at `https://github.com/SanskarSontakke/sgl-time-domain-imaging` and permanently archived under the release tag of the same repository; the code and the archive are released together under version 1.0.0; no DOI is quoted here because the version-of-record identifier is minted from the release tag at publication. The entire suite of numerical experiments, forward simulations, and figure regenerations runs from a clean clone with

```
python3 -m reproduce_all -fast
python3 -m reproduce_all -full
python3 -m reproduce_all -stages "quadrature
                clouds 3"
```

(the fast tier re-runs the analytic and audit stages and re-derives every shipped archive from its checkpoints in about four and a half minutes on four cores; the full tier re-runs all 70 stage invocations — the 28 subcommands of t run_experiments.py plus the standalone target generator — including the ten-seed ensembles, from a scratch copy of the archive tree). Both tiers print a diff report against the shipped `results/audit.json` and exit non-zero on any mismatch, so the printed commands are themselves a verification of the numbers in this paper. The legacy `src/run_experiments.py` front end remains available stage by stage (`cd src && python3 run_experiments.py <stage>`; `-help` lists all subcommands). All numerical results reported in this work are stored in the accompanying `results/` directory. Summary statistics reside in `results/*.npz` and `results/*.json`, while individual per-seed runs, harmonic band decompositions, and morphology arrays reside in `results/parts/*.npz`. Every archive is regenerated from those checkpoints by the `merge` stage, so no reported number depends on a stored intermediate; the `tests/` directory additionally asserts each headline value in this paper against the archive, and `tools/verify_flat.sh` asserts that the tree contains no superseded or versioned artefacts. The extension analyses are a separate layer and are run with `python3 -m reproduce_all -fast -ext` (or individually, `cd src && python3 -m ext.<module>`); Appendix C states what each one contributes and how its provenance is recorded, and Appendix B states the reproduction contract, including the exact meaning of "identical" in the diff above and the limits of what this environment can rebuild.

Acknowledgments

This work used open-source software (NumPy, SciPy, Matplotlib, scikit-image). The author is grateful for the detailed pre-submission review recorded in the repository's `reviews/` directory; it was carried out internally, before submission, and changed several of the numbers reported here rather than only their wording. No external funding was received.

References

Anglada-Escudé, G., Amado, P. J., Barnes, J., et al. 2016, Nature, 536, 437

Astudillo-Defru, N., Forveille, T., Bonfils, X., et al. 2017, A&A, 602, A88

Bonfils, X., Astudillo-Defru, N., Díaz, R., et al. 2018, A&A, 613, A25

Chen, J., & Kipping, D. 2017, ApJ, 834, 17

Dreizler, S., Jeffers, S. V., Rodríguez, E., et al. 2020, MNRAS, 493, 536

Dreizler, S., Luque, R., Ribas, I., et al. 2024, A&A, 684, A117

Eshleman, V. R. 1979, Science, 205, 1133

Feng, F., Tuomi, M., Jones, H. R. A., et al. 2017, AJ, 154, 135

Figueira, P., Faria, J. P., Silva, A. M., et al. 2025, A&A, 700, A174 (A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level)

Friedman, L. D., Garber, D., Turyshev, S. G., et al. 2024, Exp. Astron., 57, 1

Helvajian, H., Rosenthal, A., Poklemba, J., et al. 2023, J. Spacecr. Rockets, 60, 829

Madurowicz, A., & Macintosh, B. 2022, ApJ, 930, 19

Aerospace Corporation & NASA NIAC 2022, Atomic Planar Power for Lightweight Exploration (APPLE), NIAC Phase II study

Toth, V. T. 2025, Res. Astron. Astrophys., 25, 065007, doi:10.1088/1674-4527/add3a3 (arXiv:2504.18630)

Toth, V. T., & Turyshev, S. G. 2021, Phys. Rev. D, 103, 124038

Toth, V. T., & Turyshev, S. G. 2023, MNRAS, 525, 5846

Turyshev, S. G., & Toth, V. T. 2022b, Phys. Rev. D, 106, 044059 (arXiv:2206.03037)

Turyshev, S. G., & Toth, V. T. 2023, Phys. Rev. D, 107, 104063 (arXiv:2301.07495)

Turyshev, S. G. 2026a, Phys. Rev. D, 113, 023034 (arXiv:2506.20236)

Turyshev, S. G. 2026b, arXiv:2606.14899

Turyshev, S. G. 2026c, arXiv:2602.04198

Turyshev, S. G. 2026d, arXiv:2606.29138

Turyshev, S. G., & Toth, V. T. 2017, Phys. Rev. D, 96, 024008

Turyshev, S. G., & Toth, V. T. 2019, Phys. Rev. D, 99, 024044

Turyshev, S. G., & Toth, V. T. 2020, Phys. Rev. D, 102, 024038

Turyshev, S. G., & Toth, V. T. 2021, Phys. Rev. D, 103, 064076

Turyshev, S. G., & Toth, V. T. 2022a, MNRAS, 515, 6122

Turyshev, S. G., Shao, M., Toth, V. T., et al. 2020, Direct Multipixel Imaging and Spectroscopy of an Exoplanet with a Solar Gravity Lens Mission, NASA NIAC Phase II Final Report

A Per-Seed Ensemble Statistics and Bootstrap Intervals

Table 14 presents the per-seed Pearson correlation coefficients r across the 10 paired seeds (seeds 11–20; $s = 0, \ldots, 9$), along with the ensemble mean $\pm$ standard error and the 95% percentile bootstrap confidence interval (derived from 20,000 resamples of the 10 seeds, matching the implementation in `src/expcommon.py`). All 10 seeds share the identical synthetic planetary geography (`TRUTH_SEED = 7`) and vary only in the realization of the stochastic cloud field (cloud seed $100 +$ seed) and photon noise (seed). Scene diversity across distinct planetary landforms is evaluated separately in Section 5.14.

B Reproduction Contract and Verification Gates

Every number in this paper is produced by one of two entry points, and the distinction between them is the distinction between a claim and a check.

B.1 The two entry points

The generation side is `src/run_experiments.py`, whose 28 subcommands each write one archive (or one set of per-seed checkpoints) under `results/`, plus one standalone generator, `src/targets.py`, which writes the target catalogue archive, this paper's target table and its figure; both are registered as stages of the driver, so no artifact-producing script sits outside it. The verification side is the module-level driver `python3 -m reproduce_all`, which never writes to the shipped tree: it copies `results/` to a scratch directory, points the harness at the copy through the `SGL_RESULTS_DIR` environment variable, re-runs stages there, and diffs the result against the shipped archives.

B.2 What "identical" means in the diff

A numeric reproduction claim is only as good as its comparison operator, so we state ours. Each archive is reduced to a flat leaf map: JSON objects to dotted paths, arrays to {<name>: shape, dtype, SHA-256 of the raw buffer}. Five rules then apply.

(1) *Exact by default.* The tolerance is zero; a nonzero `-tol` must be passed explicitly, and the only place we use it is 10^{-12} for quantities that depend on BLAS summation order. (2) *Timestamps are ignored by leaf name*, not by value, so `provenance.generated_utc` cannot smuggle a real change past the check, and a field that *becomes* a timestamp is still caught. (3) *Types are part of the value*: `true` and 1 differ, and are reported as a "bool" mismatch. (4) *NaN masks are compared*: two all-NaN arrays match, and an array whose NaN pattern changed does not, even if the finite entries agree. (5) *String labels are compared with their dtype*: a <U9 and a <U8

Table 14: Per-seed Pearson correlation coefficients r for key pipeline configurations across the 10 paired seeds ($s = 0, \ldots, 9$). Ensembles represent sampling variations over weather realizations and photon shot noise on a fixed planetary albedo map. Intervals are 95% percentile bootstrap confidence bounds based on 20,000 resamples.

Configuration	s_0	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9	Mean ± SE	95% Bootstrap CI
Rotating cloud-free ($f_c = 0$)	0.9786	0.9781	0.9790	0.9790	0.9780	0.9782	0.9788	0.9789	0.9799	0.9776	0.9786 ± 0.0002	[0.9783, 0.9790]
TDI fiducial ($f_c = 0.558$, Row F)	0.3314	0.3290	0.3958	0.3404	0.3358	0.3422	0.3271	0.3246	0.3143	0.3820	0.3422 ± 0.0082	[0.3285, 0.3590]
White GLS (B3 / Row A)	0.2765	0.3042	0.3253	0.2670	0.3026	0.2725	0.2461	0.2799	0.2652	0.3468	0.2886 ± 0.0097	[0.2716, 0.3075]
Exact profile, white (Row D)	0.3305	0.3386	0.3961	0.3395	0.3438	0.3390	0.3214	0.3294	0.3263	0.3897	0.3454 ± 0.0082	[0.3320, 0.3622]
Approx. deflate, OU (Row C)	0.3316	0.3289	0.3956	0.3402	0.3357	0.3421	0.3271	0.3246	0.3143	0.3819	0.3422 ± 0.0082	[0.3285, 0.3588]
Static reference ($f_c = 0$)	0.4688	0.4509	0.3650	0.5125	0.3225	0.2838	0.3555	0.3747	0.5231	0.4172	0.4074 ± 0.0254	[0.3608, 0.4547]
Surface-coupled ($\beta_{\mathrm{surf}} = 0.25$)	0.1281	0.1552	0.1074	0.1162	0.1899	0.1446	0.2254	0.1007	0.1558	0.1211	0.1444 ± 0.0124	[0.1233, 0.1694]
Key Paired Contrasts:												
Profiling gain, white ($D - A$)	+0.0540	+0.0344	+0.0708	+0.0725	+0.0412	+0.0665	+0.0753	+0.0495	+0.0611	+0.0429	$+0.0568 \pm 0.0046$	[+0.0483, +0.0651]
OU covariance effect ($F - D$)	+0.0009	−0.0096	−0.0003	+0.0009	−0.0080	+0.0032	+0.0057	−0.0048	−0.0120	−0.0077	-0.0032 ± 0.0019	[−0.0067, +0.0004]
Exact vs. approx. OU ($F - C$)	−0.0002	+0.0001	+0.0002	+0.0002	+0.0001	+0.0001	0.0000	0.0000	0.0000	+0.0001	$+4.1 \times 10^{-5} \pm 3.0 \times 10^{-5}$	$[-2.1 \times 10^{-5}, +9.0 \times 10^{-5}]$

Table 15: The reproduction driver's modes. "Stages" counts entries of `reproduce_all.full_stages()`; the fast tier is the eight stages that re-derive every shipped archive from its checkpoints and require no simulation.

Invocation	Stages	What it establishes
`-fast`	8	the analytic and diagnostic archives regenerate bit-for-bit from the shipped checkpoints
`-stages "quadrature clouds 3"`	2	an explicit subset, for a targeted audit
`-full`	70	every simulation stage re-runs from scratch, including the 10-seed ensembles
`-full -from-zero`	70	as above with all per-seed checkpoints deleted, so nothing is inherited
`-fast -ext`	8 + 14	additionally runs the extension modules at their fast tier
`-make-baseline PATH`	0	freezes a byte-level manifest of the shipped tree
`-check`	0	compares the shipped tree against that manifest without running anything

label array holding the same text differ, which is how a renamed baseline is caught even when the values are untouched.

B.3 Gates

Three independent gates run over the tree. `tools/verify_flat.sh` enforces the single-version consolidation: no revision markup or version narrative in the manuscript, no superseded directories, no version-bearing keys or labels inside the archives, version-free prose in the README and the demo app, working entry points, the test suite, and the presence of all 14 extension archives. The pytest suite (200 test functions, 347 collected checks, 326 of them in the default fast tier) checks physics and geometry against hand-derived values, checks the shipped archives against each other, and — in its `slow` tier, deselected by default — re-solves the fiducial pipeline and reproduces five metrics for two estimators to 10^{-12}. `tools/array_manifest.py` provides the byte-level manifest behind the last two rows of Table 15. All three are wired into `.github/workflows/verify.yml`, which runs the fast tier of each on every push; the simulation tier is deliberately excluded from CI because it costs hours on four cores.

B.4 Rebuilding the manuscript itself, and what stays outside CI

Two limits should be stated rather than discovered by the reader. First, nothing in this paper is hand-maintained except the prose. Every figure is regenerated from the shipped archives (`src/make_figures.py`, or `reproduce_all -figures` inside the scratch tree), the target table (`paper/targets_table.tex`) is emitted by `src/targets.py` rather than edited by hand, and `main.pdf` is rebuilt from `main.tex` in the same audit environment used for everything above: a single `latexmk -pdf` pass over a standard two-column article, needing no class or style file beyond `texlive-latex-base`, `texlive-latex-recommended`, `texlive-latex-extra` and `texlive-fonts-recommended` (pdfTeX 1.40.22, latexmk 4.76). The shipped PDF is that build — 36 pages, no undefined references or citations, no overfull boxes — and continuous integration now compiles it on every push, behind a test that checks the brace balance of every generated `.tex` fragment first, because a single stray closing brace in a generated caption is enough to abort the build and leave every cross-reference in the manuscript unresolved. What CI still does *not* run is the simulation tier: it costs hours on four cores, so it is left to the `slow` test tier and to `reproduce_all -full`. Second, reproducing the shipped numbers requires the pinned stack (Python 3.12.13, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2, scikit-image 0.26.0); the comparisons above are exact, so a different BLAS can legitimately fail them at the 10^{-12} level, which is why that tolerance exists.

C Extension Modules and Their Provenance Contract

Sixteen additional analyses are needed before the argument closes. Fourteen are implemented as separate modules under `src/ext/`; they are deliberately *not* merged into the pipeline archives, because several of them answer their question with a proxy rather than with the full model, and a reader is entitled to know which is which without reading the source. The remaining two are not analyses at all and produce no archive: one is release engineering (the single `reproduce_all` entry point whose invocations Table 15 lists, the byte-level manifest behind its last two rows, and the continuous-integration job that re-runs the fast tier and compiles this manuscript on every push), and the other is a set of small corrections applied in place while the files were open — a wrong arithmetic result here, a mis-stated unit there. Neither is counted among the fourteen in Table 16, and neither is claimed as a result.

Each module writes exactly two files, `results/ext_<module>.{npz,json}`, and its JSON always contains three blocks. `headline` is a short machine-readable summary (every string in it is capped at 80 characters, except `conclusion` and `status`). `caveats` is required to be non-empty and to state the proxy, the sample size, or the analytic assumption on which the headline rests. `provenance` records the module name, the *tier* it ran at (`fast` or `full`, stamped automatically from the command line), the full argv, the NumPy and Python versions, and a frozen copy of the fiducial configuration and physical constants used. A module that cannot compute something writes `status: "not_established"` and empty arrays rather than inventing a number; the difference is machine-checkable, because an empty array cannot be mistaken for a result. Every extension number quoted in this paper is taken from the `full`-tier archive; `-fast -ext` re-runs the same modules on their reduced seed lists, so the diff it prints against the shipped files is expected rather than an error, and the seed count that produced any given number is readable from `provenance.argv`; the shipped full-tier archives are rebuilt by `tools/run_ext_full.sh`.

These rules are enforced, not promised: `tests/test_ext_archives.py` re-checks every shipped extension archive and fails if a headline is silent about its sample size, if a proxy method is not disclosed, if an array is empty where its JSON claims a value, if the fiducial block or the constants drift from `sglsim`, or if any single-aperture correlation is reported at a value that is not the base result to within 2×10^{-4}. It also fails if a headline reproduces the fiducial cloudy correlation $r = 0.3422$ under a name that claims to be a different (loss) quantity — the specific confusion a mislabelled static-offset baseline can produce.

Table 16 lists the modules and the question each one answers. The numbers they produce are used in this paper only where the archive declares its tier and passes the parity checks above; where a module's conclusion is provisional we say so at the point of use rather than in a footnote.

Three of these deserve a sentence of their own, because all three changed a claim we had previously made. `cloud_physics` was written to test the surface-coupling collapse of Section 5.13, but the model it implements — an OU field plus a *static* orographic offset — has no rotational phase locking and therefore cannot confirm or refute that collapse; the archive now says so explicitly, quotes the spine's own surface-coupled result from `audit.json:robust_clim` as the reference being compared against, and renames its own quantity to what it measures. A second finding from that same archive changed how we read its numbers: once every arm reports its realized cloud cover, the ordering of the arm correlations is exactly the ordering of their covers ($\rho = -1$ over five arms), so the module bounds a coverage-dominated sensitivity range rather than ranking weather models. `truth_ensemble` is the one module that can move a headline: a conclusion drawn from ten weather realisations on *one* geography is a statement about weather, and only a truth ensemble shows whether it is also a statement about surfaces. `fisher_bounds` was written to quote a Cramér–Rao floor for the map, and the floor turned out to be weaker than predicting a constant albedo by three orders of magnitude (Section 4.4) — which is a property of a rank-deficient $\mathsf{A}_{\mathrm{prof}}$ and its pseudo-inverse, not of the estimator, and the archive now carries that verdict as a flag instead of printing the number as a target.

D Metric Definitions

Five numbers are quoted throughout, and four of them are computed on a restricted region. This appendix fixes every convention, because the differences between our values and those of published SGL studies are almost always differences of definition rather than of method. All of it lives in `sglsim.eval_map`, `expcommon.evaluate`, `sglsim.detection_dprime` and `expcommon.paired_stats`.

D.1 Region of application

Except where stated, every metric is evaluated on cell centres with $|\mathrm{lat}| \le 60^\circ$. The exclusion is not cosmetic: the two polar rows of the 36×72 basis are never sampled by the circular raster at $n = 64$ (Section 4.4), so including them would score regularisation artefacts in cells the data cannot constrain. *One* metric breaks this rule: the albedo `bias` is the difference of full-grid means, $\langle \hat{\mathbf{s}} \rangle - \langle \mathbf{s} \rangle$ over all 2592 cells, because it is meant to be an energy-accounting statement about the debiasing step rather than a fidelity statement about imaged terrain. Readers comparing bias with r should keep the two supports distinct.

D.2 The four fidelity metrics

- **Pearson** r is the ordinary sample correlation of the debiased map against truth over the selected cells. It is invari-

Table 16: The fourteen extension modules. "Question" is what the archive answers; "basis" records whether the computation is the same operator used in the pipeline or a declared proxy for it.

Module	Question	Basis
`fisher_bounds`	What is the Cramér–Rao floor for the albedo map under the fiducial noise model, and how much of the Fisher information does nuisance profiling remove?	analytic, same F and C; the floor it returns is degenerate, see below
`corona_screen`	Can a wavelength-dependent coronal phase screen masquerade as, or mask, the $\sigma_{\rm cl}$ term?	refractive-screen proxy, not MHD
`disk_kernel`	How do the kernel energy fractions and off-diagonal mass depend on grid size, and how large is the neglected solar-oblateness term?	point kernel plus disk-integrated proxy
`ephemeris`	What accuracy in the spin state and in $\beta_{\rm ecl}$ is required before the phase-locked geometry is wrong by a resolution element?	analytic propagation, assumed ephemeris errors
`portability`	Does reconstruction fidelity survive a non-SGL forward operator, and with it the resolution claims?	alternative kernel, declared proxy
`telemetry`	What downlink, storage and timing chain does the campaign actually require, derived from the ledger rather than asserted?	derived from `ledger.npz`
`truth_ensemble`	Do the headline conclusions survive a different planetary geography, rather than one fixed truth map?	full re-simulation, N truth maps
`evidence_lambda`	Does a marginal-likelihood criterion select the regularisation weight that image quality prefers?	Laplace approximation to the evidence; it has no interior optimum on the tested grid, Section 5.16
`scheduling`	How much is lost when dwells are opportunistic rather than uniformly spaced?	resampled cadence, same operator
`priors`	Would a modern prior (total variation, learned dictionary) change the method comparison?	declared surrogate priors, Section 5.7
`observable_roc`	At what detectability d' does each scientifically meaningful feature enter, and against which nulls?	matched-filter d' on the reconstructed maps
`cloud_physics`	Does the collapse under surface-coupled clouds survive a topographically anchored cloud model?	OU noise plus static orographic offset (not the spine's surface-coupling mechanism); the arms realize different cloud covers, so the module returns a sensitivity range, Section 5.13
`swarm`	How much does a two-aperture joint inversion gain over the single spacecraft?	joint solve of the same scene, $N = 2$
`target_uncertainty`	Which entries of the target table survive propagated range, photon-flux and $D_{\rm img}$ uncertainty?	analytic error propagation on the nominal proxy distances

ant to any affine rescaling of the map, which is precisely why it is paired with contrast and bias rather than reported alone.

- **SSIM** is computed on the two-dimensional latitude–longitude grid with `data_range` set to the truth range over the selection. Because it uses a fixed square window in *index* space, its cells are not equal area: it weights high latitudes more than an area-corrected statistic would. We keep it because it is the field's convention, and note the distortion rather than silently correcting it.

- **NRMSE** is $\sqrt{\langle(\hat{s}-s)^2\rangle}$ divided by the same truth range, so it is a fraction of the albedo dynamic range, not of the RMS signal.

- **Contrast** is the ratio of the retrieved land–ocean mean separation to the true one, with land defined by a truth albedo threshold of 0.15. It equals one when the reconstruction preserves the amplitude of the dichotomy, and it is the metric that distinguishes "sharper" from "more accurate" in the large-λ control of Section 5.16.

D.3 Detectability and its nulls

The land–ocean statistic is $d' = (\mu_{\rm land} - \mu_{\rm ocean})/\sqrt{(\sigma^2_{\rm land} + \sigma^2_{\rm ocean})/2}$ computed on the *retrieved* map with the *true* labels, the same $|\mathrm{lat}| \le 60^\circ$ support, and a 10^{-12} guard in the pooled variance. Two nulls are reported against it and they answer different questions, which is why both are printed and neither is called "the" p-value. The *label-permutation* null reassigns the class labels at random on the same map and asks whether this labelling is unusual; it is a statement about the labelling, not about detection. The *cloud-only physical* null runs the whole pipeline on scenes with no surface contrast at all and asks whether the instrument produces this separation from weather alone; that is the null a detection claim must clear. With 2000 and 40 members respectively, the two floors are $1/2001 = 5.0 \times 10^{-4}$ and $1/41 = 0.024$, and both are quoted as saturation statements in Section 5.17. A Fourier phase-scrambling surrogate used in

an earlier draft is retained in the code only for reproducibility and is deprecated: it ignores cos(lat) cell areas, does not respect the sphere, and contains neither photon noise nor the reconstruction operator, so it is not a null for any statistic reported here.

D.4 Spectral fidelity

$r(\ell)$ is computed by projecting the map onto orthonormal *real spherical harmonics* sampled at the cell centres, not by taking FFT magnitudes. The distinction matters: on an equirectangular grid the longitude cells are not equal area, the map is periodic in longitude but not in latitude, and the index $\sqrt{k_x^2+k_y^2}$ has no reason to coincide with degree ℓ. Band means are reported both unweighted and $(2\ell+1)$-weighted, because the unweighted version hides a gain and a loss of similar size inside one band (Section 5.16).

D.5 Ensemble statistics

A paired comparison of two configurations over the $N = 10$ seeds reports the mean difference, the standard error with `ddof=1`, a paired t-test, an exact two-sided sign test using the central binomial tail $\min(1, 2\sum_{i\le k}\binom{N}{i}/2^N)$ with k the smaller of the positive and negative counts, and a 95% percentile bootstrap over 20,000 resamples with a fixed seed. The sign-test convention is stated because the alternative (twice the point probability at the observed split) is anti-conservative: at $N = 10$ with an 8/2 split it returns 0.088 where the central tail returns 0.109. The attainable floor is $2/2^{10} = 1.95\times10^{-3}$, which is why no sign-flip p-value in this paper is printed below it. Effective independent looks are $N_{\mathrm{eff}} = M_p^2/\left(M_p + 2\sum_k (M_p-k)\rho_k\right)$ with $\rho_k = e^{-k\Delta t/\tau}$, i.e. pair-weighted by the temporal covariance of the weather; dropping the weights collapses N_{eff} from 12.1 to 59.8-looking-like-independent at $M_p = 64$, which is the error an earlier draft made.